UNIT - 10

Representation of Alternating Quantities, Concept of Phasor and Phasor diagram

Study A.C. through Pure Resistance alone, A.C. through pure Inductance alone, & A.C. through pure Capacitance alone

ACS

A.C. Series Circuits

INTRODUCTION. In this chapter, we shall study simple circuits containing resistance, inductance and capacitance in series. In actual practice, a.c. circuits contain two or more than two components in series. An a.c. series circuit may be

(i) R–L series circuit.

(ii) R–C series circuit.

(iii) R–L–C series circuit.

ACS–1. RESISTANCE AND INDUCTANCE IN SERIES

The figure ACS–1 shows a simple circuit containing resistance R ohms and pure inductance of L Henry in series.

A voltage of *R.M.S.* value V is applied across the combination. Let the current flowing through the circuit is I amperes.

Let V_R be the voltage across R and V_L is the voltage across inductance L.

The circuit can be easily solved by drawing the phasor diagram.

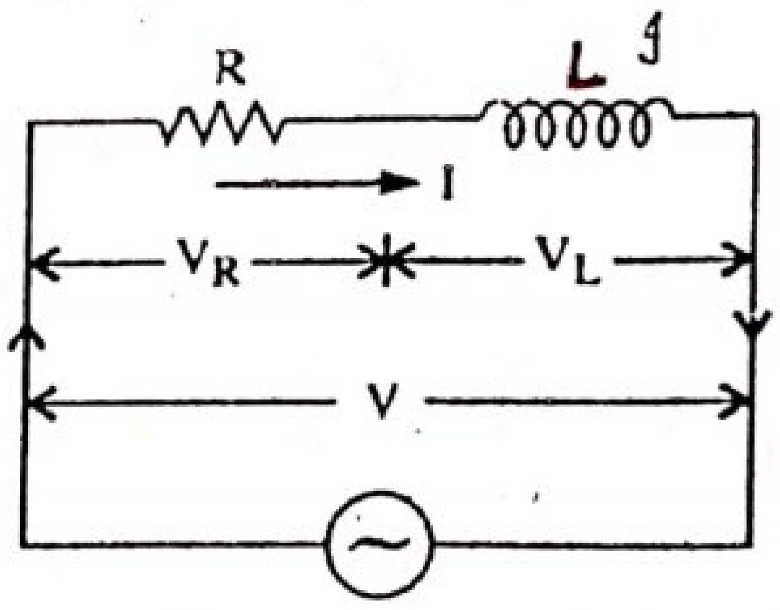

Fig. ACS—1

Drawing Phasor diagram. Since in series circuit, the current is passing through both resistance and inductance, so it should be chosen as reference vector.

The phasor diagram is shown in fig. ACS–2.

Taking I, as the reference vector, it is shown as OB drawn along X–axis. The voltage across pure resistance R is ($V_R = I.R$) and acts in phase with the current as shown in fig. ACS–2.

It is taken as $\overrightarrow{OA}$ in the same line of current.

The voltage across pure inductance is ($V_L = I \cdot X_L$) and leads the current by 90° (shown in figure) and it is represented by $\overrightarrow{AC}$ to the same scale.

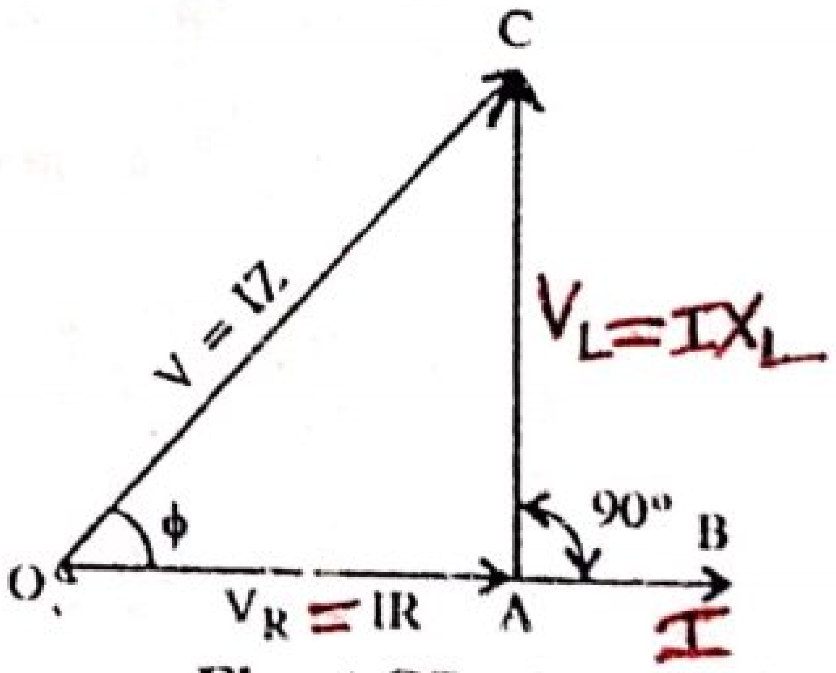

Fig. ACS—2

The applied voltage V is the sum of phasors OA and AC. This is given by OC. $\overrightarrow{OC}$ can be measured and checked as per scale.

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(Resistance) X_L = for Inductance
$X_L = \frac{V_L}{I}$

From right angled triangle OAC.

$$OC^2 = OA^2 + AC^2$$

or
$$V^2 = (V_R)^2 + (V_L)^2$$
$$(V)^2 = (I.R)^2 + (I.X_L)^2$$
$$V^2 = I^2 (R^2 + X_L^2)$$
$$V = I\sqrt{R^2 + X_L^2}$$
$$V = I.Z. \quad \text{or} \quad I = V/Z, \quad Z = V/I$$

where $Z = \sqrt{R^2 + X_L^2}$ is known as impedence of the circuit and is measured in ohms.

The impedence triangle. Since the current is common to all V_R, V_L and V, so it may be nitted and *impedence* triangle constructed with only R, X_L and Z is as shown in fig. ACS–3.

From the impedence triangle *(removing I from impedance triangle)*, impedence of the circuit $= \sqrt{R^2 + X_L^2}$, where $X_L = \omega L = 2\pi f.L$ is the inductive reactance.

$$\therefore \quad \tan\phi = \frac{X_L}{R}$$

or
$$\phi = \tan^{-1}\frac{X_L}{R} \text{ is known as the phase angle.}$$

The current lags behind the voltage by this angle ϕ.

$\cos\phi = \dfrac{R}{Z}$ is known as the power factor of the circuit and is ly a ratio.

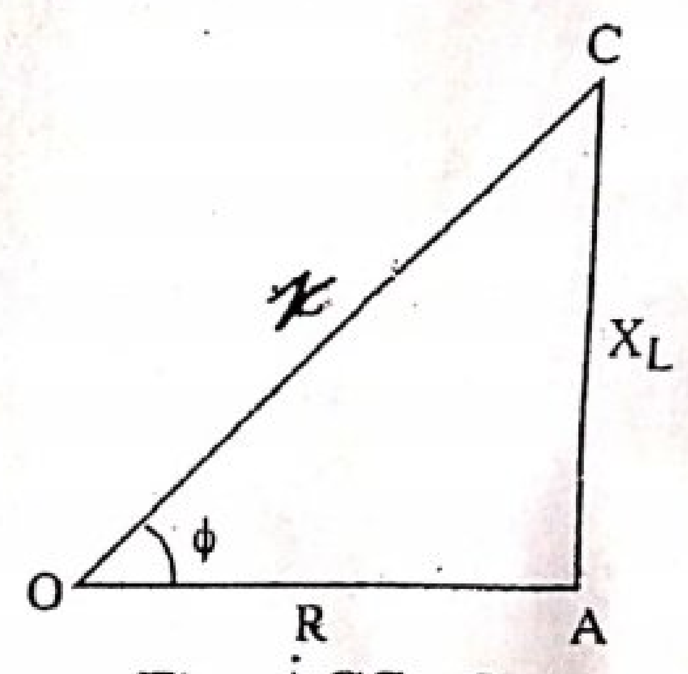

Fig. ACS—3

Power : The instantaneous value of applied voltage is given by the expression

$$v = V_m \sin\omega t.$$

then
$$i = I_m \sin(\omega t - \phi)$$

$\therefore$ Instantaneous power, $p = vi$

$$= V_m \sin\omega t.\, I_m \sin(\omega t - \phi)$$
$$= \frac{V_m . I_m}{2}\, 2.\sin\omega t \sin(\omega t - \phi)$$
$$= \frac{V_m}{\sqrt{2}} . \frac{I_m}{\sqrt{2}}\left[\cos\phi - \cos(2\omega t - \phi)\right]$$
$$= \frac{V_m}{\sqrt{2}} . \frac{I_m}{\sqrt{2}} . \cos\phi - \frac{V_m}{\sqrt{2}} . \frac{I_m}{\sqrt{2}} . \cos(2\omega t - \phi)$$

cos

Average power consumed in the circuit over a complete cycle,

$$P = \text{average of } \frac{V_m}{\sqrt{2}} . \frac{I_m}{\sqrt{2}} . \cos\phi - \text{average of } \frac{V_m}{\sqrt{2}} . \frac{I_m}{\sqrt{2}} . \cos(2\omega t - \phi)$$

or $$P = \frac{V_m}{\sqrt{2}} \cdot \frac{I_m}{\sqrt{2}} \cdot \cos\phi - \text{zero}$$

or $$P = V_{r.m.s} I_{r.m.s.} \cos\phi$$

or $$P = V.I. \cos\phi$$

Here $\cos\phi$ is called power factor of the circuit.

From phasor diagram,

$$\cos\phi = \frac{V_R}{V} = \frac{I \cdot R}{I \cdot Z} = \frac{R}{Z}$$

So, power factor is defined as the cosine of the angle between voltage and current in an a.c. :ircuit.

Or

Power factor is also defined as the ratio of resistance to impedence of an a.c. circuit.

Alternatively.

$$\text{Power} = V.I. \cos\phi$$

$$= (I.Z.).I.\left(\frac{R}{Z}\right)$$

$$= I^2.\ R$$

From the above, it is clear that power is consumed in resistance only and inductance does not onsume any power.

Power Curve. The power curve for an R–L series circuit is shown in fig. ACS–4. To make ie treatment more clear it is assumed that the current lags the voltage by 30°. *i.e.* ϕ = 30°. It is een that power is negative between 0°and 30°and between 180° to 210°. During the rest of the ycle the power is positive. Since the area under the positive loop is greater than the area under ie negative loop, the net power over the cycle is positive. Hence a definite quantity of power is onsumed by this circuit.

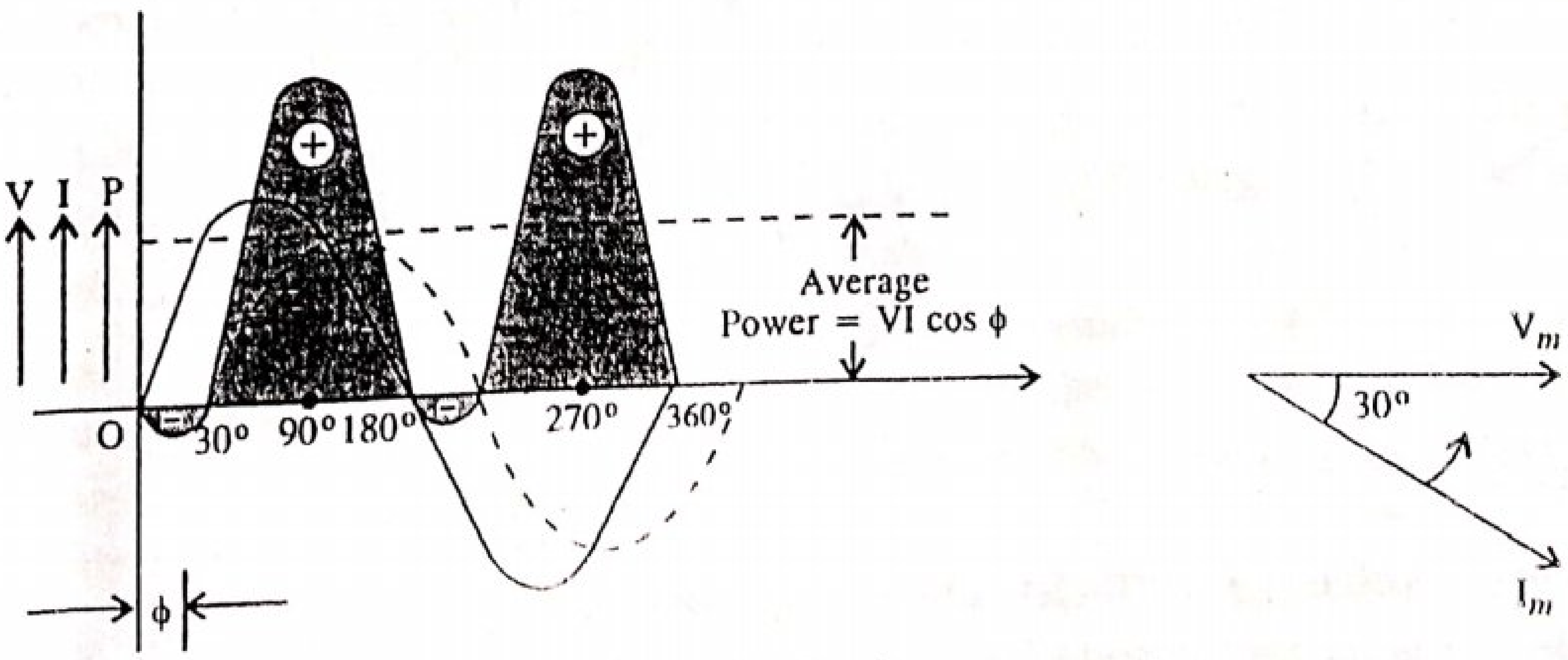

Fig. ACS—4

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Power factor. It may be defined as :

(i) Cosine of the angle between voltage and current –

(ii) The ratio $\frac{R}{Z} = \frac{\text{Resistance}}{\text{Impedence}}$

(iii) The ratio $\frac{\text{True Power}}{\text{Apparent Power}} = \frac{V.I.\cos\phi}{VI}$

The phase angle between voltage and current in a pure resistive circuit is zero,i.e. $\phi = 0°$. Therefore, power factor of a pure resistive circuit is $\cos 0° = 1$. Similarly for pure inductive and capacitive circuits the phase angle between voltage and current is 90°. Therefore, power factor for pure inductive and capacitive circuits is $\cos 90° = 0$, For other circuits containing R, L and C in varying values, the power factor will lie between 0 and 1. This may be noted that power factor can never have a value greater than 1.

It has also been seen that power factor may be lagging or leading depending upon whether the current lags behind or leads the voltage. Thus if a circuit has a power factor 0·8 and the current lags the voltage, we generally write power factor as 0·8 lagging.

Sometimes power factor is expressed as a percentage. Thus 0·8 lagging power may be expressed as 80% lagging.

ACS-2. TRUE POWER AND REACTIVE POWER

The power which is really or actually consumed in the circuit is called **True Power or Real power or Active power.** As already discussed that power is consumed only in resistance. A pure inductor or capacitor do not consume any power. The power received from a source in half cycle is returned to the source in the next half cycle. This circulating power is called Reactive power. It does not do any useful work in the circuit.

In pure resistive circuit, current and voltage are in phase whereas they are 90° out of phase in pure L and C circuits.

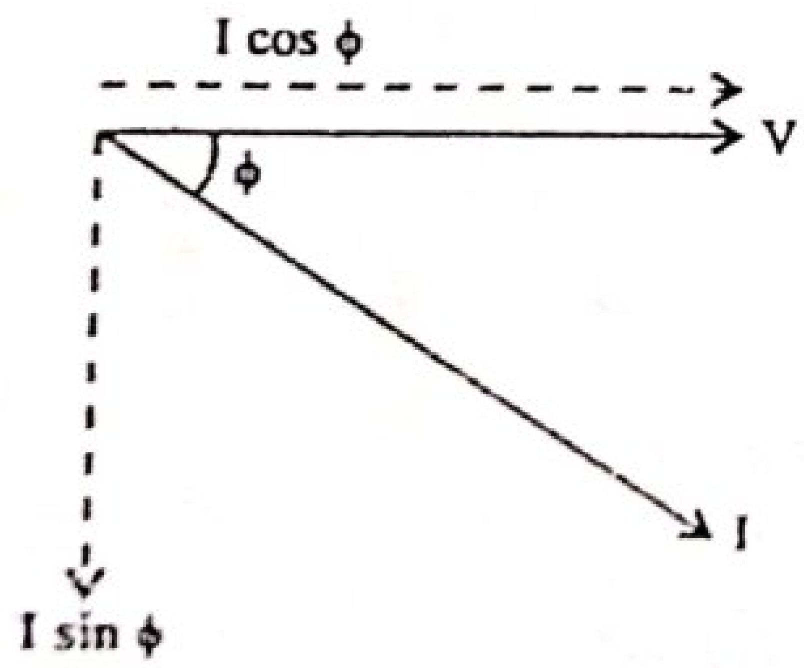

Fig. ACS—5

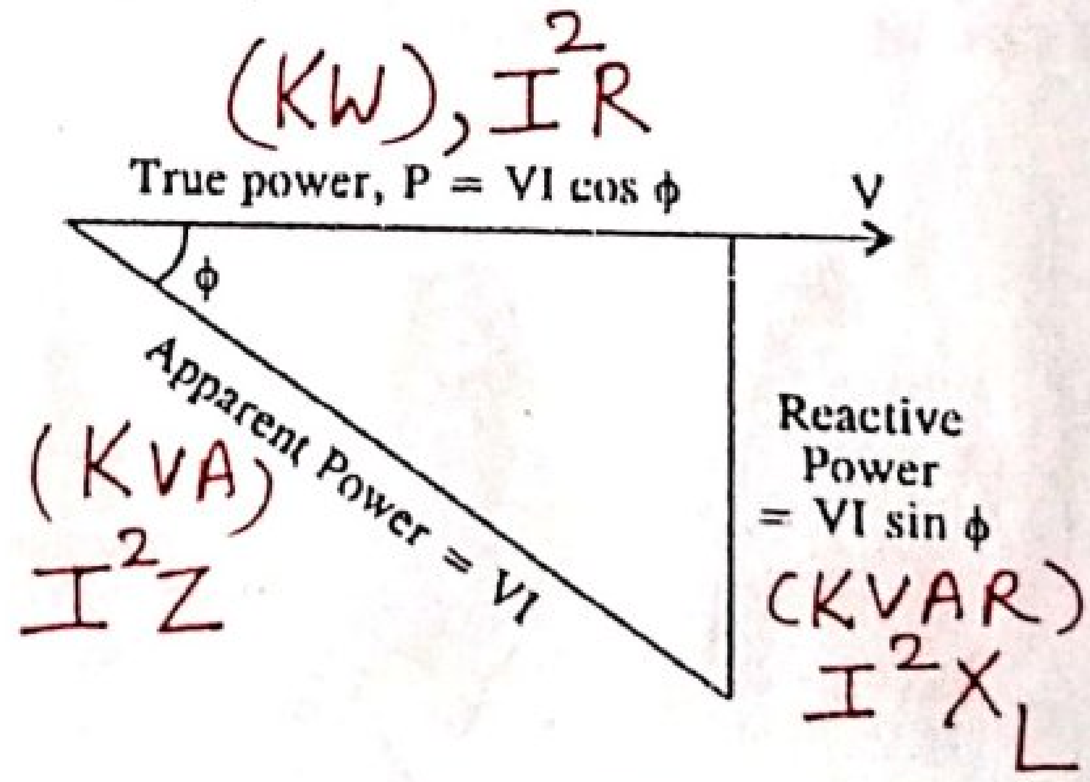

Fig. ACS—6

True Power = voltage × current in phase with voltage.

Reactive Power = voltage × current 90° out of phase with voltage.

The phasor diagram for an inductive circuit is shown in fig. ACS-5. Here, current I lags behind the voltage by an angle ϕ. The current I can be resolved into two rectangular components

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(i) $I \cos \phi$ in phase with the voltage V and

(ii) $I \sin \phi$; 90° out of phase with voltage V.

$\therefore$ *True Power*, $P = V \times I \cos \phi = VI \cos \phi$ *watts or kW*.

Reactive Power, $P_r = V \times I \sin \phi = VI \sin \phi$ *VAR or kVAR*.

Apparent Power, $P_a = V \times I = VI$ *VA or kVA*.

From fig. ACS-6 the following relations can be deduced

$$kVA = \sqrt{(kW)^2 + (kVAR)^2} \rightarrow \text{Imp.}$$

$$kW = kVA \cos \phi$$

$$kVAR = kVA \sin \phi$$

It should be noted that lagging *kVAR* have been taken as negative.

ACS–3. IMPORTANCE OF POWER FACTOR

The apparent power drawn by a circuit has two components (Sec fig. ACS–7).

(i) True power and *(ii)* Reactive power.

Since true power component does the useful work in the circuit, therefore, this component of power should be as large as possible. This is possible only, if the reactive power component is small. As seen from the fig. ACS–7 the smaller the phase angle ϕ (*i.e.* greater the p.f. $\cos \phi$), the smaller is the reactive power component. Thus when $\phi = 0°$ (i.e. $\cos\phi = 1$). Then the reactive power component is zero and then true power is equal to apparent power. This means that whole of the apparent power drawn by the circuit is being utilized by it. Thus power factor of a circuit is a measure of its utilizing the apparent power drawn by it. The greater the power factor, greater is its ability to use apparent power. For this reason we wish that the power factor of the circuit to be as near to 1 as possible.

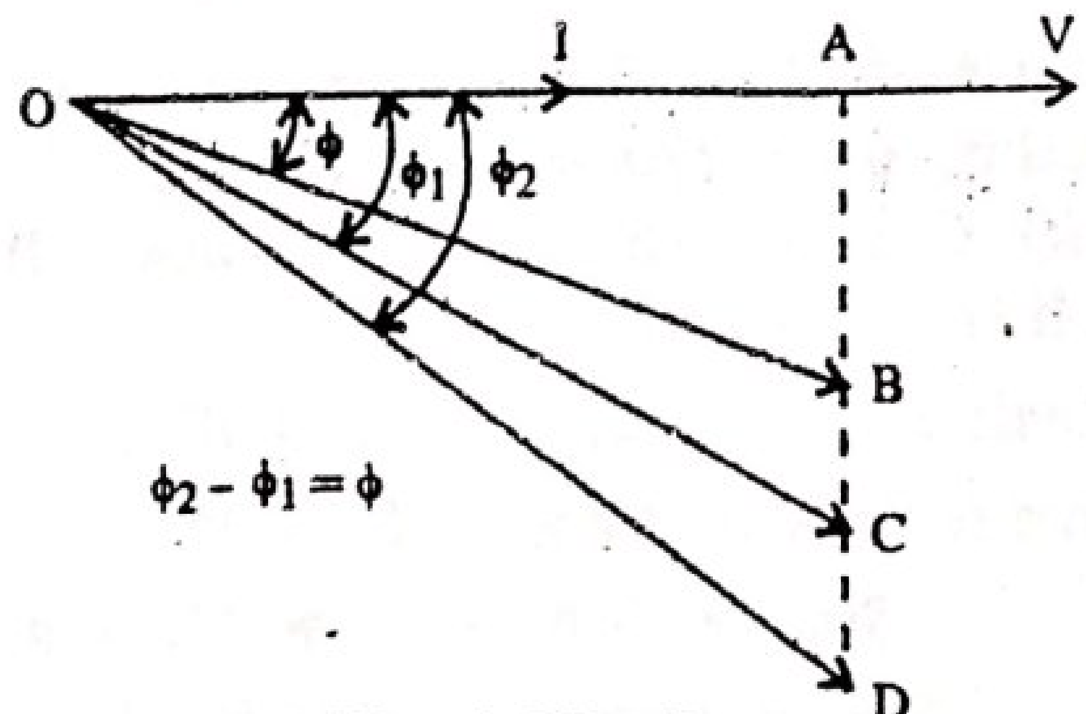

Fig. ACS—7

Fig ACS–7 shows that for the same power component *OA*, the current drawn from mains may be *OB* for angle ϕ, *OC* for angle ϕ_1 or much larger *OD* for angle ϕ_2 . Thus larger the angle the lower is the value of power factor and vice versa.

It does not mean that only lagging power factor represents wastage of energy. In fact, the leading power factor is also equally harmful and is to be avoided. It may be recalled that whereas inductive circuit cause lagging power factors, the leading power factors are due to capacitive circuits.

In order to correct the effect of lagging power factor, a capacitor may be used. The capacitor will take leading current and so to a particular extent neutralize the effect of lagging power factor. It is always economical to raise the power factor to unity.

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Problem ACS-1. A coil having a resistance of 10 ohms and inductance of 31·8 mH is nnected to 230 V, 50 Hz supply. Calculate :

(i) the circuit current **(ii) phase angle**

(iii) power factor **(iv) power consumed.**

Sol. Resistance of coil, $R = 10$ ohms

Inductive Reactance, $X_L = 2\pi fL = 2 \times 3{\cdot}14 \times 50 \times 31{\cdot}8 \times 10^{-3} = 10\ \Omega$

$\therefore$ Coil Impedence, $Z = \sqrt{10^2 + 10^2} = 14\cdot14\,\Omega$

(i) Circuit Current $I = \frac{V}{Z} = \frac{230}{14\cdot14} = 16\cdot26$ A **(Ans.)**

(ii) $\tan\phi = \frac{X_L}{R} = \frac{10}{10} = 1$

$\therefore$ $\phi = \tan^{-1} 1 = 45°$ (lagging) **(Ans.)**

(iii) Power factor $= \cos\phi = \cos 45° = 0{\cdot}707$ **(Ans.)**

(iv) Power consumed, $P = V.I.\cos\phi.$

$= 230 \times 16{\cdot}26 \times 0{\cdot}707 = 2644$ Watts **(Ans.)**

Problem ACS-2. A coil having a resistance of 10 ohms and inductive reactance of 25 ohms onnected across a 220 volt, 50 Hz supply. Find (i) Inductance of the coil (ii) Impedence of coil (iii) Current (iv) Angle of phase difference (v) Power factor (vi) True power) Sketch the vector diagram.

Sol. Resistance of coil, $R = 10\ \Omega$

Inductive reactance of coil $X_L = 25\ \Omega$

Supply Voltage, $V = 220$ volts.

Supply frequency, $f = 50$ Hz.

(i) Since, $X_L = \omega L$ $\therefore$ $L = \frac{X_L}{\omega} = \frac{X_L}{2\pi f}$

$\therefore$ $L = \frac{25}{2 \times 3{\cdot}14 \times 50} = 79{\cdot}58$ mH **(Ans.)**

(ii) Impedence, $Z = \sqrt{R^2 + X_L{}^2} = \sqrt{(10)^2 + (25)^2} = 26{\cdot}925\ \Omega$ **(Ans.)**

(iii) Current, $I = \frac{V}{Z} = \frac{220}{26\cdot925} = 8{\cdot}17$ A **(Ans.)**

(iv) $\tan\phi = \frac{X_L}{R} = \frac{25}{10} = 2{\cdot}5$

$\therefore$ $\phi = \tan^{-1} 2{\cdot}5 = 68{\cdot}198°$ **(Ans.)**

(v) Power factor $= \frac{R}{Z} = \frac{10}{26{\cdot}925} = 0{\cdot}3714$ (lagging) **(Ans.)**

(iv) True Power $= V.\ I.\ \cos\phi = 220 \times 8{\cdot}17 \times 0{\cdot}3714$

$= 667{\cdot}55$ Watts **(Ans.)**

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(vii) Vector diagram :—

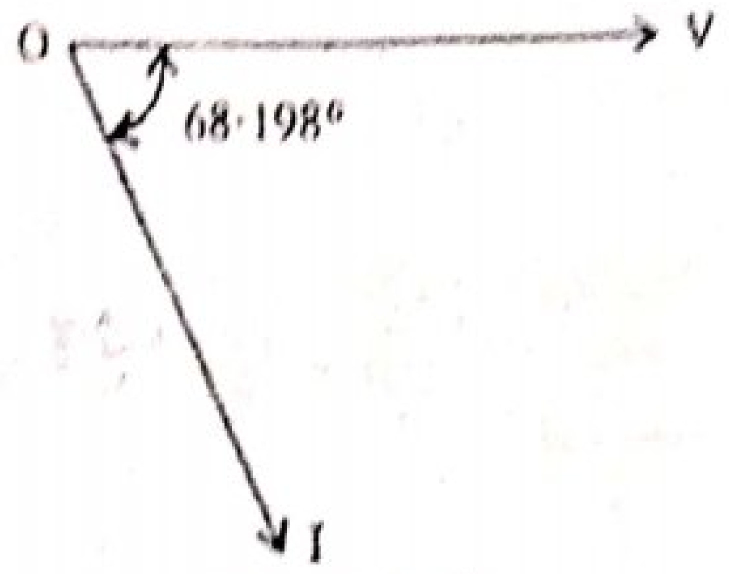

Fig. ACS—8

Problem ACS-3. A coil when connected across a 100 V *d.c.* supply dissipates 800 watts of power. When connected across a 100 V a.c. supply of frequency 50 Hz, it dissipates 400 watts. Calculate the value of resistance and inductance of the coil.

Sol. (For *d.c.* supply, because frequency is 0 ∴ inductive reactance $X_L = 2\pi f \cdot L = 0$).

Hence for *d.c.* supply, we have to take into account the resistance only. But for *a.c.* supply both resistance and inductance of the coil offer opposition to the flow of current.

For D.C. supply :

Resistance of the coil, $R = \dfrac{V^2}{P} = \dfrac{100 \times 100}{800} = 12{\cdot}5\ \Omega$. **(Ans).**

For A.C. supply :

Power consumed $P = V.I. \cos\phi$

$$\therefore \quad P = V.\frac{V}{Z}.\frac{R}{Z} = \frac{V^2.R}{Z^2}$$

$$\therefore \text{ Impedence of the coil, } Z = \sqrt{\frac{V^2.R}{P}} = \sqrt{\frac{100^2 \times 12\cdot5}{400}} = 17\cdot67\,\Omega$$

$$\text{Reactance of the coil } X_L = \sqrt{Z^2 - R^2} = \sqrt{(17\cdot67)^2 - (12\cdot5)^2}$$

$$= 12{\cdot}48\ \Omega$$

$$\text{Inductance of the coil, } L = \frac{12\cdot48}{2\pi \times 50} = 0\cdot039\text{ H } \textbf{(Ans.)}$$

Problem ACS-4. A 120 volt, 60 watt lamp is to be operated on 220 V, 50Hz mains. Find what value of *(i)* non inductive resistance *(ii)* pure inductance that is required in order that the lamp should work safely on the correct voltage. Which method is preferred and why ?

Sol. Rated power of the lamp, $P = 60$ watts

Rated voltage of the lemp, $V_L = 120$ Volts

Total supply voltage, $V = 220$ Volts.

Supply frequency, $f = 50$ Hz.

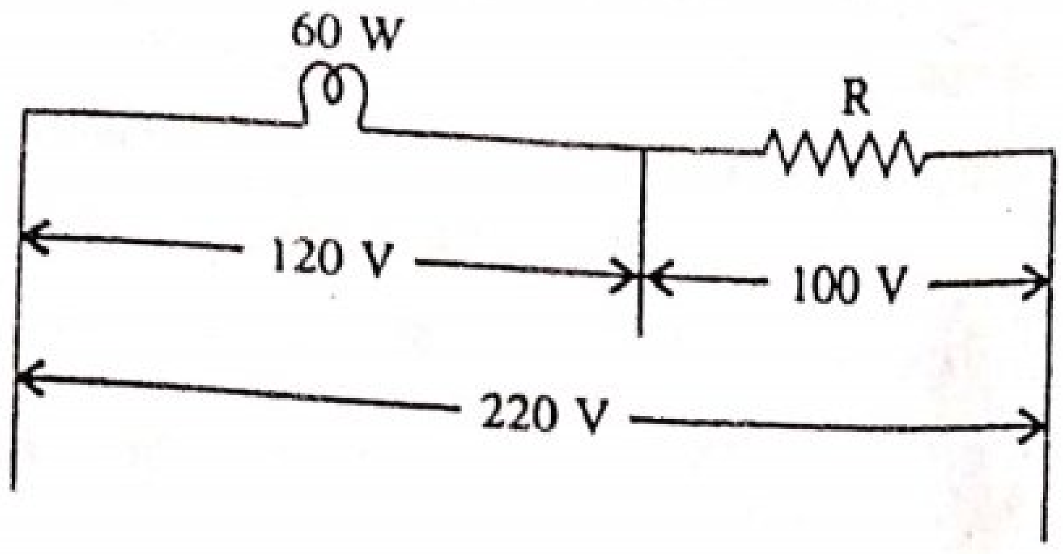

Fig. ACS—9(a)

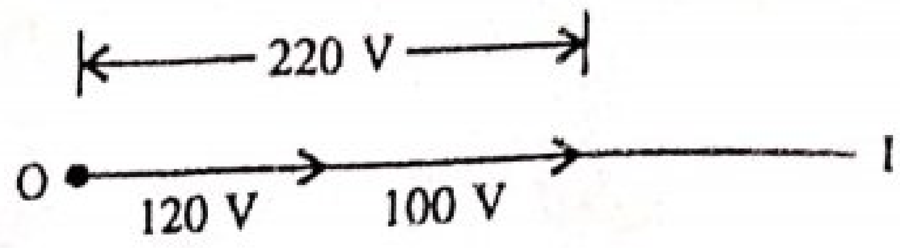

Fig. ACS—9(b)

(1)The voltage across the lamp (pure resistance) is to remain 120 volts in each case. Then the value of current flowing in each case

$$I = \frac{\text{Power}}{\text{Voltage}} = \frac{60}{120} = 0 \cdot 5 \text{ ampere.}$$

Fig. ACS–9 *(a)* shows the required value of $R\ \Omega$ in series with the lamp. The voltage drop across the lamp is 120 volts and the excess voltage (220 – 120 = 100 V) is across the resistance *R*.

$\therefore$ Voltage across resistance, $V_R = 220 - 120 = 100$ V

Current in lamp, $I = 0 \cdot 5$ A.

$\therefore$ Requires value of $R = \dfrac{100}{0 \cdot 5} = 200$ ohms **(Ans.)**

*(ii)*Fig. ACS–10 *(a)* shows the required inductance to be connected in series with the lamp. The phasor diagram of the circuit is shown in fig.ACS–10 *(b)*.

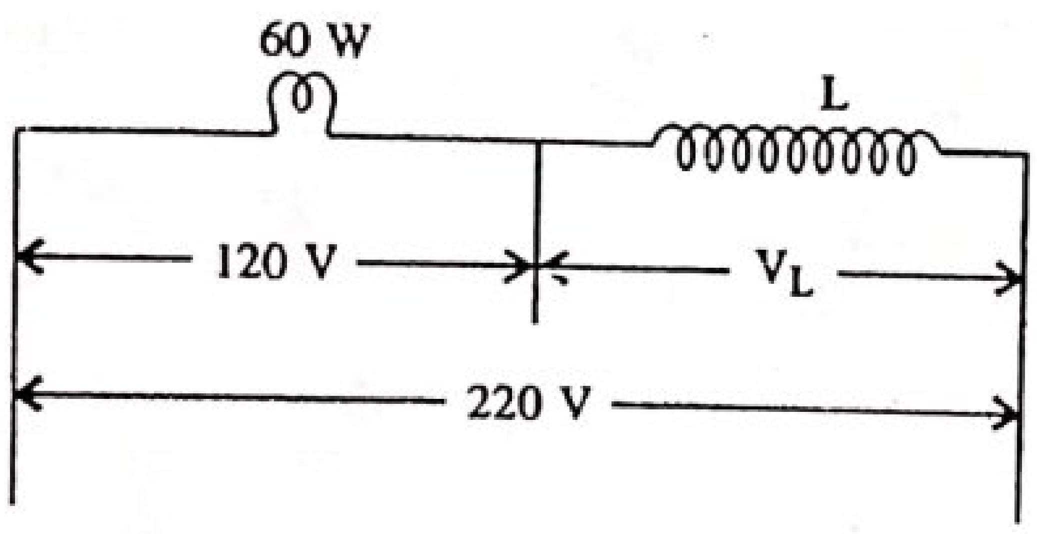

Fig. ACS—10(a)

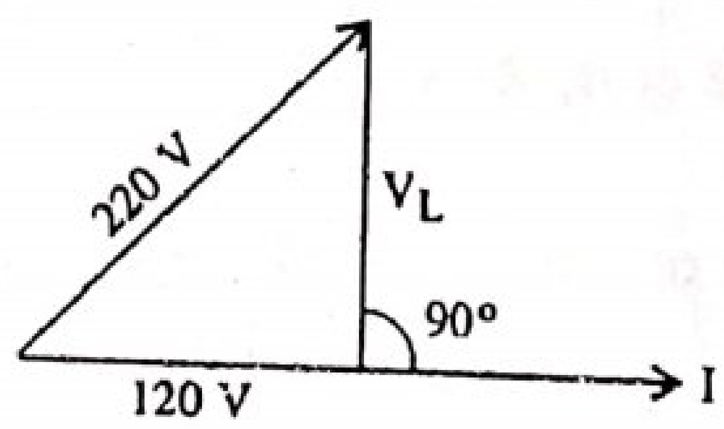

Fig. ACS—10(b)

From the phasor diagram $V_L = \sqrt{(220)^2 - (120)^2} = 184 \cdot 4$ volts

Current in the lamp, $I = 0 \cdot 5$ ampere.

$\therefore$ Inductive reactance, $X_L = \dfrac{X_L}{1} = \dfrac{184 \cdot 4}{0 \cdot 5} = 368 \cdot 8\ \Omega$

$\therefore$ Required inductance, $L = \dfrac{368 \cdot 8}{2\pi \times 50} = 1 \cdot 17$ Henry **(Ans.)**

Problem ACS–5. A circuit consists of a pure resistance and a coil connected in series. Power dissipated in the resistance and in the coil are 1000 watts an 250 watts respectively. Voltage drop across the resistance and the coil are 200 V and 300 V respectively. Determine the reactance of the coil and the supply voltage.

Sol. The pure resistance R and coil have been shown connected in series in figure ACS –11 *(a)*. The phasor diagram showing different voltages is shown in fig. ACS–11 *(b)*.

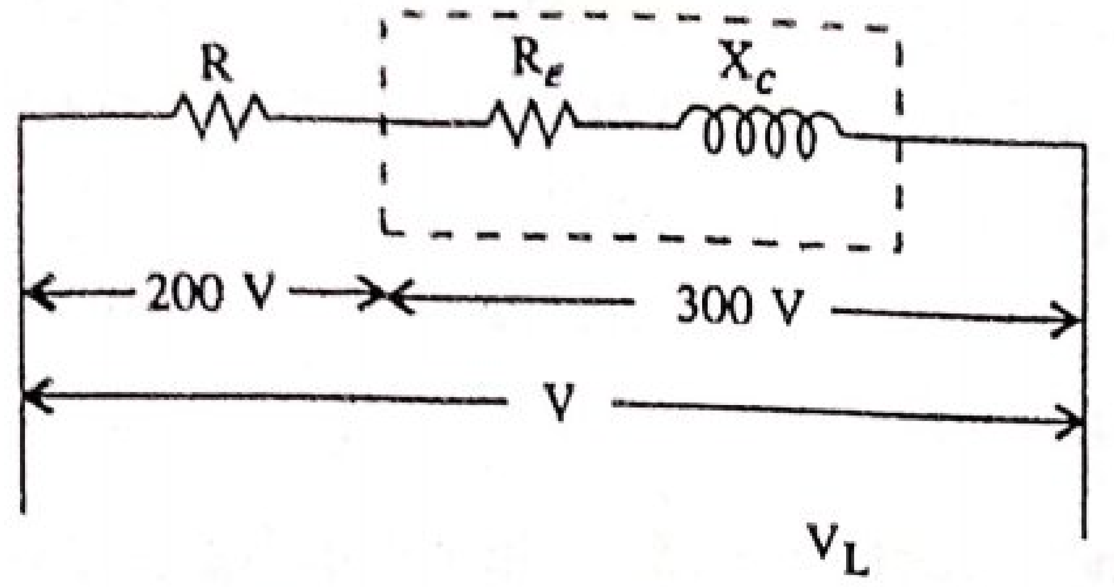

Fig. ACS—11*(a)*

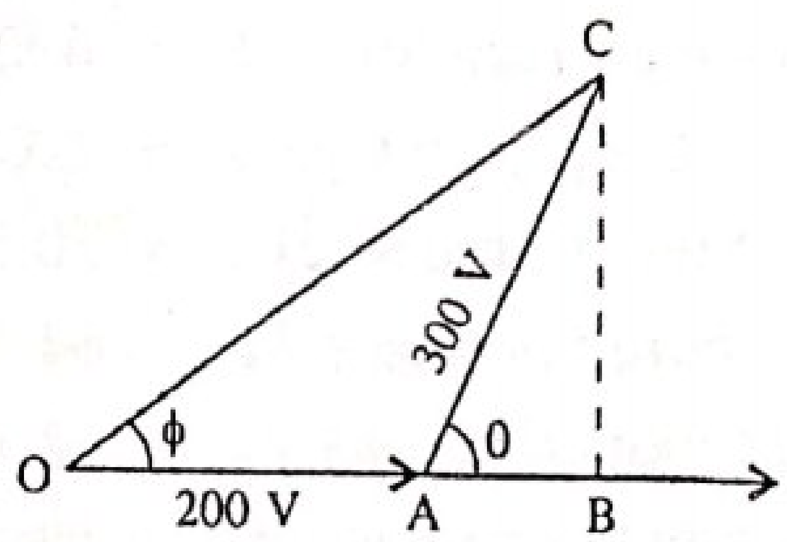

Fig. ACS—11*(b)*

Power consumed in resistance = 1000 W

Power consumed in coil = 250 W

Voltage across resistance R = 200 V.

Voltages across the coil V_c = 300 V.

$$\therefore \quad R = \frac{V^2}{P} = \frac{200 \times 200}{1000} = 40\,\Omega$$

Voltage across pure resistance = 200 V.

$$\therefore \quad \text{Current in resistance, } I = \frac{200}{40} = 5 \text{ amperes}$$

$$\text{Power consumed in coil, } P = I^2 \cdot R_c$$

$$\therefore \quad 250 = (5)^2 \times R_c$$

$$\therefore \quad R_c = \frac{250}{25} = 10\,\Omega.$$

$$\text{Coil impedence , } Z_{coil} = \frac{300}{5} = 60\,\Omega.$$

Inductive reactance of the coil $= \sqrt{60^2 - 10^2} = 59{\cdot}16\,\Omega$. **(Ans.)**

$$\text{Total impedence of the circuit } Z = \sqrt{(R + R_c)^2 + (E_L)^2}$$

$$= \sqrt{(40 + 10)^2 + (59 \cdot 16)^2}$$

$$= 77{\cdot}46 \text{ ohms}$$

Total voltage across the circuit $V = I\,Z$.

$$\therefore \quad V = 5 \times 77{\cdot}46 = 387{\cdot}3 \text{ Volts (Ans.)}$$

Problem ACS–6. An inductive load is connected in series with a non inductive resistance of 8 ohms. The combination is connected across an a.c. supply of 100 volts, 50 Hz. The voltage across the non– Inductive resistor is 64 volts and across the inductive load is 48 volts respectively. Calculate : *(i)* Impedence of the load *(ii)* Impedence of the combination *(iii)* Power absorbed by the load *(iv)* Total power taken from the supply *(v)* Power factor of the load *(vi)* Power factor of the whole circuit.

Sol. Value of pure resistance, $R = 8\ \Omega$

Supply voltage $V = 100$ volts

Supply frequency $f = 50$ Hz.

Voltage across pure resistance $V_R = 64$ Volts

Voltage across inductive load $V_L = 48$ Volts

The circuit diagram containing pure resistance R and inductive load is shown in fig. ACS– 12 *(a)* The phasor diagram showing different values of voltages is shown in fig. ACS–12 *(b)*.

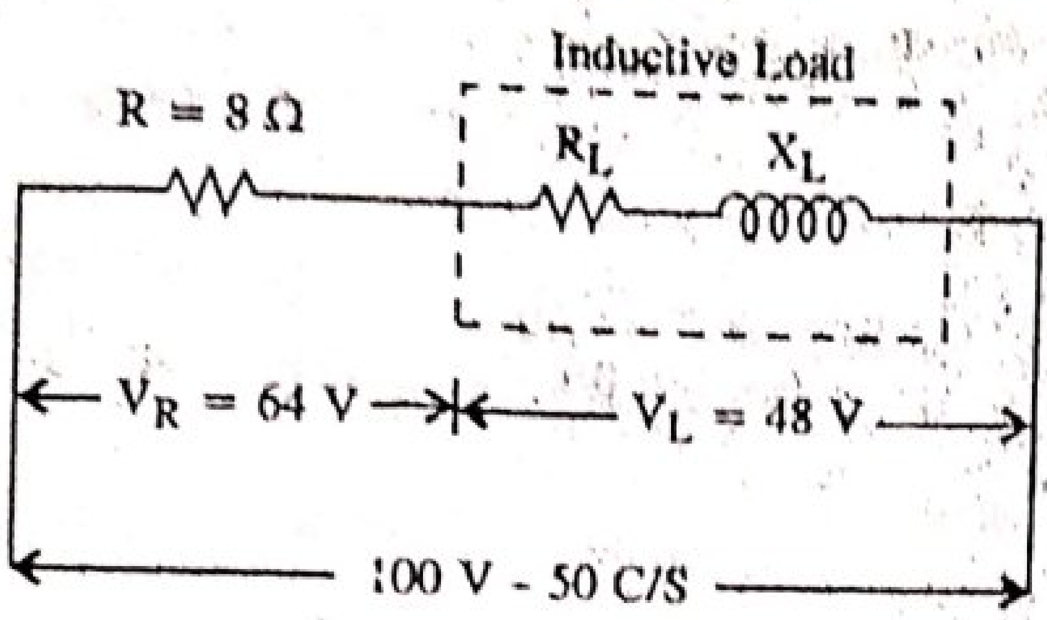

Fig. ACS—12*(a)*

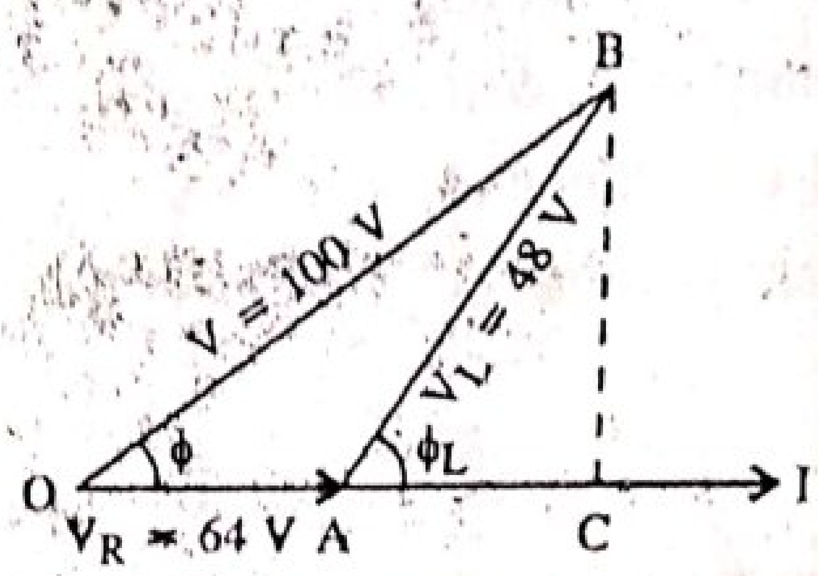

Fig. ACS—12*(b)*

Current in the circuit = Current in 8 ohm resistor $= \dfrac{64}{8} = 8$ amperes

Since inductive load is in series with 8 ohm resitor, therefore same current of 8 A will pass hrough the inductive load.

(i) $\therefore$ Impedence of the load $Z_L = \dfrac{48}{8} = 6$ ohms

(ii) Impedence of the combination $Z = \dfrac{V}{I} = \dfrac{100}{8} = 12 \cdot 5\Omega.$

(iii) From the vector diagram

$$OB^2 = OA^2 + AB^2 + 2\ OA \cdot AB \cdot \cos \phi_L$$

$\therefore$ Power factor of the inductive load $\cos \phi_L$ is given as,

$$\cos \phi_L = \frac{OB^2 - OA^2 - AB^2}{2 \cdot OA \cdot AB}$$

$$= \frac{100^2 - 64^2 - 48^2}{2 \cdot 64 \cdot 48} = 0 \cdot 586 \text{ (Ans.)}$$

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Since $\cos \phi_L = \dfrac{R_L}{Z_L}$ $\therefore$ $R_L = Z_L \cos\phi_L$

$\therefore$ $R_L = 6 \times 0{\cdot}586 = 3{\cdot}516$ ohms

(iv) Power absorbed by the load, $P_L = V_L \cdot I . \cos\phi_L$

$= 48 \times 8 \times 0{\cdot}586 = 225$ watts

Total resistance of the circuit $= 8 + 3{\cdot}516 = 11{\cdot}516$ ohms.

(v) Power absorbed by the entire circuit $= I^2 \cdot R$

$= (8)^2 \times 11{\cdot}516 = 737$ watts **(Ans.)**

(vi) Power factor of the entire circuit,

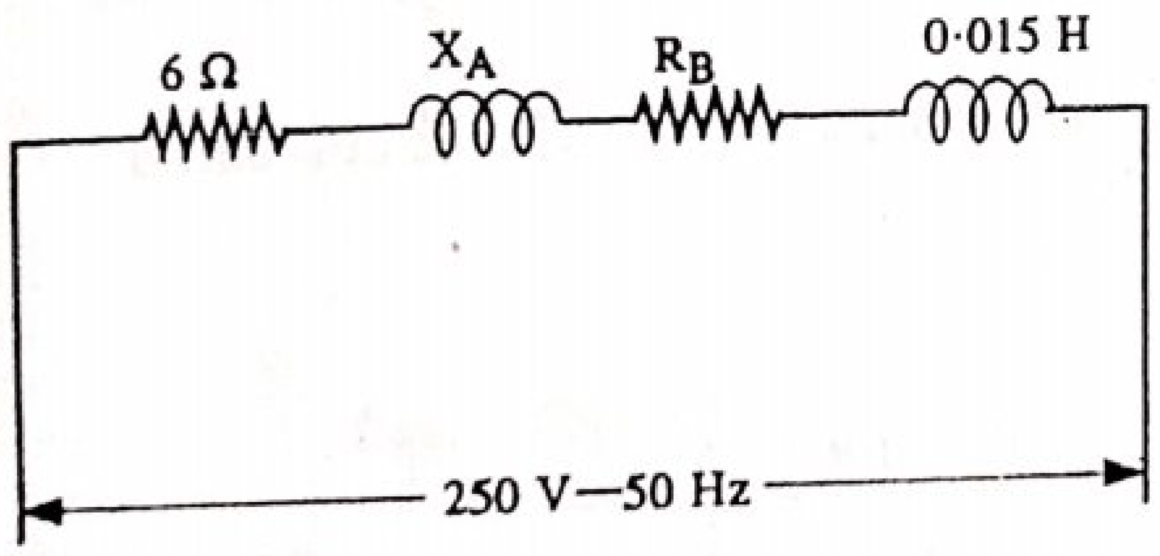
$$\cos \phi = \frac{8 + 3{\cdot}516}{12{\cdot}5} = \frac{11{\cdot}516}{12{\cdot}5} = \textbf{0·92 lagging (Ans.)}$$

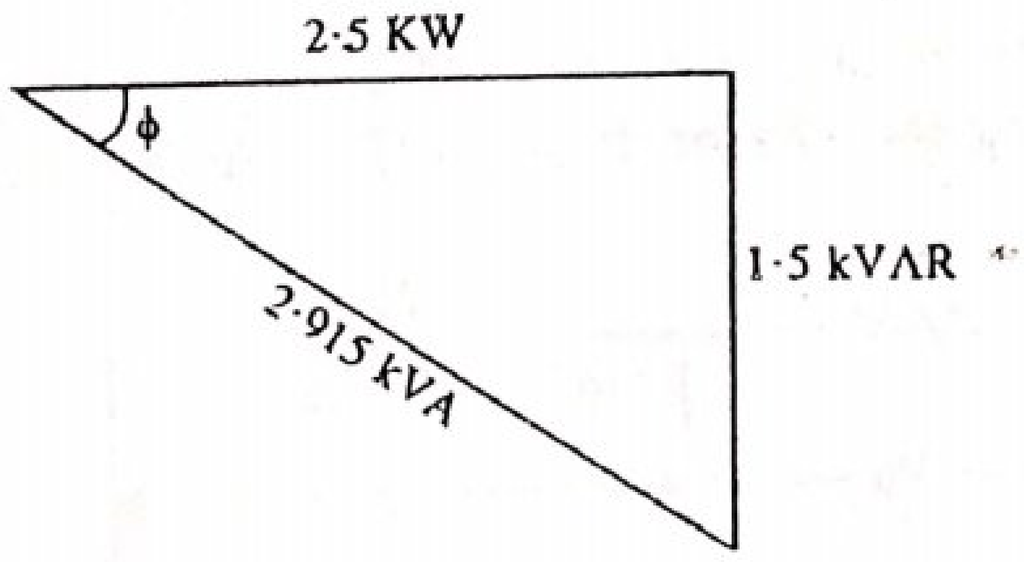
Alternatively :

$$\cos \phi = \frac{P}{VI} = \frac{737}{100 \times 8} = 0{\cdot}92 \text{ lagging } \textbf{(Ans.)}$$

Problem ACS–7 : Two coils A and B are connected in series across 250 V, 50 Hz supply. The resistance of A is 6 ohms and the inductance of B is 0·015 H. If the input from the supply is 2·5 kW and 1·5 kVAR. Find the inductance of A and the resistance of B. Calculate the voltage across each coil.

Sol. The circuit diagram of two coils A and B is shown in fig. below and the kVA triangle is also shown.





The circuit kVA is given by kVA $= \sqrt{(2{\cdot}5)^2 + (1{\cdot}5)^2} = 2{\cdot}915$ [$\because$ kVA $= \sqrt{\text{kW}^2 + \text{kVAR}^2}$]

The circuit current, $I = \dfrac{2{\cdot}915 \times 1000}{250} = 11{\cdot}66$ Amp.

Power consumed in the circuit, $P = I^2 (R_A + R_B)$

$\therefore$ $2{\cdot}5 \times 1000 = (11{\cdot}66)^2 (R_A + R_B)$

$\therefore$ $R_A + R_B = \dfrac{2500}{(11{\cdot}66)^2} = 18{\cdot}38$ ohms

$R_B = 18{\cdot}38 - R_A = 18{\cdot}38 - 6 = 12{\cdot}38$ ohms **(Ans.)**

Impedance of the entire circuit, $Z = \frac{V}{I} = \frac{250}{11 \cdot 66} = 21 \cdot 44$ ohm

Then

$$[X_A + X_B] = \sqrt{Z^2 - (R_A + R_B)^2} = \sqrt{(21 \cdot 44)^2 - (18 \cdot 38)^2}$$

$\therefore$

$$X_A + X_B = 11 \cdot 03\ \Omega$$

Now

$$X_B = 2 \times 3 \cdot 14 \times 50 \times 0 \cdot 015 = 4 \cdot 713\ \Omega$$

$\therefore$

$$X_A = 11 \cdot 03 - 4 \cdot 713 = 6 \cdot 323\ \Omega$$

or

$$2 \times 3 \cdot 14 \times 50 \times L_A = 6 \cdot 323$$

$\therefore$

$$L_A = \frac{6 \cdot 323}{314} = \mathbf{0 \cdot 02\ Henry\ (Ans.)}$$

Now

$$Z_A = \sqrt{(6)^2 + (6 \cdot 323)^2} = 8 \cdot 7 \text{ ohms}$$

Potential difference across coil A $= I.\, Z_A = 11 \cdot 66 \times 8 \cdot 7 =$ **101·44 volts (Ans.)**

Impedance of coil B, $Z_B = \sqrt{(12 \cdot 38)^2 + (4 \cdot 713)^2} = 13 \cdot 24$ ohms

Potential difference across coil B $= I.\, Z_B = 11 \cdot 66 \times 13 \cdot 24 =$ **154·38 volts (Ans.)**

S–4 RESISTANCE AND CAPACITANCE IN SERIES

The figure ACS–13 *(a)* shows a circuit containing pure resistance R ohms and a pure acitance in series.

A voltage of r.m.s. value V is applied across the combination. Let the current flowing ugh the circuit is I amperes.

Let V_R be the voltage across R and V_C is the voltage across capacitance C. The circuit can be vn by drawing the phasor diagram (Fig. ACS–13 *(b)*).

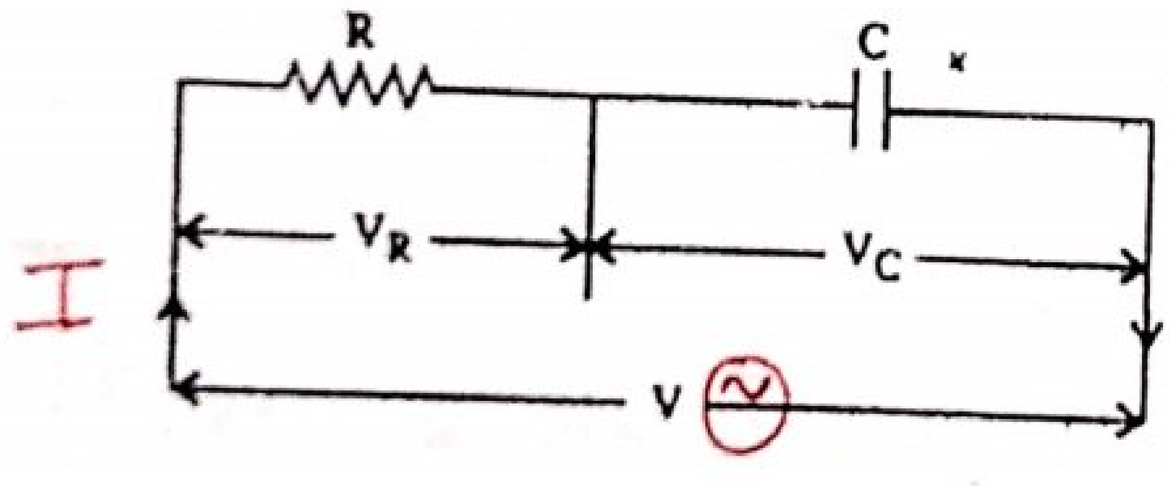


Fig. ACS—13*(a)*

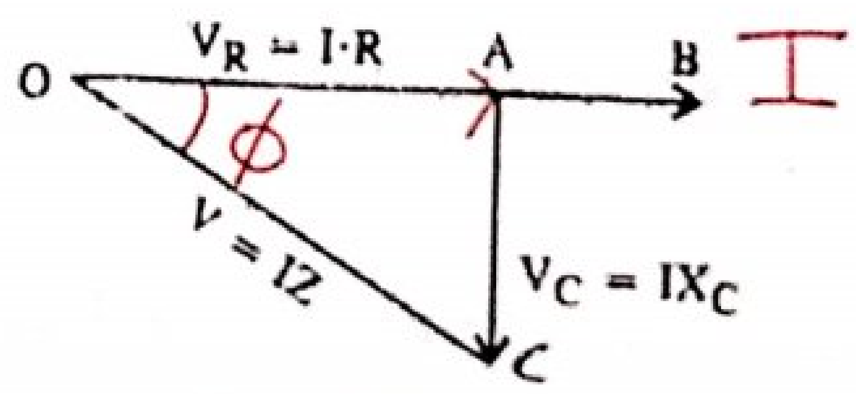


Fig. ACS—13*(b)*

)rawing of Phasor Diagram : In series circuits, the current is common to both resistance apacitance, so it should be taken as reference vector. The phasor diagram is shown in fig. -13 *(b)*. Taking I, as the reference vector, it is shown as OB drawn along X–axis. The ge across pure resistance R is ($V_R = I\,R$) and acts in phase with the current as shown in fig. – 13 *(b)*. It is taken as $\overrightarrow{OA}$ in the same line of the current.

he voltage across pure capacitance is ($V_C = I \times X_C$) lags the current by 90° (shown in fig.) represented by $\overrightarrow{AC}$ to the same scale

The applied voltage V is the sum of phasors OA and AC. This is given by $\overrightarrow{OC}$. OC can be measured and checked as per scale.

From right angled triangle OAC :

$$OC^2 = OA^2 + AC^2$$

or

$$V^2 = (V_R)^2 + (V_C)^2$$

$$V^2 = (IR^2 + (IX_C)^2$$

$$V = I\sqrt{R^2 + X_C^2}$$

$$\therefore \quad V = I \cdot Z \quad \text{or } I = \frac{V}{Z}, \text{ and } Z = \frac{V}{I}$$

Here $\sqrt{R^2 + X_C^2}$ is known as Impedence of the circuit. It is denoted by Z and is measured in ohms.

The Impedence Triangle. Since the current is common to all V_R, V_C and V, so it may be omitted and impedence triangle constructed with only R, X_C and Z as shown in figure ACS–14 *(a)*.

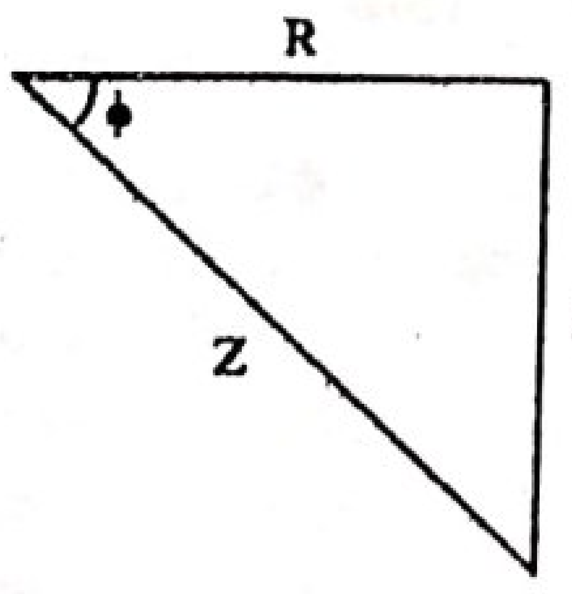


Fig. ACS—14*(a)*

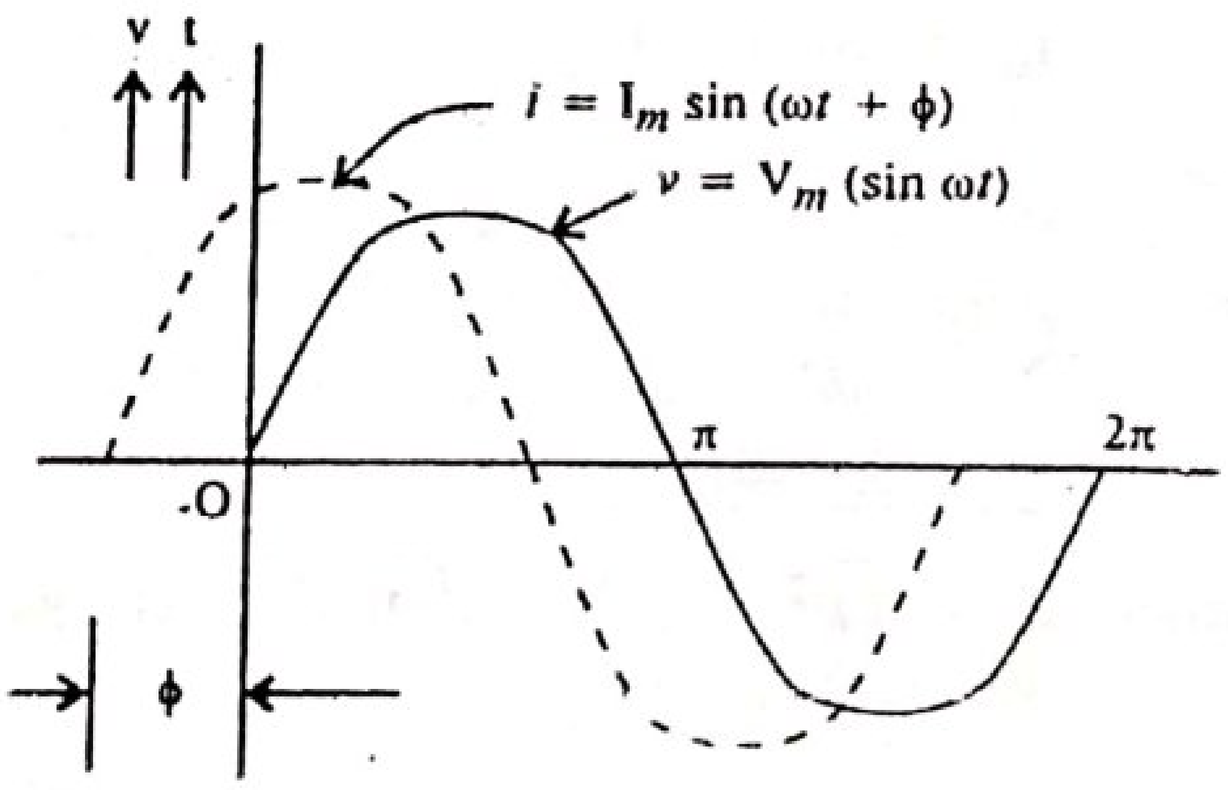


Fig. ACS—14*(b)*

From the Impedence triangle :

Impedence of the circuit, $Z = \sqrt{R^2 + X_C^2}$

Where $X_c = \dfrac{1}{\omega C} = \dfrac{1}{2\pi f C}$ is the capacitive reactance.

$X_c = \dfrac{1}{\omega_c} = \dfrac{1}{2\pi f c}$

$$\tan\phi = \frac{X_C}{R}, \phi = \tan^{-1}\frac{X_C}{R} \text{ is known as the phase angle.}$$

The current leads the voltage by this angle ϕ. $-\pi/2 < \phi < 0$

Power : The instantaneous value of v is given by the expression

$$v = V_m \sin \omega t$$

$$i = I_m \sin(\omega t + \phi)$$

Average power $\quad P =$ average of vi

$= VI \cos \phi$ (By same way as in article)

Problem ACS–8. A resistance of 10 ohms and a capacitor of 100 μF (mirco farad) capacitance are connected in series across a 250 volts, 50 Hz supply. Calculate *(i)* Capacitive reactance *(ii)* Impedence *(iii)* Current flowing *(iv)* Angle of phase difference *(v)* Power factor *(vi)* True power *(vii)* Phasor diagram.

Sol. Value of resistance, $R = 10$ ohms.

Value of capacitance, $C = 100\ \mu F = 100 \times 10^{-6}$ F.

Supply voltage, $V = 250$ Volts

Supply frequency, $f = 50$ Hz.

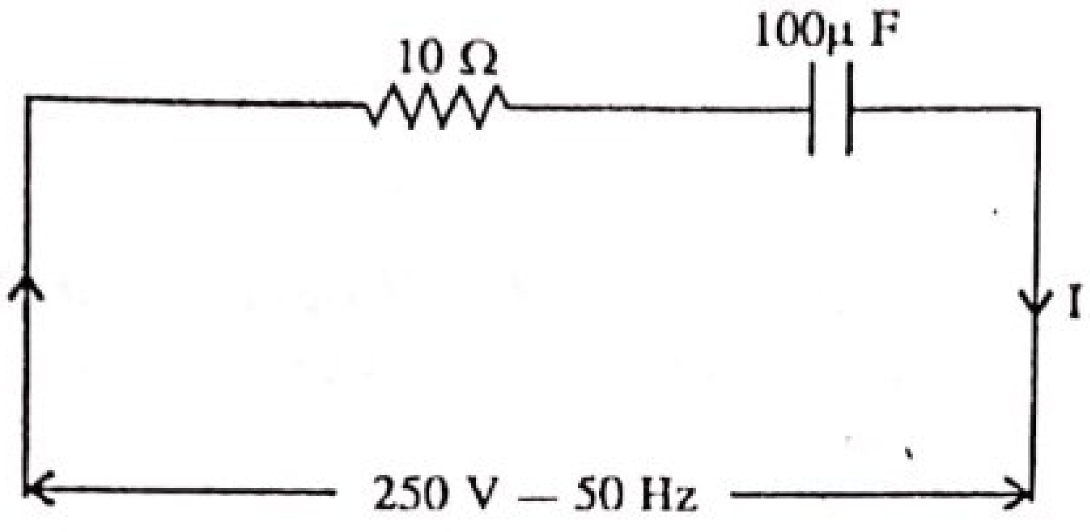


Fig. ACS—15*(a)*

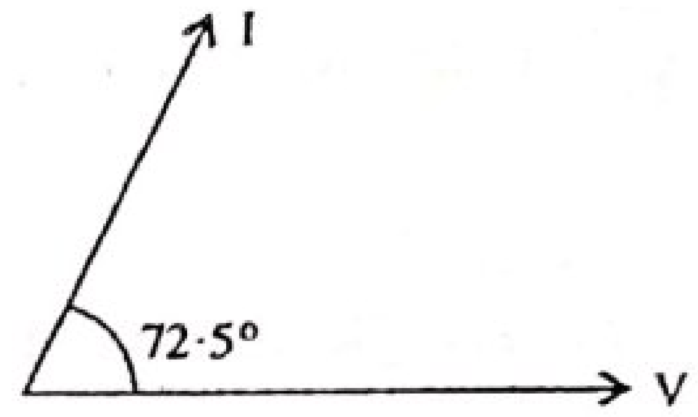


Fig. ACS—15*(b)*

(i) Capacitive reactance :

$$X_c = \frac{1}{\omega C} = \frac{10^6}{2 \times 3 \cdot 14 \times 50 \times 100} \qquad \frac{1}{2\pi f c}.$$
$$= 31{\cdot}84\Omega$$

(ii) Impedence, $Z = \sqrt{R^2 + X_C^2} = \sqrt{(10)^2 + (31 \cdot 84)^2}$

$= 33{\cdot}37$ ohms

(iii) Current flowing, $I = \dfrac{V}{Z} = \dfrac{250}{33 \cdot 37} = 7 \cdot 489\text{A}$ **(Ans.)**

(iv) Phase angle $\phi = \tan^{-1} \dfrac{X_C}{R}$

$\therefore \quad \tan \phi = \dfrac{31 \cdot 84}{10} = 3 \cdot 184$

$\therefore \quad \phi = \tan^{-1} 3{\cdot}184 = 72{\cdot}5°$

(v) Power factor , $\cos \phi = \cos 72{\cdot}5° = 0{\cdot}3$ **(leading) (Ans.)**

(vi) True Power $= V.I. \cos \phi$

$= 250 \times 7{\cdot}489 \times 0{\cdot}3$

$= 561{\cdot}5$ **Watts (Ans.)**

(vii) The phasor diagram is as shown above [Fig. ACS 15 *(b)*]

Problem ACS–9. A capacitor and resistor are connected in series across a 120 V, 50 Hz supply. The circuit takes a current of 1·144 amperes from the mains. If power loss in the

circuit is 130·8 watts , find the values resistance and capacitance.

Sol. The figure ACS–16 shown an $R - C$ series circuit.

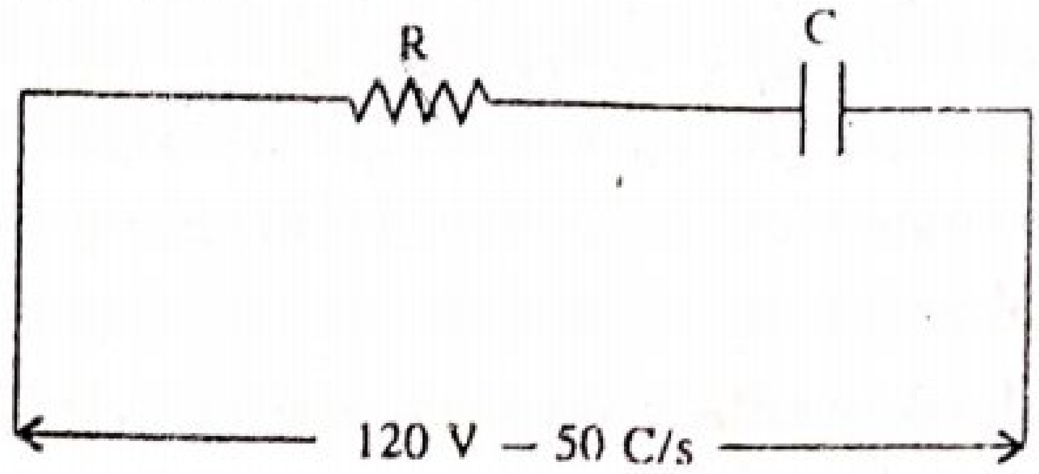


Fig. ACS—16

Voltage across the circuit, $V = 120$ volts.

Power consumed by the circuit, $P = 130{\cdot}8$ watts

Supply frequency, $f = 50$ Hz

Circuit current, $I = 1{\cdot}144$ amperes.

Since power loss occurs in resistance only as capacitor does not consume power.

If R and C be the resistance and capacitance respectively.

Then $P = I^2 \times R$

or $130{\cdot}8 = (1{\cdot}144)^2 \times R$

$$\therefore \quad R = \frac{130 \cdot 8}{(1 \cdot 144)^2} = 100\text{ohms} \quad \textbf{(Ans.)}$$

$$\text{Circuit Impedence, } Z = \frac{120}{1{\cdot}144} = 104{\cdot}9\text{ohms}$$

$$\text{Capacitive reactance } X_C = \sqrt{Z^2 - R^2} \;\; ; X_C = \sqrt{(104{\cdot}9)^2 - (100)^2} = 31{\cdot}7\text{ohms}$$

$$\therefore \quad \text{Capacitance, } C = \frac{1}{2\pi f X_C} = \frac{1}{2\pi \times 50 \times 31{\cdot}7} = 100\,\mu F$$

Problem ACS –10. A metal filament lamp, rated 750 W, 100 volts is to be connected in series with a capacitor across a 230 V, 60 Hz supply. Calculate *(i)* the capacitance required. *(ii)* The power factor.

Sol. Supply voltage, $v = 230$ V.

Supply frequency $f = 60$ Hz .

Rated power of lamp, $P = 750$ watts

Voltage across lamp, $V_R = 100$ V.

Let C farads be the capacitance connected in series with the lamp.

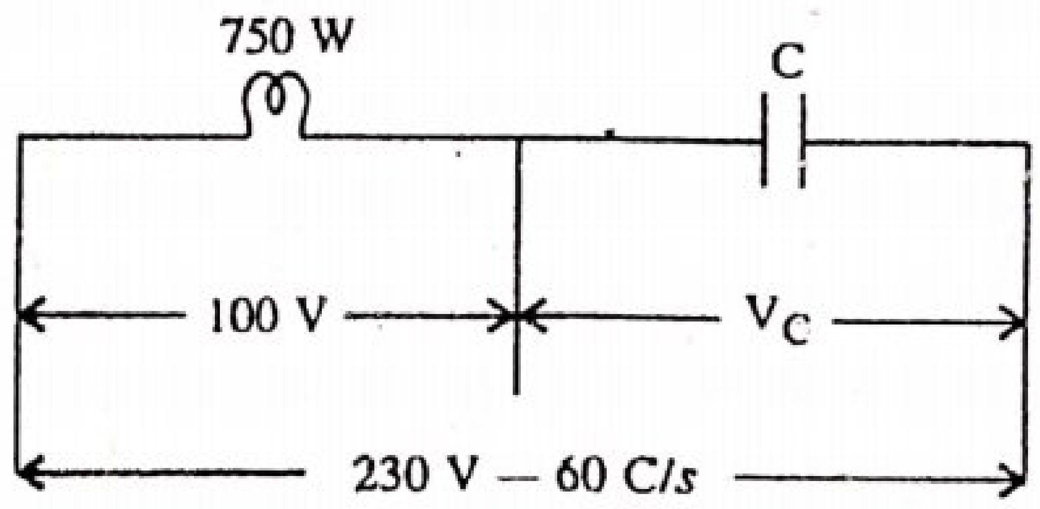


Fig. ACS—17

$$\therefore \quad \text{Resistance of the lamp, } R = \frac{V^2}{P} = \frac{(100)^2}{750} = 13 \cdot 33 \text{ ohms}$$

$$\text{Current in the lamp, } I = \frac{P}{V} = \frac{750}{100} = 7 \cdot 5 \text{ A}$$

$$\text{Circuit Impedence, } Z = \frac{V}{I} = \frac{230}{7 \cdot 5} = 30 \cdot 67 \text{ ohms}$$

$$\therefore \quad X_C = \sqrt{Z^2 - R^2} = \sqrt{(30{\cdot}67)^2 - (13{\cdot}33)^2} = 27{\cdot}618\ \Omega$$

$$\therefore \quad C = \frac{I}{2\pi f X_C} = \frac{10^6}{2\pi \times 60 \times 27{\cdot}618} = 96\,\mu\text{F} \quad \text{(Ans.)}$$

$$\text{Power factor of the circuit , } \cos\phi = \frac{R}{Z} = \frac{13{\cdot}33}{30{\cdot}67} = 0{\cdot}434 \quad \text{(Ans.)}$$

Problem ACS-11 : An iron cored choking coil takes 5 A at a power factor of 0·6 when supplied at 100 V 50 H_z. When the iron core is removed and the supply is reduced to 15 V, the current rises to 6 A at power factor of 0·9. Determine *(i)* The iron loss in the core *(ii)* The copper loss at 5 A *(iii)* The inductance of the choking coil with core when carrying a current of 5 A.

Sol. When the iron core is removed ·

Supply voltage, V = 15 V

Current in the coil, I = 6 A

$\therefore$ Impedence of the coil, $Z = \frac{V}{I} = \frac{15}{6} = 2{\cdot}5$ ohms

Power factor the coil *i.e.*, $\frac{R}{Z} = 0{\cdot}9$

$\therefore$ Resistance of the coil, $R = Z\cos\phi = 2{\cdot}5 \times 0{\cdot}9 = 2{\cdot}25$ ohms

With iron Core

Power supplied, $P = V.\ I.\ \cos\phi = 100 \times 5 \times 0{\cdot}6 = 300$ watts

Power lost in the coil *i.e.*, Copper loss $= (5)^2 \times 2{\cdot}25 =$ **56·2 watts (Ans.)**

Iron losses in the core $= 300 - 56{\cdot}2 =$ **243·8 watts (Ans.)**

Impedence of the coil when carrying current of 5A

i.e., $Z = \frac{V}{I} = \frac{100}{5} = 20\ \Omega$

$X_L = Z\sin\phi = 20 \times 0{\cdot}8 = 16\ \Omega$

Since $X_L = 2\pi f.\ L \qquad \therefore 16 = 2 \times 3{\cdot}14 \times 50 \times L$

$\therefore \quad L = \frac{16}{2 \times 3{\cdot}14 \times 50} =$ **0·0509 H (Ans.)**

ACS-5. R-L-C SERIES CIRCUIT

A circuit which contains a pure resistance R, a pure inductance L and a pure capacitance C connected in series is called an R–L–C series circuit. The figure ACS-18 *(a)* shown an R–L–C series circuit.

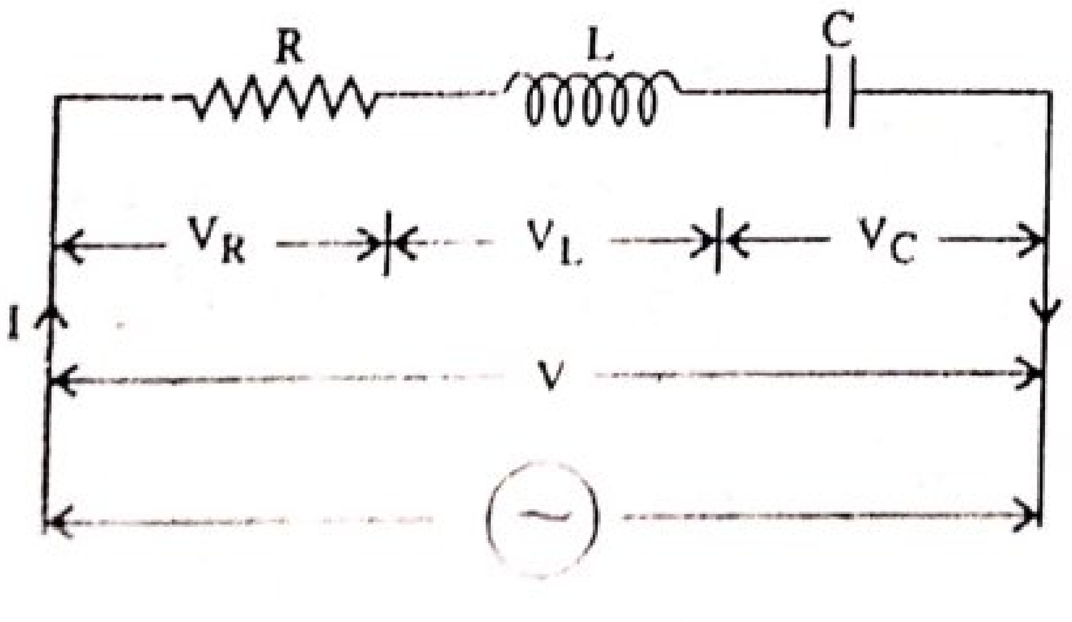


Fig. ACS–18*(a)*

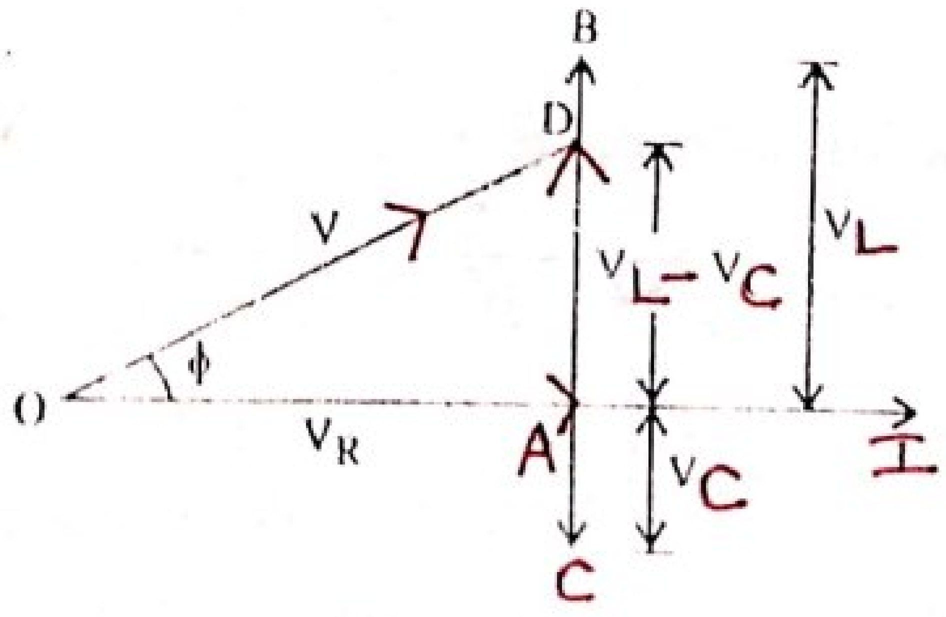


Fig. ACS–18*(b)*

If V (r.m.s.) is the voltage across the circuit and I (r.m.s) is the current flowing in the circuit Then the voltages across each component will be as below

Voltage across R, $V_R = I.R.$, where V_R is in phase with I

Voltage across L, $V_L = IX_L$, where V_L leads I by 90°

Voltage across C, $V_C = IX_C$, where V_C lags I by 90°.

The phasor diagram is drawn again taking current as the reference vector.

In the phasor diagram (shown in fig. ACS-18(b). OA represents V_R, AB represents V_L and AC represents V_C. It is seen from the phasor diagram that phasor V_L is in phase opposition to V_C. It is clear from the phasor diagram, that the circuit can either be effectively inductive or capacitive, depending upon the voltage drop (*i.e.* when $V_L > V_C$) it is inductive and on the other hand (*i.e.* when $V_C > V_L$) it is capacitive. For the case considered, $V_L > V_C$, so that net voltage drop across L–C combination is $(V_L - V_C)$ and is represented by AD. Therefore the applied voltage V is the the phasor sum of V_R and $(V_L - V_C)$.

$$\therefore \quad V = \sqrt{(V_R^2) + (V_L - V_C)^2} = \sqrt{(I.R)^2 + (IX_L - IX_C)^2}$$

$$V = I\sqrt{R^2 + (X_L - X_C)^2} \;;\; V = I.Z$$

Where $Z = \sqrt{R^2 + (X_L - X_C)^2}$ and offers opposition to the flow of current and is called impedence of the circuit.

Phase angle : From phasor diagram :—

Circuit power factor, $\cos\phi = \dfrac{R}{Z} = \dfrac{R}{\sqrt{R^2 + (X_L - X_C)^2}}$

$$\tan\phi = \frac{V_L - V_c}{V_R} = \frac{(X_L - X_C)}{R}$$

since X_L, X_C and R are known, therefore, phase angle ϕ of the circuit can be determined.

Power : Instantaneous power, $p = v.i$

Here, $\quad v = V_m \sin\omega t$; $i = I_m \sin(\omega t - \phi)$

$i = I_m \sin(\omega t \pm \phi)$
+ve sign is used when current leads i.e. $X_C > X_L$

$\therefore$ Instantaneous power, $p = V_m \sin(\omega t) \; I_m \sin(\omega t - \phi)$

or $\quad p = V_m . I_m . \sin\omega t \sin(\omega t - \phi)$

Average power consumed in the circuit can be derived as in case of R–L circuit.

$\therefore$ Average Power, $P = V.I \cos\phi$.

Three cases of R–L–C Series Circuit. In the previous article, we have seen that the impedence of an R–L–C series circuit is given by :

$$Z = \sqrt{R^2 + (X_L - X_C)^2}$$

Case I. When $(X_L - X_C)$ is positive *i.e* if $X_L > X_C$, then phase angle ϕ is positive and the circuit will be inductive. In such cases, current lags the voltage by an angle ϕ.

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Case II. When $(X_L - X_C)$ is negative *i.e.* $(X_L < X_C)$, then the phase angle ϕ is negative and e circuit is capacitive. In such circuits, current leads the voltage by an angle ϕ.

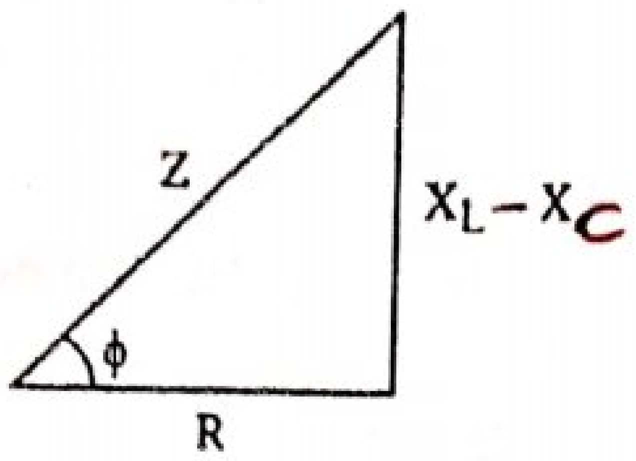

Fig. ACS--19*(a)*

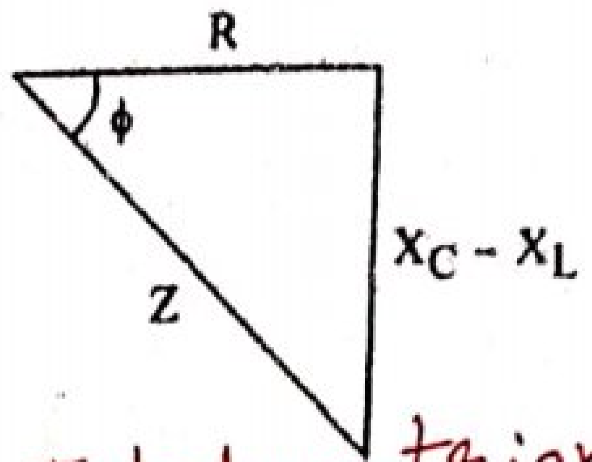

Fig. ACS--19*(b)* Impedance triangle for series R-L-C circuit when $X_L < X_C$

Case III. When $(X_L - X_C)$ is zero (i.e. $X_L = X_C$), then the phase angle $\phi = 0°$ and the circuit pure resistive , In such cases, the current is in phase with the voltage. The power factor of ch circuits is unity. If the equation for the applied voltage is $v = V_{max} \sin \omega t$.

Then the equations of the currents can be :

(i) $i = I_{max} \sin(\omega t - \phi)$... case *(i)* ; $X_L > X_C$

(ii) $i = I_{max} \sin(\omega t + \phi)$... case *(ii)* ; $X_L < X_C$

(iii) $i = I_{max} \sin \omega t$ ... .case *(iii)* ; $X_L = X_C$

Problem ACS – 12. An a.c. circuit having a resistance of 10 ohms, an inductance of 2 H and a capacitance of 100 μF in series, is connected across a single phase 110 volts, Hz supply . Calculate the following for this circuit.

(i) Resultant reactance.

(ii) Impedence.

(iii) Current .

(iv) Voltage across R, L and C.

(v) Phase difference between supply voltage and current.

(vi) Draw the phasor diagram.

Sol. Value of resistance, $R = 10$ ohms.

Value of inductance, $L = 0{\cdot}2$ Henry

Value of capacitance, $C = 100 \times 10^{-6}$ Farads

Supply frequency, $f = 50$ *Hz*.

Supply voltage, $V = 110$ V.

Inductive reactance, $X_L = \omega L = 2\pi \times 50 \times 0{\cdot}2 = 62{\cdot}8$ ohms.

$$\text{Capacitive reactance, } X_c = \frac{1}{\omega C} = \frac{10^{-6}}{2\pi \times 50 \times 100} = 31{\cdot}8 \text{ ohms} \quad \textbf{(Ans.)}$$

(i) Resultant reactance $= (X_L - X_C) = 62{\cdot}8 - 31{\cdot}8 = 31$ ohm **(Ans.)**

(ii) $$\text{Impedence, } Z = \sqrt{R^2 + (X_L - X_C)^2} = \sqrt{10^2 + 31^2} = 32 \cdot 573 \text{ ohms} \quad \textbf{(Ans.)}$$

(iii) $$\text{Current , } I = \frac{V}{Z} = \frac{110}{32 \cdot 573} = 3 \cdot 377 \text{ A (Ans.)}$$

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(iv) Voltage across R, $V_R = I\ R = 3{\cdot}377 \times 10 = 33{\cdot}77$ volts.

Voltage across L, $V_L = I.\ X_L = 3{\cdot}377 \times 62{\cdot}8 = 212$ volts

Voltage across C, $V_C = I\ X_C = 3{\cdot}377 \times 31{\cdot}8 =$ **107·388 volts (Ans.)**

(v) Phase difference, $\phi = \text{Tan}^{-1}\ \dfrac{X_L - X_C}{R} = \tan^{-1}\dfrac{31}{10} = 72{\cdot}12^\circ$ (lagging)

(vi) **Phasor diagram :**

$OA = 33{\cdot}77$ Volts ; $AB = 212$ Volts ; $AC = 107{\cdot}388$ Volts.

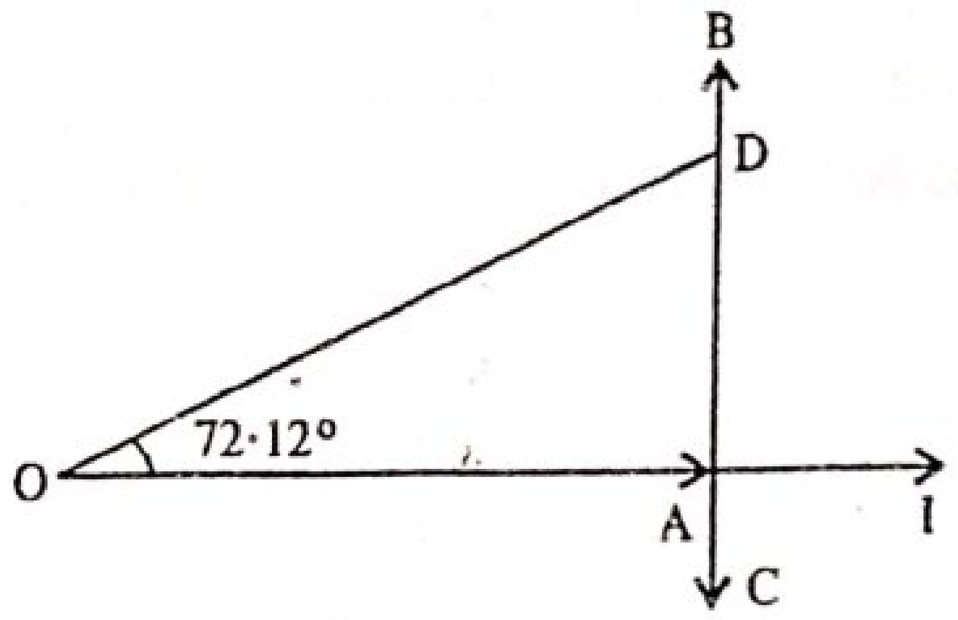

Fig. ACS—19

Problem ACS– 13. Find the active and reactive components of current taken by a series rcuit consisting of a coil of inductance 0·1 Henry and a resistance of 10 ohms and a ıpacitor of 240 μF connected to 230 V, 50 Hz supply mains.

Sol. Value of resistance, $R = 10$ ohms.

Value of inductance, $L = 0{\cdot}1$ Henry

Value of capacitance, $C = 240\ \mu F = 240 \times 10^{-6}$ F

Supply voltage, $V = 230$ volts

Supply frequency, $f = 50$ Hz.

Inductive reactance, $X_L = \omega L = 2\pi f L$

$= 2\pi\ .\ 50 \times 0{\cdot}1 = 31{\cdot}4$ ohms

Capacitive reactance, $X_c = \dfrac{1}{\omega C} = \dfrac{1}{2\pi f.C}$

$= \dfrac{10^6}{2\pi \cdot 50 \cdot 240} = 13{\cdot}26$ ohms

Resultant reactance, $X_L - X_C = 31{\cdot}4 - 13{\cdot}26 = 18{\cdot}14\Omega$

Impedence, $Z = \sqrt{R^2 + (X_L - X_C)^2}$

$= \sqrt{10^2 + 18{\cdot}14^2} = 20{\cdot}71\Omega$

$\therefore$ Current $I = \dfrac{V}{Z} = \dfrac{230}{20{\cdot}71} = 11{\cdot}105$ A.

Power factor, $\cos\phi = \dfrac{R}{Z} = \dfrac{10}{20{\cdot}71} = 0{\cdot}4828$

$\sin\phi = \left(\dfrac{X_L - X_C}{Z}\right) = \dfrac{18{\cdot}14}{20{\cdot}71} = 0{\cdot}8759$

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Active component of current $= I\cos\phi = 11{\cdot}105 \times 0{\cdot}4828 =$ **5·36 A (Ans.)**

Reactive component of current $= I\sin\phi = 11{\cdot}105 \times 0{\cdot}8759 =$ **9·72 A (Ans.)**

Problem ACS–14 Two impedences Z_1 and Z_2 when connected separately across a 230 V, 50 Hz, supply, consume powers of 100 W, 60 W at power factors of 0·5 lagging and 0·6 leading respectively. If the two impedences are now connected together in series across the same supply. Calculate *(i)* current *(ii)* power absorved*(iii)* circuit power factor.

Sol. Let Z_1 and Z_2 be the two impedences. Since the power factor of Z_1 is lagging and that of Z_2 leading, therefore Z_1 consists of an R– L circuit whereas Z_2 consists of an R - C circuit.

Case No. 1. Let Z_1 consists of R– L series circuit.

Voltage across $Z_1 = 230$ V.

Power consumed $P_1 = 100$ W.

Power factor, $\cos\phi_1 = 0{\cdot}5$ (lagging)

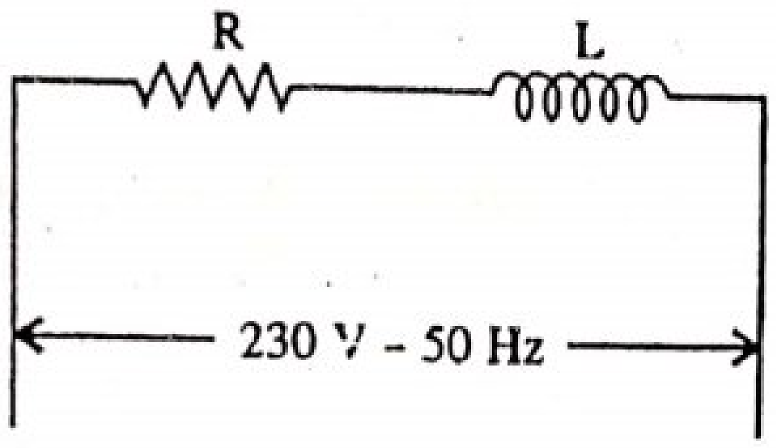


Fig. ACS—20*(a)*

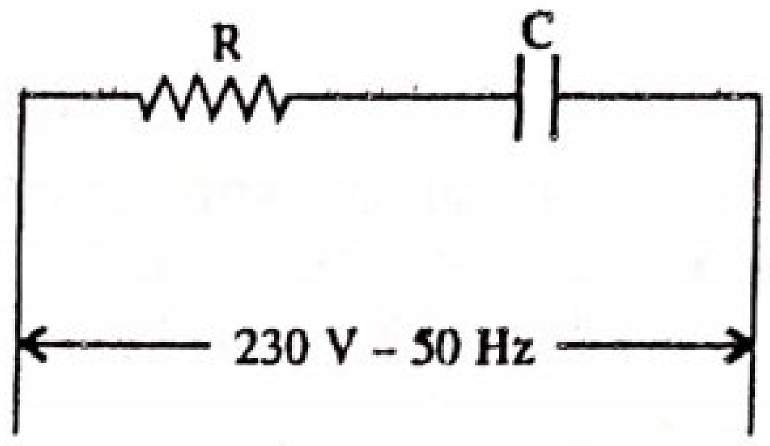


Fig. ACS—20*(b)*

Since power, $P = V.\ I.\ \cos\phi$

or $100 = 230 \times I \times 0{\cdot}5$

$$\therefore \quad I = \frac{100}{230 \times 0{\cdot}5} = 0{\cdot}8695 \text{ A}$$

Also $$P = I^2R \quad \therefore \quad R = \frac{P}{I^2} = \frac{100}{({\cdot}8695)^2} = 132{\cdot}26\,\Omega$$

$$\cos\phi = \frac{R}{Z} \quad \therefore \quad Z = \frac{R}{\cos\phi} = \frac{132{\cdot}26}{{\cdot}5} = 264{\cdot}52\,\Omega$$

Inductive reactance, $X_L = \sqrt{Z^2 - R^2} = \sqrt{(264{\cdot}52)^2 - (132{\cdot}26)^2}$

$= \mathbf{229{\cdot}0\ \Omega}$

Case No. 2. Let Z_2 consists of R–C circuit. [Fig ACS–20 ***(b)***]

Voltage across $Z_2 = 230$ V.

Power consumed, $P_2 = 60$ W.

Power factor $\cos\phi_2 = 0{\cdot}6$ (leading).

Power $= V.I.\ \cos\phi$ or $60 = 230 \times I \times 0{\cdot}6$

$$I = \frac{60}{230 \times 0{\cdot}6} = 0{\cdot}4347 \text{ A}.$$

Impedence, $Z_2 = \dfrac{230}{0.4347} = 529\,\Omega$

Power factor, $\cos\phi = \dfrac{R}{Z}$ $\quad\therefore\quad R = Z\cos\phi = 529 \times 0{\cdot}6 = 317{\cdot}4\,\Omega$

Capacitive reactance $X_C = \sqrt{Z^2 - R^2}$ or $X_C = \sqrt{529^2 - 317{\cdot}4^2} = 423{\cdot}2\Omega$

The circuit can be re-arranged in series when the two impedence are connected in series, across the same voltage.

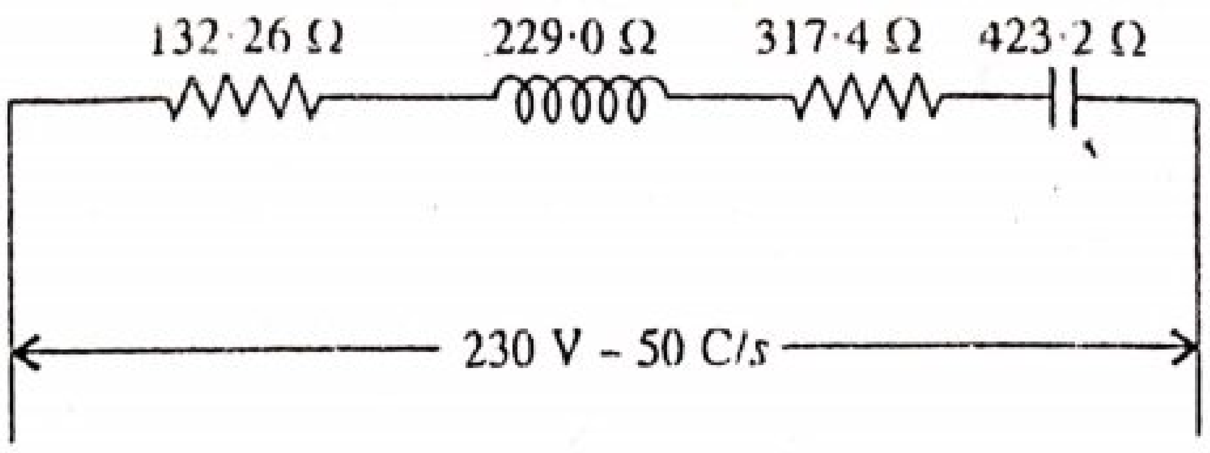


Fig. ACS—20(*c*)

Total resistance $R = 132{\cdot}26 + 317{\cdot}4 = 449{\cdot}66\ \Omega$

Net reactance $X = X_C - X_L = 423{\cdot}2 - 229 = 194{\cdot}2\Omega$

$\therefore$ Total impedence, $Z = \sqrt{(449{\cdot}66)^2 + (194{\cdot}2)^2} = \sqrt{202194{\cdot}12 + 37713{\cdot}64} = 489{\cdot}8$ ohms

Current, $I = \dfrac{V}{Z} = \dfrac{230}{489{\cdot}8} = 0{\cdot}47\text{ A}$ **(Ans.)**

Power absorbed $I^2 R = (0{\cdot}47)^2 \times 449{\cdot}66 =$ **99 watts (Ans.)**

Circuit power factor, $\cos\phi = \dfrac{R}{Z} = \dfrac{449{\cdot}66}{489{\cdot}8} = 0{\cdot}918$ **(Ans.)**

ACS-6. RESONANCE IN R–L–C SERIES CIRCUIT

A series circuit is said to be in electrical resonance when its net reactance ($X_L - X_C$) is zero. The frequency at which this resonance occurs is known as resonant; frequency and is written as f_r.

In *R-L-C* series circuit, the net reactance is ($X_L - X_C$) and $Z = \sqrt{R^2 + X^2}$

where X = Net reactance

At resonance $X_L - X_C = 0 \;\therefore\; X_L = X_C$ $\qquad \omega l = \dfrac{1}{\omega c}$

or $\qquad \omega L = \dfrac{1}{\omega C}$ $\quad$ or $\quad \omega^2 = \dfrac{1}{LC}$

or $\qquad \omega = \dfrac{1}{\sqrt{LC}}$ $\quad$ or $\quad 2\pi f_r = \dfrac{1}{\sqrt{LC}}$ $\quad$ or $\quad \boxed{f_r = \dfrac{1}{2\pi\sqrt{LC}}}$

If L and C are in Henries and Farads respectively then f_r is in Hz. Under resonant conditions $X = 0$, $\boxed{\therefore\ Z = R.}$

This is the minimum possible value of impedence. Hence the circuit behaves like a pure resistive circuit and then maximum current $I_m = \dfrac{V}{R} = \dfrac{V}{Z}$ The current will be in phase with voltage and so the power factor is unity under resonant conditions.

As the current passing through the circuit is maximum the voltage drops across L and C would be large. But these voltage drops being equal and opposite, will cancel each other. While taking L and C together, it forms a part of the circuit across which no voltage is developed but the current flowing is large. It is only due to the presence of resistance of the circuit. If it were not present, such a circuit would act like a short circuit to the currents of the frequency to which it resonates. Therefore sometimes it is known as acceptor circuit and series resonance as voltage resonance.

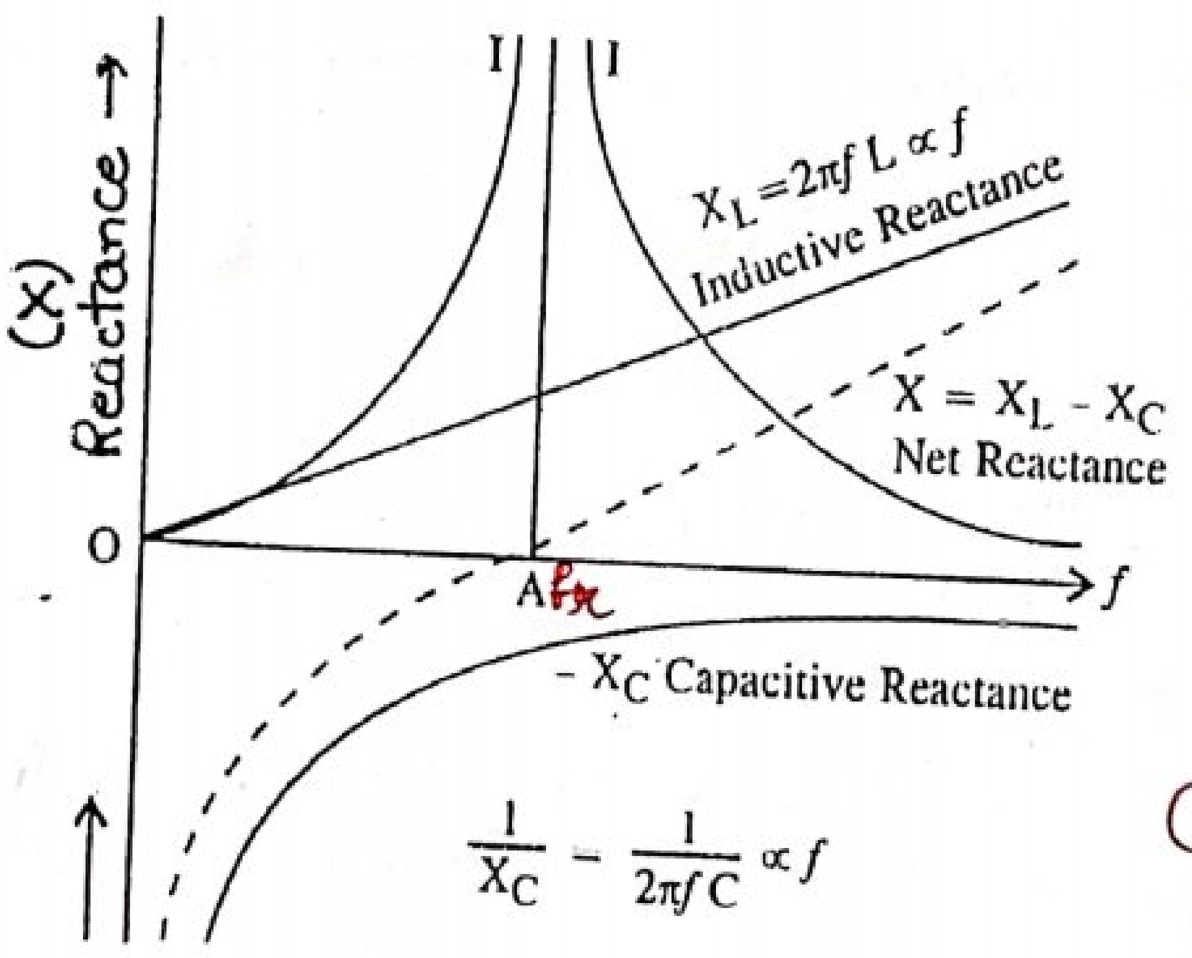
ACS -7. GRAPHICAL REPRESENATION OF RESONANCE

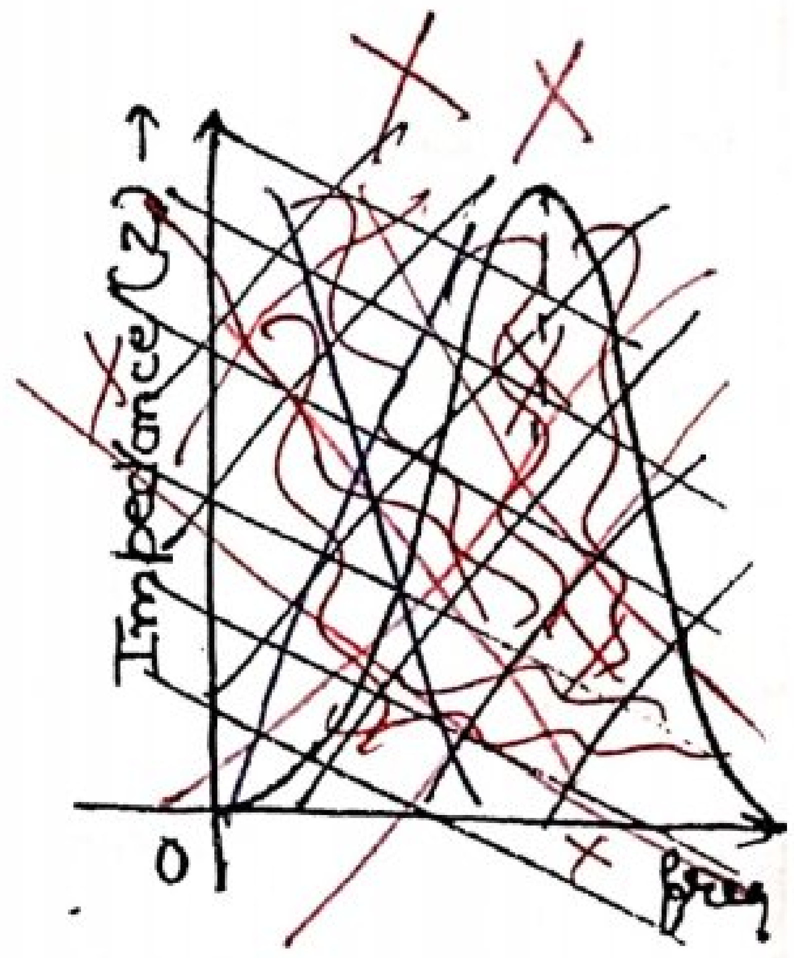
Here $$X_L = 2\pi f . L \quad i.e.\ X_L \propto f$$

Fig. ACS- 21 shows X_L for various values of frequency. It is a straight line through the origin.

$$-X_C = -\frac{1}{2\pi f \cdot C}$$

i.e. $$X_C \propto \frac{1}{f}$$





(Point A represents resonant frequency)

Fig. ACS—21
Reactance Curve

The graph for X_C is a rectangular hyperbola and in fourth quadrant because X_C is regarded negative.

Net reactance $X = X_L - X_C$

where $f = f_r$

Its graph is again hyperbola which crosses the frequency axis at A. At this point the total actance X is zero. Since R is zero (assumed), the total impedence Z is zero. Therefore, the lue of current I is infinite at this point as shown. For frequencies more than OA. The inductive reactance, X_L predominates and the current lags. For frequencies less than OA, the capacitive reactance, X_C predominates and therefore the current leads.

The following points are worth noting in series resonance.

(1) The net reactance, ($X = X_L - X_c$) in the circuit is zero.

(2) The impedence of the circuit at resonance $Z = R$

(3) The current I at resonance will reach its maximum value and it is in phase with the applied voltage.

(4) The power factor *i.e.* $\cos \phi$ is unity.

(5) The voltage across the inductance is equal to the voltage of and capacitance *i.e.*, $E_L = E_C$.

ACS-8. Q–FACTOR OF A SERIES CIRCUIT

In R-L-C, circuit, it is also referred as voltage magnification of resonance.

At resonance, the current is maximum

i.e. $$I_m = \frac{E}{R}$$

$$f_r = \frac{1}{2\pi\sqrt{LC}}$$

Voltage across either coil or capacitor $= I_m \cdot X_L$

Supply voltage $E = I_m \cdot R$

$$\text{Voltage magnification} = \frac{I_m \cdot X_L}{I_m \cdot R} = \frac{X_L}{R} = \frac{\omega L}{R}$$

$$\therefore \quad \text{Q-factor} = \frac{\omega L}{R} = \frac{2\pi f_r L}{R}$$

$$Q = \frac{1}{R}\sqrt{\frac{L}{C}}$$

Since resonant frequency, $f_r = \dfrac{1}{2\pi\sqrt{LC}}$ or $2\pi f_r = \dfrac{1}{\sqrt{LC}}$

$$\therefore \quad \boxed{\text{Q-factor} = \frac{L}{R}\cdot\frac{1}{\sqrt{LC}} = \frac{1}{R}\cdot\sqrt{\frac{L}{C}}}$$

$\sqrt{L} \times \sqrt{L}$ $\quad \frac{1}{\sqrt{L}} \times$

In series resonance, high value of Q means high voltage magnification and also high selectivity of the tuning coil. The Q-factor can be increased by having a coil of large inductance and small resistance.

Resonance Curve. The curve obtained by plotting a graph between current and frequency is known as resonance curve. The shape of such a curve is shown in fig. ACS-22. For smaller value of R, the current frequency curve is sharply peaked, but for larger values of R, the curve is flat. At resonant frequency f_r, the current reaches its maximum value, when the frequency is lower than the resonance frequency, $X_C > X_L$ and if frequency is higher than resonant frequency $X_C < X_L$. In both cases, the impedence of the circuit increases and the value of current decreases.

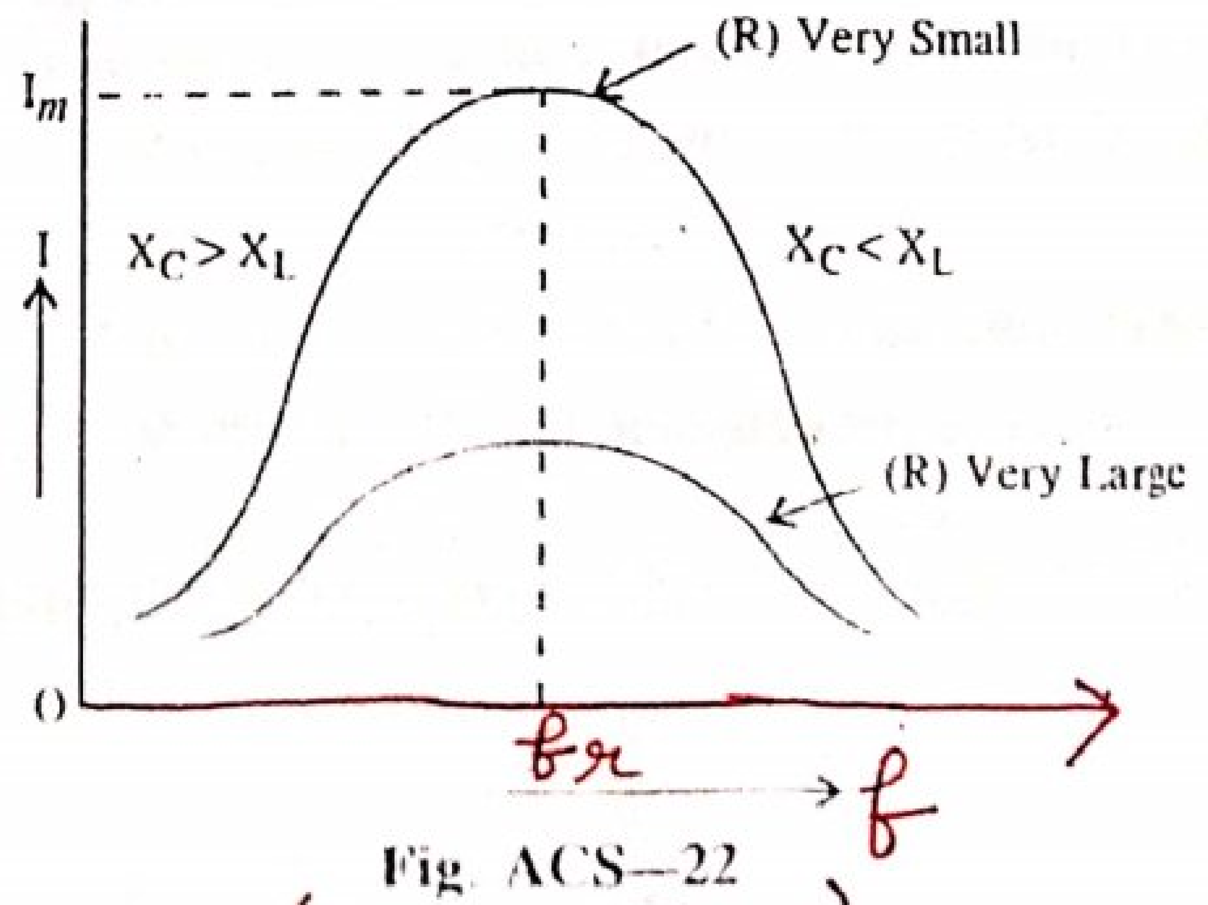


Fig. ACS—22
(Resonance Curve)
or response curve

Band Width. The range of frequency over which circuit current is equal to or more than 70.7% of maximum current is known as *band width* of a series resonant circuit.

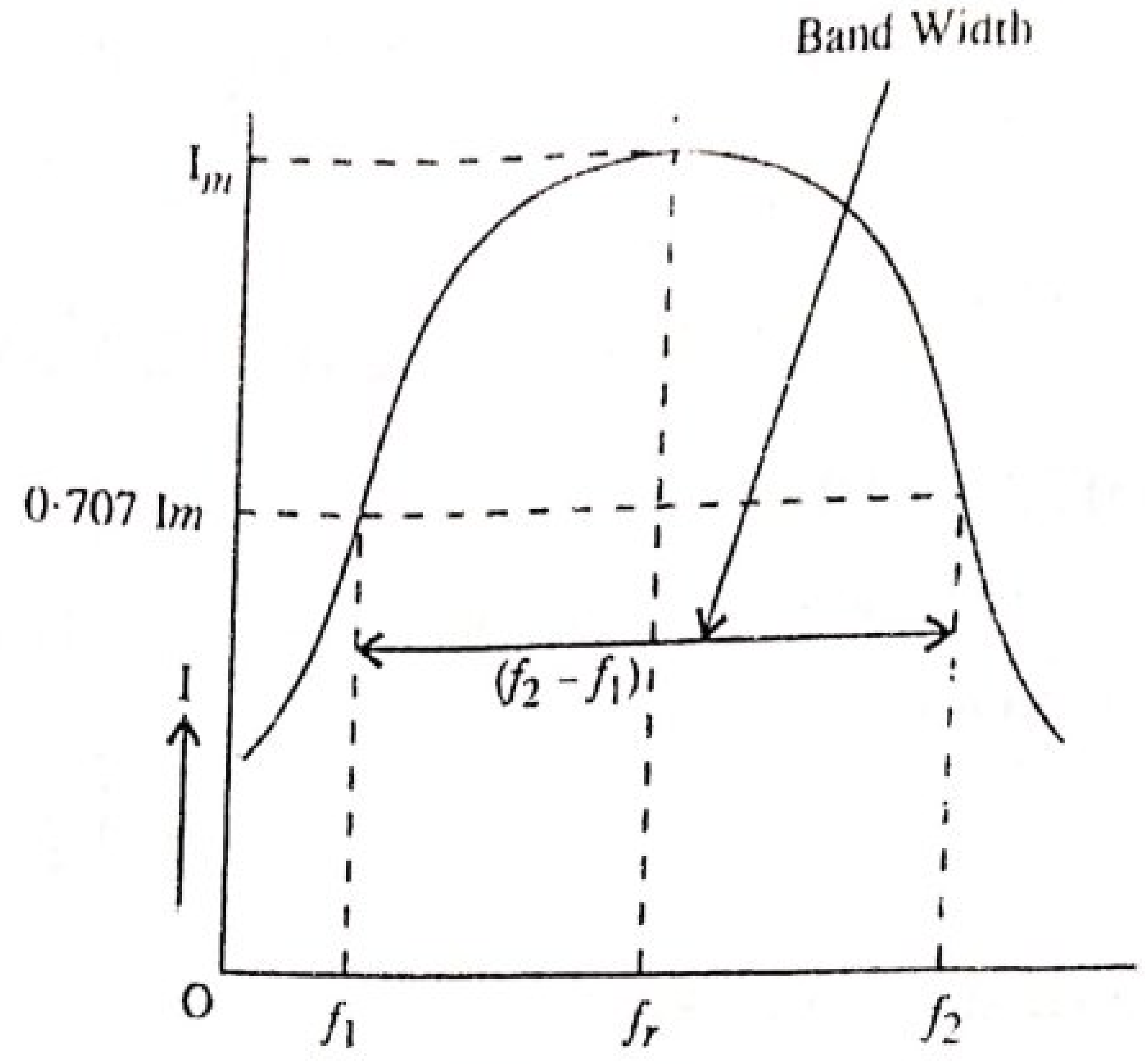


Fig. ACS—23

Figure ACS–23 shows a resonance curve of an *R-L-C* circuit, where the circuit current is equal to or greater than 70·7% of maximum current between frequency f_1 and f_2.

$\therefore$ Band Width $= f_2 - f_1$.

Here, the frequency f_1 is called lower cut–off frequency and the frequency f_2 is called upper cut-off frequency. The band width represents the frequency range at which the circuit offers low impedence to circuit current.

The ability of a circuit to select any particular frequency is decided by the sharpness of the resonant circuit and therefore by the smallness of resistance *R*. The selectivity is therefore inversely proportional to resistance *R*.

Problem ACS–15. A resistance, a capacitor and a variable inductor are connected in series across a 200 V, 50 Hz, supply. The maximum current which can be obtained by varying the inductance is 314 mA and the voltage across capacitance is then 300 V. Calculate the capacitance of a capacitor and the values of resistance and inductance.

Sol. Voltage across the circuit, $V = 200$ V.

Supply frequency, $f = 50$ Hz

Maximum value of current, $I_m = 314$ mA

Voltage across capacitance $V_C = 300$ V

At resonance, the current is maximum and is equal to $\frac{V}{R}$

$$\therefore \quad I_m = \frac{V}{R}$$

or $\quad 314 \times 10^{-3} = \frac{200}{R} \quad \therefore \quad R = \frac{200}{314 \times 10^{-3}} = 636.9$ ohms (Ans.)

Also, at resonance, the voltage drop across capacitance is equal to voltage drop across inductance.

Voltage drop across capacitance, $V_C = I X_C$

or $\quad 300 = 314 \times 10^{-3} X_c$

$$\therefore \quad X_C = \frac{300}{314 \times 10^{-3}}$$

$$X_C = \frac{1}{\omega C} \quad \therefore \quad C = \frac{1}{\omega X_C} = \frac{314 \times 10^{-3} \times 10^6}{2\pi \cdot 50 \cdot 300}$$

or $\quad C = 3{\cdot}33\ \mu F$ **(Ans.)**

Voltage drop across inductance, $V_1 = V_C$

$$\therefore \quad 300 = I.X_L \quad \text{or} \quad X_L = \frac{300}{314 \times 10^{-3}}$$

$$\therefore \quad L = \frac{X_L}{\omega} = \frac{300}{314 \times 10^{-3} \times 314} = 3{\cdot}04 \text{ Henry} \quad \textbf{(Ans.)}$$

Problem ACS–16. A resistance of 30 Ω, an inductance of 4 H and a capacitance of 25 μF are connected in series across 230 V supply. Calculate *(i)* the frequency at which current will be maximum *(ii)* the current at this frequency *(iii)* the potential difference across the inductance.

Sol. Value is resistance, $R = 30\ \Omega$.

Value of inductance, $L = 4$ H.

Value of capacitance, $C = 25\ \mu F$.

Supply voltage, $V = 230$ V.

The current will be maximum, when resonance occurs and the value of that resonant frequency, $f_r = \dfrac{1}{2\pi\sqrt{LC}}$.

$$\therefore \quad f_r = \frac{10^3}{2\pi.\sqrt{4 \times 25}} = \frac{10^3}{20\pi} = 15{\cdot}9 \text{ Hz.} \quad \textbf{(Ans.)}$$

The value of max. current, $I_m = \dfrac{V}{R} = \dfrac{230}{30} = 7{\cdot}67$ A **(Ans.)**

The potential difference across inductance

$$V_L = I.\ X_L = I.\ \omega L = 7{\cdot}67 \times 2\pi \times 15{\cdot}9 \times 4 = \textbf{3063·46 Volts (Ans.)}$$

Problem ACS–17. A choking coil of resistance 5 ohms and inductance 0·6 H is in series with a capacitance of 10 μF. If a voltage of 230 V is applied and the frequency is adjusted to resonance, find the voltage across the coil and capacitor.

Sol. Inductance, $L = 0.6$ H

Capacitance, $C = 10\ \mu F$

Resonant frequency, $f_r = \dfrac{1}{2\pi\sqrt{LC}}$

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$$= \frac{1}{2 \times 3{\cdot}14\sqrt{0{\cdot}6 \times 10 \times 16^{-6}}} = 65 \text{ Hz}$$

$$\text{Current at resonance} = \frac{230}{5} = 46\text{A}$$

Inductive reactance $X_L = \omega L = 2\pi f L = 2 \times 3{\cdot}14 \times 65 \times 0{\cdot}6 = 245\ \Omega$

$$Z_L = \sqrt{R^2 + X_L{}^2} = \sqrt{(5)^2 + (245)^2} = 245\ \Omega$$

Voltage across coil $= 46 \times 245 = 11270$ volts

$$\text{Capacitive reactance } X_C = \frac{1}{2\pi f C} = \frac{10^6}{2 \times 3 \cdot 14 \times 65 \times 10} = 245\ \Omega$$

Voltage across capacitor $V_C = 245 \times 46 =$ **11270 volts.**

ACS-9. POWER FACTOR and its practical importance

As already explained power factor may be defined as the cosine of the angle between voltage and current in an *a.c.* system.

For inductive circuit the current lags the voltage by some angle, say ϕ, then power factor is known as lagging power factor. But for capacitive circuit, the current leads the voltage by some angle then the power factor is known as leading power factor.

Fig. ACS–24 shows a phasor diagram for an inductive circuit. Here voltage, V is taken as reference vector and current I lags the voltage by an angle ϕ. The horizontal component of current $I \cos\phi$ (known as active component of current) is in phase with the voltage, whereas the vertical component of current $I \sin\phi$ (known as reactive component of current) is perpendicular to the voltage vector. If all the three sides are multiplied by voltage, V then a power triangle is obtained.

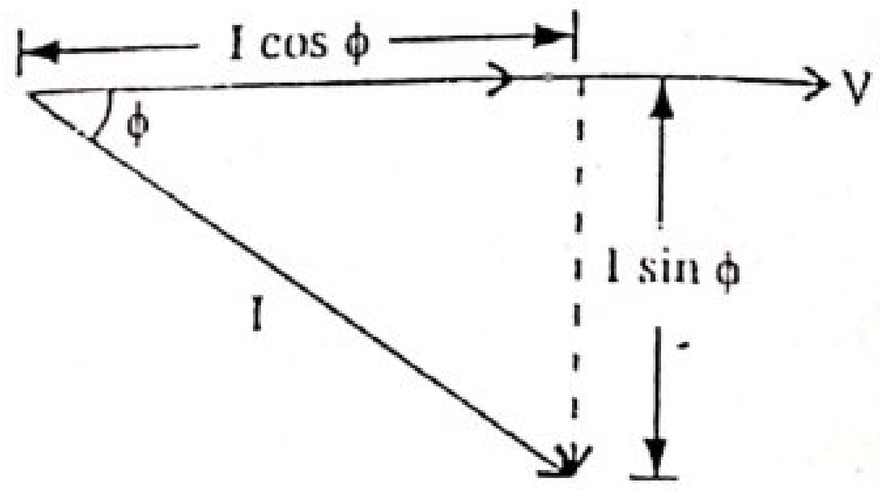

Fig. ACS—24

$VI\cos\phi$ is called real power in kW or W.

$VI\sin\phi$ is reactive power in kVAR or VAR

VI is apparent power in kVA or VA.

From the above three power equations,

$$\text{Power factor} = \frac{\text{Real Power}}{\text{Apparent Power}} = \frac{V.I.\cos\phi}{VI} = \cos\phi$$

Imp So the ratio of real power to the apparent power is called power factor.

Disadvantages of Low Power factor. As power consumed in any single phase *a.c.* circuit, $P = VI\cos\phi$.

$$\therefore \quad I = \frac{P}{V\cos\phi} \qquad (i)$$

But power consumed in three phase a.c. circuit is

$$P = \sqrt{3} \cdot V_L I_L \cos\phi$$

or

$$I_L = \frac{P}{\sqrt{3} V_L \cos\phi} \qquad \ldots(ii)$$

From equations *(i)* and *(ii)*, we see that for a fixed value of power and voltage, the load currents are inversely proportional to the power factor. Lower the power factor, higher is the current, and vice versa. The higher current results in the following disadvantages.

1. **Greater Conductor Size.** To transmit or distribute a fixed amount of power at a fixed voltage at low power factor, the conductor will have to carry large amount of current which will necessitate a large conductor size.

2. **Large Copper Losses.** Due to more currents carried by the conductors, I^2R losses are increased which results in poor efficiency of transmission or distribution.

3. **Poor Voltage Regulations.** Due to low power factor and increased currents, the voltage in transformers, alternators and transmission distribution lines drops more which again results in poor voltage regulation.

4. **Large kVA rating of the equipment.** The output power of electrical machine e.g. alternators and transformers is always rated in apparent power *i.e.* kVA

$$\text{Because} \quad \text{kVA} = \frac{\text{kW}}{\cos\phi}$$

Thus, we see that the rating of the equipment is inversely proportional to the power factor. At low power factor, the value of kVA will increase. Therefore at low power factor, equipment of higher kVA has to be installed which will increase the size and cost of the equipment.

Causes of Low power factor. The following are the main causes of low power factor.

1. Transformers and chokes are the inductive devices which causes the current to lag behind the voltage *i.e.* they operate at a lagging power factor. However, at normal loads, the magnetizing current does not effect too much, but at low loads or light loads, the magnetising current effects too much and lowers the power factor.

2. The motors which we use in our industry are mostly induction motors i.e. having inductive loads, which have low power factor. Normally the power factor of induction motors vary from 0·8 to 0·85 when operating at full load and 0·3 to 0·5 when running lightly i.e. at small loads.

3. Arc lamps, heating furnaces (Induction and arc furnaces) also operate at a low power factor (lagging).

Power Factor Improvement. The low power factor is generally due to the inductive loads in nature and also take the current at lagging power factor. In order to improve (to raise) the power factor, a static capacitor is connected in parallel with the load as shown in the fig. ACS 25*(a)* The static capacitor draws a current which leads the voltage and thus neutralizes the lagging wattless component partly or completely

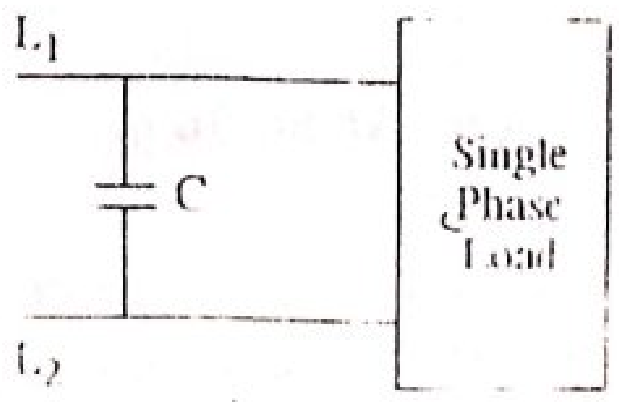

Fig. ACS—25*(a)*

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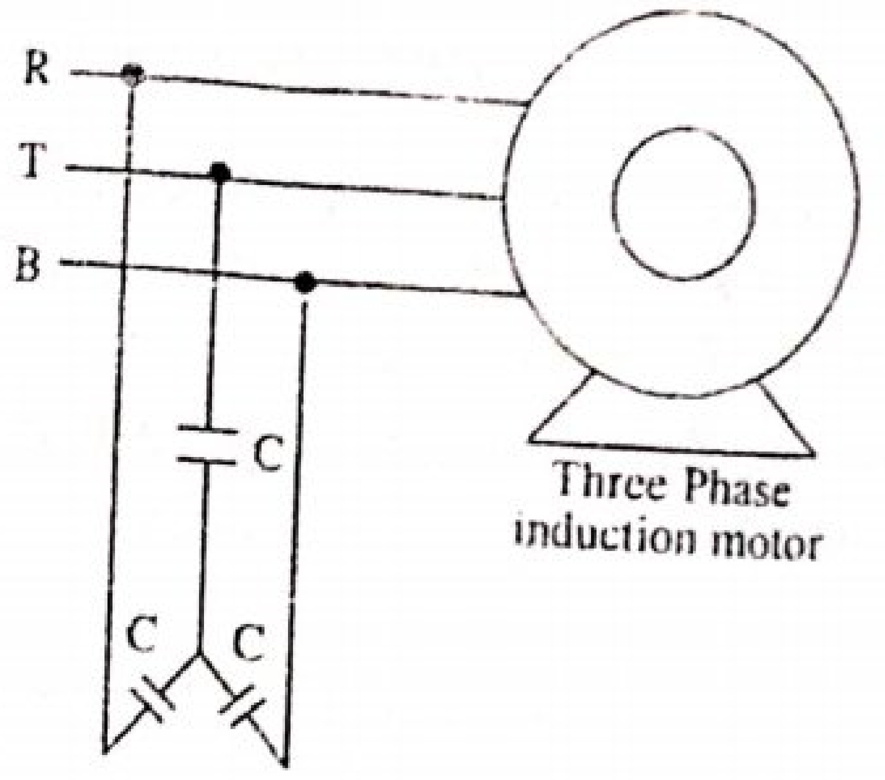
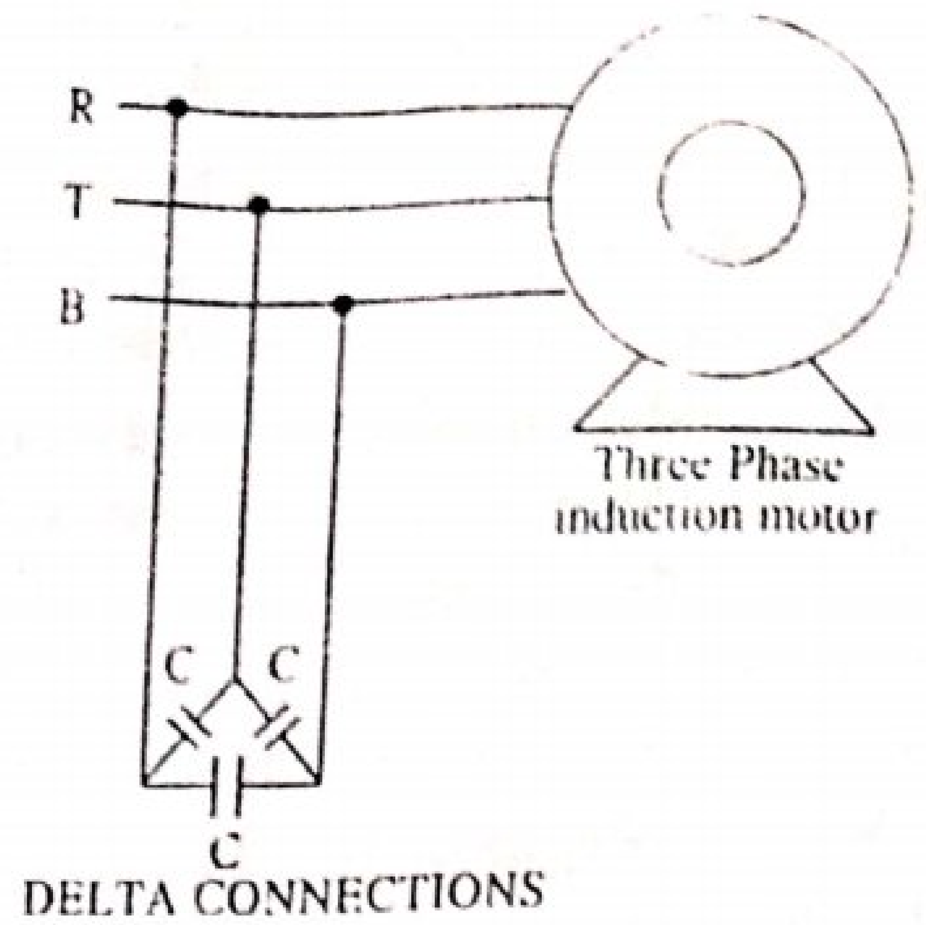
For three phase loads or 3 phase induction motors, the capacitors can be connected in star or delta as shown in the fig. ACS -25 *(b)* and ACS-25 (c) respectively.



STAR CONNECTIONS

Fig. ACS—25 *(a)*



DELTA CONNECTIONS

Fig. ACS—25 *(b)*

Generally, in practice the capacitors are connected in delta because in delta connections, the size of the capacitors will be one third of the size in star connections. There is not much power lost in the capacitors and also they need not much maintenance and are also available in small sizes.

Consider an inductive load as shown in the circuit diagram. The load current I lags by an angle ϕ_1 .

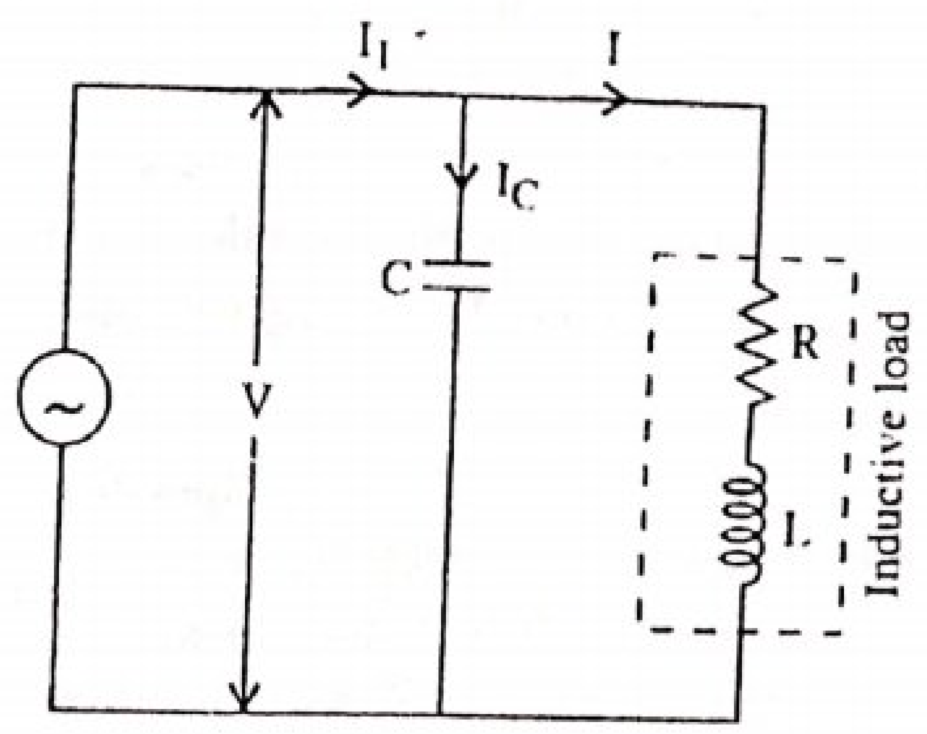


Fig. ACS—26 *(a)*

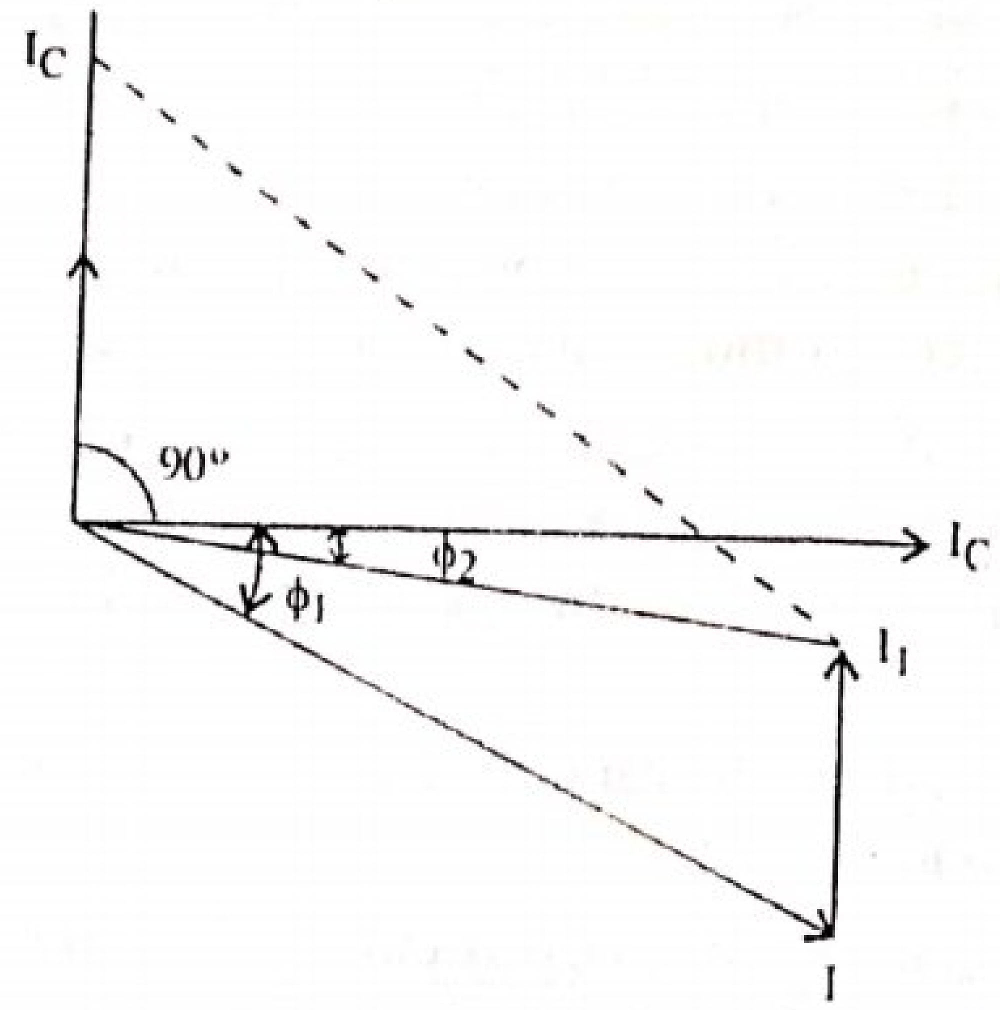


Fig. ACS—26 *(b)*

A capacitor C is connected in parallel with the load. The capacitor current I_C leads the voltage by 90°. The resultant current I_1 is the vector sum of the two currents I and I_C and its angle of lag is now ϕ_2, which is less than ϕ_1. The value of cos ϕ_2 is more than cos ϕ_1. Hence the power factor of the load is improved from cos ϕ_1 to cos ϕ_2.

Advantages of Improved Power Factor. When the power factor of the system is improved, then the current flowing through whole of the system decreases. Due to decreased current, the following are the advantages.

1. Small conductor size.
2. Small copper losses, better efficiency.
3. Less voltage drop, better regulation.
4. For the same kW out put, smaller is the kVA rating of the equipment.

Problem ACS–18. A single phase 220 volts, 50 Hz, motor takes a supply current of 50 amperes at a power factor of 0·5 lagging. The motor power factor has been improved to 0·9 lagging by connecting a condensor in parallel. Calculate the capacitance of the capacitor required..

Sol. Applied voltage, $V = 220$ volts

Current taken from supply, $I = 50$ A

Supply frequency, $f = 50$ Hz

Power factor of the motor $\cos \phi_1 = 0{\cdot}5$ (lagging)

$\therefore \quad \phi_1 = \cos^{-1} 0{\cdot}5 = 60°$

Active component of current, $I_a = I \cos \phi = 50 \times \cdot 5 = 25$ A.

Since load on the motor is constant,

The active component of current remains the same.

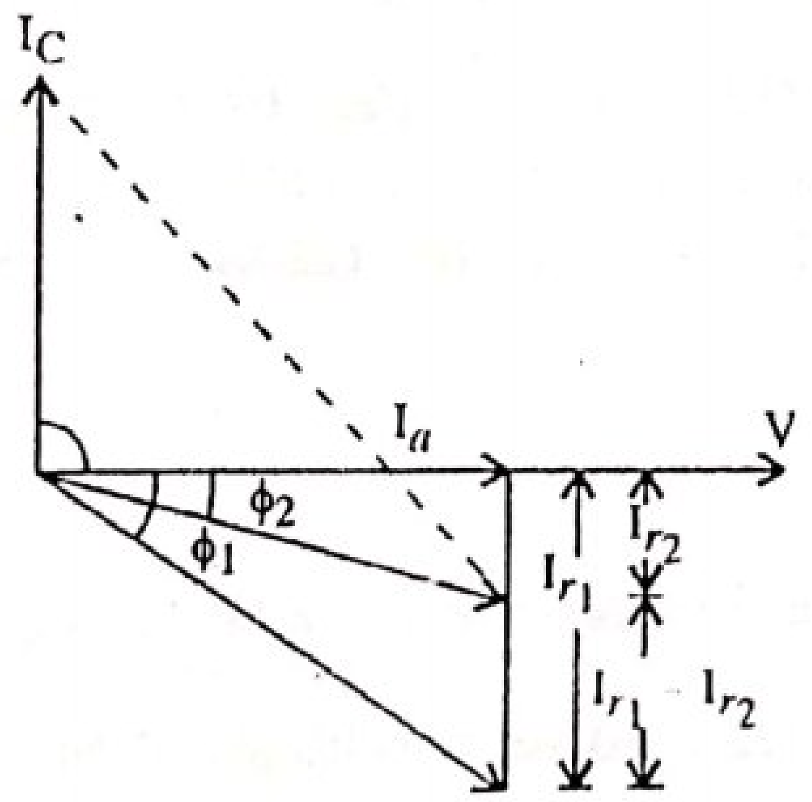


Fig. ACS—27 *(a)*

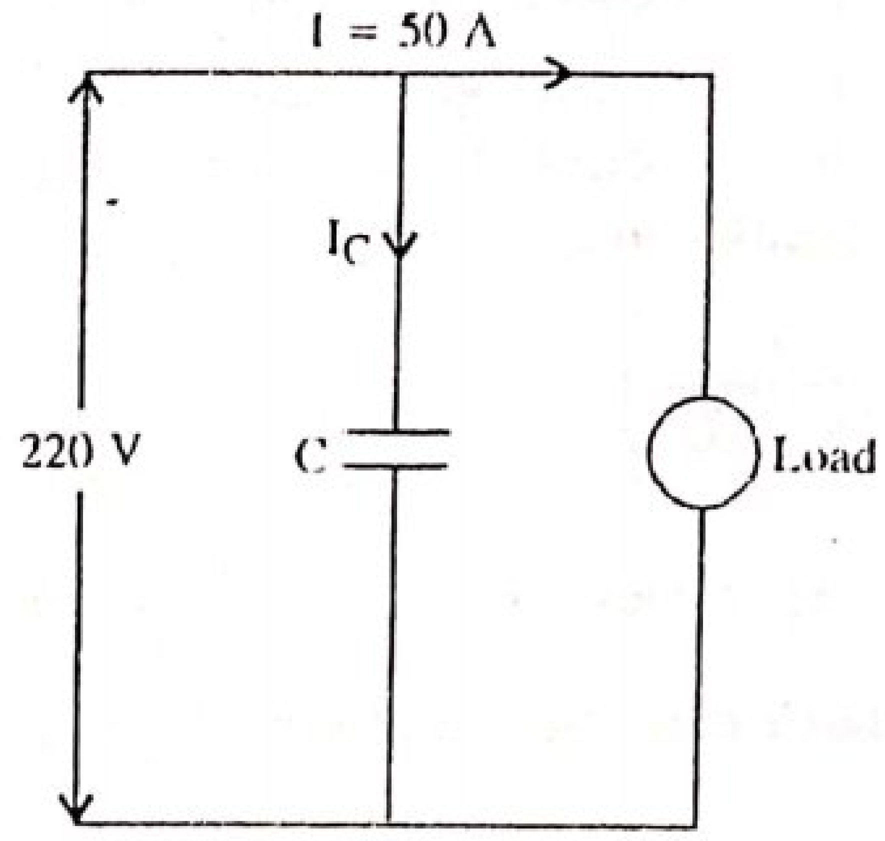


Fig. ACS—27 *(b)*

$$\tan \phi_1 = 1{\cdot}732$$

i.e. $\tan 60° = 1{\cdot}732$

Reactive component of current,

$$I_{r1} = I_a \tan \phi_1$$

i.e. $= 25 \times 1{\cdot}732 = 43{\cdot}3$ A

Let a capacitor of capacitance C farads be connected in parallel with the load as shown in fig ACS-27 *(b)* to improve the power factor from $\cos \phi_1$ to $\cos \phi_2$ as shown in the vector diagram

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Now $\cos \phi_2 = 0{\cdot}9$

$\therefore$ $\tan \phi_2 = \tan \cos^{-1} 0{\cdot}9 = 0{\cdot}4843$

Reactive component I_{r2} of the current $= I_a \tan \phi_2$

$= 25 \times 0{\cdot}4843 = 12{\cdot}1\ A$

$\therefore$ Current through capacitor $I_C = (I_{r1} - I_{r2})$

$I_C = (43{\cdot}3 - 12{\cdot}1) = (31{\cdot}2)$ A.

Now
$$I_C = \frac{V}{X_c} = 2\pi f\, C\, V$$

$\therefore$
$$C = \frac{I_c}{2\pi f . V} = \frac{31 \cdot 2 \times 10^6}{2 \times 3 \cdot 14 \times 50 \times 220} = 451\, \mu F.$$

ACS-10. TANK CIRCUIT OR LC OSCILLATORS

A resonator which consists of an LC circuit is known as tank circuit.

These oscillators are particularly suited for the radio frequency (*R.F.*) range. There are many types of *L–C* oscillators and they use a conventional type of *L–C* tank circuit. Their principle of operation is simple and almost similar for any of the many variations of the basic circuit. The operatiom of the basic circuit is as follows.

An *L–C* oscillator uses a tank circuit which comprises of inductor *L* in parallel with a capacitor. The parallel combination of *L* and *C* is excited into oscillations and the output voltages of the tank circuit is amplified by either a transistor or vacuum tube amplifier.

A fraction of the amplified voltage is fed back into the tank circuit either by an inductor or capacitor coupling. This is done to compensate the power loss in the tank circuit. This regenerative feed back results in a constant amplified out put voltage of the resonal frequency of the tank circuit $\left(f_r = \frac{1}{2\pi\sqrt{LC}} \right)$.

It is clear from the expression in $f_r = \frac{1}{2\pi\sqrt{LC}}$ that since the value of inductance required to produce. The *LC* oscillator can operate at very high frequencies upto several hundred MHz.

SUMMARY

1. The cosine of the angle between voltage and current in an a.c. circuit is known as power factor or it is defined as the ratio of resistance to impedence of an a.c. circuit. It ranges from 0 to 1.

2. In R-L series circuit, impedence, $Z = \sqrt{R^2 + X_L^2}$

Current, $I = \frac{V}{Z}$

Power $= I^2 R$

The power is consumed only in pure resistance, a pure inductor does not consume any power

3.(a) $kVA = \sqrt{(kW)^2 + (kVAR)^2}$

(b) The power which is actually consumed in the circuit is called true power.

(c) The power received from a source in half cycle is returned to the source in the next half cycle. is circulating power is known as reactive power. It does not do any useful work in the circuit.

$$\text{Active power or true power, } P = VI \cos\phi \text{ watts or } kW$$

$$\text{Reactive power, } P_r = VI \sin\phi \text{ VAR or } kVAR$$

$$\text{Apparent power, } P_a = VI \text{ VA or } kVA$$

$$kW = kVA \cos\phi$$

$$kVAR = kVA \sin\phi, \text{ Lagging } kVAR \text{ is taken - ve.}$$

4. Smaller the phase angle ϕ between voltage and current, greater will be the value of power :tor cos ϕ, if $\phi = 0$, power factor = Max *i.e.*, '1', and reactive power *VIS*-ϕ will be zero. It :ans the whole of the apparent power drawn by the circuit is being utilized by it. Thus power :tor is a measure of its utilizing the apparent power drawn by it. Hence power factor to 1 must as near to 1 as possible.

5. (a) The inductive circuit causes lagging power factor.

(b) The capacitive circuit causes leading power factor.

6. In a series R-C circuit, impendence $Z = \sqrt{R^2 + X_C^2}$

where, $X_C = \dfrac{1}{\omega_C}$ = Capacitive reactance

$$I = \frac{V}{Z}$$

$$\text{Average power, P} = VI \cos\phi$$

7.(a) In an *R-L-C* series circuit at resonance, $X_L = X_C$

and resonance frequency, $f_r = \dfrac{1}{2\pi\sqrt{LC}}$

(b) *Q*-factor of a series circuit $= \dfrac{1}{R}\sqrt{\dfrac{L}{C}}$

(c) The range of frequency over which circuit current is equal to or more than 70·7% max rrent is known as band width of a series resonant circuit.

8. Low power factor results in greater conductor size, large Cu loss, poor voltage regulation id large kVA rating of equipment.

9. Power factor can be improved by using a static capacitor in parallel for a 1-ϕ load or in ar or delta for a 3 ϕ load

10.A resonator which consists of an LC circuit is known as tank circuit. These oscillations e particularly suited for Radio Frequency (R.F.) range

REASONING QUESTIONS

(i) The power factor of an a.c. circuit is never more than unity, why ?

(ii) The power factor of a purely inductive circuit is zero, why ?

(iii) The power factor of a purely capacitive circuit is zero, why ?

(iv) With the reduction of frequency, the inductive reactance of the circuit decreases why ?

(v) The true power in an a.c. circuit is given by; kW = kVA cos θ but not by kVA, why ?

(vi) The power consumed by a pure inductance connected in a.c. source is zero, why ?

(vii) The power factor of an R-L-C circuit of which $X_L > X_C$ is always lagging, why ?

(viii) In an a.c. series circuit at resonant frequency, the value of circuit impedence reduces to the resistance of the circuit, why ?

(ix) With the increase of frequency, the capacitive reactance of the circuit decreases, why ?

(x) In case of delta connected circuit, when one resistor is open, the power will be reduced by $\frac{1}{3}$ rd, why ?

EXAMINATION QUESTIONS

(Appeared & Expected)

Q. 1. It is usually said that power factor should be as near to 1 as possible. Why ?

Ans. Smaller the phase angle ϕ between voltage and current, greater will be the value of power factor $\cos\phi$, if $\phi = 0$, power factor = Max *i.e.*, '1', and reactive power $VI S\phi$ will be zero. It means the whole of the apparent power drawn by the circuit is being utilized by it. Thus power factor is a measure of its utilizing the apparent power drawn by it. Hence power factor to 1 must be as near to 1 as possible.

Q. 2. In a circuit, the applied voltage is 100 V and is found to lag the current of 10A by 30°.

(i) Is the P.F. lagging or leading ?

(ii) What is the value of P.F. ?

(iii) Is the circuit inductive or capacitive ?

(iv) What is the value of active and reactive power ?

Ans. $V = 100$ V, $I = 10$ A

V lags I by 30° or current leads applied voltage by 30°

(i) Hence P.F. is leading (Ans.)

(ii) P.F. *i.e.*, $\cos\phi = \cos 30° = \frac{\sqrt{3}}{2} = 0.866$ (*leading*) (Ans.)

(iii) As P.F. is leading one, so the circuit is capacitive. **(Ans.)**

(iv) Active power, $P_a = VI \cos\phi = 100 \times 10 \times 0.866 = 866$ watts **(Ans.)**

Reactive power, $P_r = VI \sin\phi = 100 \times 10 \times \sin 30^\circ$ i.e., 0.5

$= 500$ VAR $= 0.5$ kVAR **(Ans.)**

Q. 3. What is resonance ? What is the condition for resonance in series circuit ?

Ans. A series circuit is said to be in electrical resonance when its net reactance $(X_L - X_C)$ is zero. The frequency at which this resonance occurs is known as resonant frequency and is written as f_r.

In R-L-C series circuit, the net reactance is $(X_L - X_C)$ and $Z = \sqrt{R^2 + X^2}$

where X = Net reactance

At resonance $X_L - X_C = 0 \quad \therefore \quad X_L = X_C$

or $\quad \omega L = \dfrac{1}{\omega C} \quad$ or $\quad \omega^2 = \dfrac{1}{LC}$

or $\quad \omega = \dfrac{1}{\sqrt{LC}} \quad$ or $\quad 2\pi f_r = \dfrac{1}{\sqrt{LC}} \quad$ or $\quad f_r = \dfrac{1}{2\pi\sqrt{LC}}$

If L and C are in Henries and Farads respectively then f_r is in Hz. Under resonant conditions $X = 0, \quad \therefore \quad Z = R$.

This is the minimum possible value of impedence. Hence the circuit behaves like a pure resistive circuit and then maximum current $I_m = \dfrac{V}{R} = \dfrac{V}{Z}$. The current will be in phase with voltage and so the power factor is unity under resonant conditions.

As the current passing through the circuit is maximum the voltage drops across L and C would be large. But these voltage drops being equal and opposite, will cancel each other. While taking L and C together, it forms a part of the circuit across which no voltage is developed but the current flowing is large. It is only due to the presence of resistance of the circuit. If it were not present , such a circuit would act like a short circuit to the currents of the frequency to which it resonates. Therefore sometimes it is known as acceptor circuit and series resonance as voltage resonance.

Q. 4. What do you mean by inductive reactance and capacitive reactance ?

Ans. **Inductive reactance :** The opposition offered to the flow of alternating current by an inductance is called inductive reactance. Its unit is Ω and denoted by X_L

$$X_L = \omega L = (2\pi f)L \quad (\because \omega = 2\pi f)$$

where f is the frequency of alternating current in Hz, L is the inductance of the coil in henry.

Capacitive reactance : The opposition offered to the flow of alternating current by a capacitive circuit is known as capacitive reactance. It is denoted by X_C. Its unit is also ohm (Ω). It is given by, $X_C = \dfrac{1}{\omega C} = \dfrac{1}{(2\pi f)C}$ where f is the supply frequency of alternating current and C is the capacitance of capacitor in farad.

Q. 5. **(a) What is power factor ? Give its importance.**

(b) What is active reactive and apparent power ?

Ans. *(a)* The cosine of the angle between voltage and current in an a.c. circuit is known as power factor or it is defined as the ratio of resistance to impedence of an a.c. circuit. It ranges from 0 to 1.

(b) The power which is actually consumed in the circuit is called true power.

The power received from a source in half cycle is returned to the source in the next half cycle. This circulating power is known as reactive power. It does not do any useful work in the circuit.

$$\text{Active power or true power, } P = VI \cos\phi \text{ watts or kW}$$

$$\text{Reactive power, } P_r = VI \sin\phi \text{ VAR or kVAR}$$

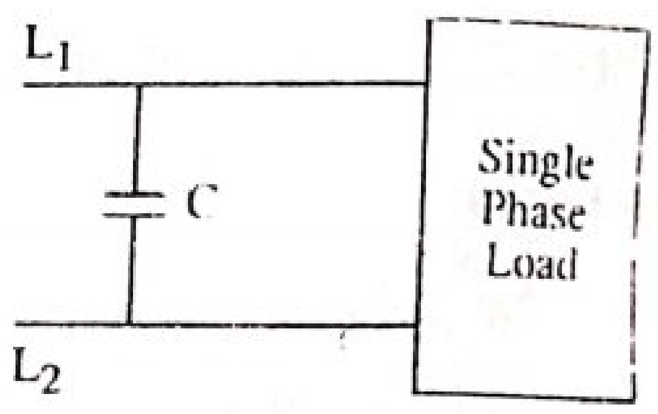
$$\text{Apparent power, } P_a = VI \text{ VA or kVA}$$

$$kW = kVA \cos\phi$$

$$kVAR = kVA \sin\phi, \text{ Lagging } kVAR \text{ is taken } -\text{ve.}$$

Q. 6. **How will you improve power factor of an inductive load ?**

Ans. The low power factor is generally due to the inductive loads in nature and also take the current at lagging power factor. In order to improve (to raise) the power factor, a static capacitor is connected in parallel with the load as shown in the fig. The static capacitor draws a current which leads the voltage and thus neutralizes the lagging wattless component partly or completely.



Q. 7. **What are the dis-advantages of low power factor ? Explain.**

Ans. The various dis-advantages of low power factor are explained below.

Disadvantages of Low Power factor. As power consumed in any single phase *a.c.* circuit, $= VI\cos\phi$

$$\therefore \quad I = \frac{P}{V\cos\phi} \qquad \ldots(i)$$

But power consumed in three phase a.c. circuit is

$$P = \sqrt{3}\,.\,V_L I_L \cos\phi$$

$$\text{or} \quad I_L = \frac{P}{\sqrt{3}V_L \cos\phi} \qquad \ldots(ii)$$

From equations *(i)* and *(ii)* we see that for a fixed value of power and voltage, the load currents are inversely proportional to the power factor. Lower the power factor, higher is the current, and vice versa. The higher current results in the following disadvantages.

1. **Greater Conductor Size.** To transmit or distribute a fixed amount of power at a fixed voltage at low power factor the conductor will have to carry large amount of current which will necessitate a large conductor size.

2. **Large Copper Losses.** Due to more currents carried by the conductors, I^2R losses are increased which results in poor efficiency of transmission or distribution.

3. **Poor Voltage Regulations.** Due to low power factor and increased currents, the voltage in transformers, alternators and transmission distribution lines drops more which again results in poor voltage regulation.

4. **Large kVA rating of the equipment.** The out put power of electrical machine e.g alternators and transformers is always rated in apparent power *i.e.* kVA

$$\text{Because} \quad \text{kVA} = \frac{\text{kW}}{\cos\phi}$$

Thus, we see that the rating of the equipment is inversely proportional to the power factor. At low power factor, the value of kVA will increase. Therefore at low power factor, equipment of higher kVA has to be installed which will increase the size and cost of the equipment.

Q. 8. Discuss the causes of low power factor.

Ans. Causes of Low power factor. The following are the main causes of low power factor.

1. Transformers and chokes are the inductive devices which causes the current to lag behind the voltage *i.e.* they operate at a lagging power factor. However, at normal loads, the magnetizing current does not effect too much, but at low loads or light loads, the magnetising current effects too much and lowers the power factor.

2. The motors which we use in our industry are mostly induction motors i.e. having inductive loads, which have low power factor. Normally the power factor of induction motors vary from 0·8 to 0·85 when operating at full load and 0·3 to 0·5 when running lightly i.e. at small loads.

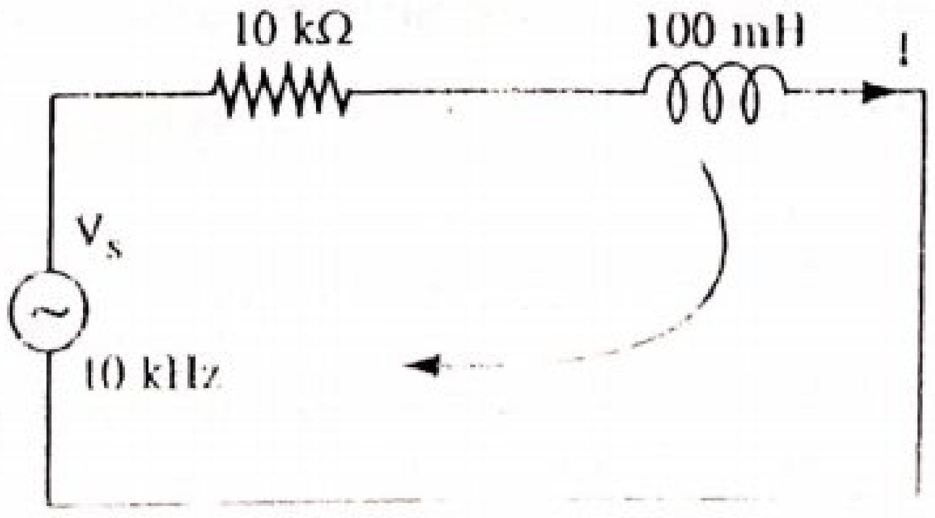
3. Arc lamps, heating furnaces (Induction and arc furnaces) also operate at a low power factor (lagging).

Q. 9. State the advantages of improved power factor.

Ans. Advantages of Improved Power Factor. When the power factor of the system is improved, then the current flowing through whole of the system decreases. Due to decreased current, the following are the advantages.

1. Small conductor size.
2. Small copper losses, better efficiency.
3. Less voltage drop, better regulation.
4. For the same kW out put, smaller is the kVA rating of the equipment.

Q. 10. In figure current, $i = 0.2\angle 0^\circ$ mA. Find the source voltage and express it in polar form.



Ans. Here,

$$R = 10\ \text{K}\Omega$$

$$X_L = 2\pi f L$$

$$= 2\pi \times 10 \times 10^3 \times 100 \times 10^{-3}$$

$$= 6.283\ k\Omega$$

Impedence, $\vec{Z} = \sqrt{R^2 + X_L^2} = \sqrt{(10)^2 + (6{\cdot}283)^2} = 11{\cdot}81\ k\Omega$

$$\tan\phi = \frac{6{\cdot}283}{10} \quad \therefore\ \phi = \tan' \frac{6{\cdot}283}{10} = 32{\cdot}142°$$

$$\vec{I} = 0.2\underline{|0°}$$

Source voltage, $\vec{V} = \vec{I}\,\vec{Z} = 0.2\underline{|0°} \times 10^{-3} \times 11{\cdot}81 \times 10^3\ \underline{|32.142°}$

$$\therefore \quad V = 2{\cdot}362\ \underline{|32.142°}\ \text{volts.}$$

OBJECTIVE TYPE QUESTIONS

A. Fill in the blanks

1. At resonance in series *RLC* circuit, *XL* = ...
2. Power factor of an inductive circuit will be always ....
3. The ability of a resonant circuit to discriminate between one particular frequency and all other is called ....
4. The frequency at which an *R-L-C* network behaves as a purely resistive network is called the .... frequency.
5. The maximum value of power factor in an a.c. circuit can be ....
6. The power consumed by a pure inductive circuit is ....
7. In an a.c. circuit, the ratio of resistance to impedence is called ....
8. The power factor of a pure capacitive circuit is ....
9. The power factor of a pure resistive circuit is ....
10. At resonance, the power factor of the circuit is ....
11. For a given load, if reactive power is low, the power factor will be ....
12. In a series *R-C* circuit, current .... with the increase in frequency.
13. The power factor of an a.c. circuit is equal to cosine of ....
14. In an *R-L* series circuit, the power factor is ....
15. In a series R-C circuit excited by a d.c. voltage, the steady state current is .... A.
16. The units of time constant $\frac{L}{R}$ is ....
17. Two alternating quantities are added ....
18. The average value of an unsymmetrical alternating quantity should be calculated over ....
19. In an RCL series circuit, during resonance the .... will be minimum
20. In an R-L circuit, the resistance and reactance are 4 ohms each. In this circuit, current .... the voltage by 45°

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B. Multiple Choice Questions

Select the correct answers :

1. In an *R-L* series circuits, the power factor is
   (*a*) leading (*b*) lagging
   (*c*) zero (*d*) unity.

2. A circuit component that opposes the change in circuit voltage is
   (*a*) resistance (*b*) capacitance
   (*c*) inductance (*d*) All of the above.

3. The power factor of a practical inductor is
   (*a*) unity (*b*) zero
   (*c*) lagging (*d*) leading.

4. The power factor of an ordinary electric bulb is
   (*a*) zero (*b*) unity
   (*c*) slightly more than unity. (*d*) slightly less than unity.

5. The power factor of an a.c. circuit is equal to
   (*a*) cosine of the phase angle (*b*) sine of the phase angle
   (*c*) unity for a reactive circuit. (*d*) None of these.

6. Power loss in an electrical circuit can take place in
   (*a*) inductance only (*b*) capacitance only
   (*c*) resistance only (*d*) inductance and resistance.

7. While drawing phasor diagram for a series circuit, the reference vector is
   (*a*) voltage (*b*) current
   (*c*) resistance (*d*) power.

8. At very high frequencies, a series *R-C* circuit behaves as almost purely ……… circuit
   (*a*) capacitive (*b*) resistive
   (*c*) inductive (*d*) None of these

9. For a series R-C circuit, V_R is measured to be 4 V and V_C is measured as 3 V. The a.c. source voltage will be
   (*a*) 7 V (*b*) 4 V
   (*c*) 5 V (*d*) 3 V

10. The power factor at resonance in R-L-C series circuit is
    (*a*) zero (*b*) unity
    (*c*) 0·5 lagging (*d*) 0·5 leading

11. In a series *R-C* circuit, current ……… with the increase in frequency
    (*a*) increases (*b*) decreases
    (*c*) remains same (*d*) None of these

12. When an inductor and capacitor are in series in an a.c. circuit, the phase angle between voltage drops across them will be

(a) 0°
(b) $\pi/2$ radians
(c) π radians
(d) $\pi/4$ radians.

13. At very low frequencies, a series R-C circuit behaves as almost purely .... circuit

(a) inductive
(b) resistive
(c) capacitive
(d) None of these.

14. In an a.c. circuit, the ratio of active power to apparent power is called the .... factor.

(a) form
(b) load
(c) power
(d) Q.

15. The product of apparent power and sin of the phase angle between circuit voltage and circuit current is called the .... power.

(a) true
(b) reactive
(c) wattless
(d) Any of (b) and (c)

16. A coil when connected on d.c. 100 V supply will draw .... current as compared to 100 V a.c. supply.

(a) more
(b) less
(c) same
(d) None of these.

17. If the frequency of the pure inductive circuit is halved the current of the circuit will be

(a) same
(b) doubled
(c) halved
(d) four times.

18. Which of the following statement is correct ?

(a) W = V × I
(b) W = V I sin θ
(c) kW = kVA cos θ
(d) kW = kVA sin θ.

19. Q-factor of a circuit is given by ;

(a) ratio of R/Z
(b) ratio of Z/R
(c) ratio of X_L/Z
(d) ratio of X_C/Z.

20. For an RLC series circuit, current at series resonance is

(a) maximum at unity power factor
(b) maximum at leading power factor
(c) minimum at leading power factor
(d) minimum at unity power factor

C. Whether the following statements are true or False

1. Ohmic resistance is termed as effective resistance.
2. All the rules and laws which apply to d.c. network also apply to a.c. networks consisting of resistance only.
3. In a pure capacitive circuit, the current will lead the voltage by 90°

4. Capacitive reactance, $X_C = W_C$.
5. In a pure capacitive circuit, the current will lag the voltage by 90°.
6. The power factor of an a.c. circuit is equal to cosine of the phase angle.
7. A circuit of zero lagging power factor behaves as inductive circuit.
8. The power factor of an a.c. circuit is equal to the ratio of resistance and impedence $\left(\frac{R}{Z}\right)$.
9. Skin effect occurs when a conductor carries current at low frequencies.
10. The power factor of an a.c. circuit lies between 0 and 1.
11. In a series resonant circuit, the current and voltage are in phase.
12. The quality factor of *R-L-C* circuit will increase, if *R*-increases.
13. In a series resonant circuit, the voltage across the circuit is the same as the voltage across the resistance.
14. The power in a single phase a.c. circuit is given by $VI \sin \phi$.
15. The voltage applied across on *R-L* series circuit is equal to the algebraic sum of voltage drops across *R* and *L* (V_R and V_L).

ANSWERS

A	1. X_C	2. lagging	3. selectivity	4. resonant	5. one
	6. zero	7. power factor	8. zero	9. unity	10. unity
	11. high.	12. increases	13. the phase angle		14. lagging.
	15. zero	16. second	17. vectorially	18. one cycle	19. impedence
	20. lags.				
B	1. (b)	2. (b)	3. (c)	4. (b)	5. (a)
	6. (c)	7. (b)	8. (b)	9. (c)	10. (b)
	11. (a)	12. (c)	13. (b)	14. (c)	15. (d)
	16. (a)	17. (b)	18. (c)	19. (d)	20. (a)
C	1. True	2. False	3. False	4. False	5. False
	6. True	7. True	8. True	9. False	10. True
	11. True	12. False	13. True	14. False	15. False

TEXT QUESTIONS

1. Derive the relationship between the voltage and current for a purely inductive circuit. Also show that the average power consumed by a circuit is zero.
2. Explain, inductive reactance and capacitive reactance.

ACP

A.C. Parallel Circuits

Introduction. As in d.c. parallel circuits, the voltage across each resistor (branch) is the same, but the value of current in any resistor (branch) depends upon the value of branch resistance. The total line current is the algebraic sum of all branch currents. Similarly in a.c. parallel circuits, the voltage across each branch is same, the branch current depends upon the branch impedence and the total line current is equal to the phasor sum of all the branch currents. The parallel circuits are used more frequently in our electrical systems because of the following points.

The parallel circuits are used because each equipment or device can be operated independently of the other. The other point is that most of the electrical devices (equipments) require different currents at the same voltage, so these equipments are connected across the same bus bars.

ACP–1. METHODS OF SOLVING PARALLEL A.C. CIRCUITS

An a.c. parallel circuit consists of two or more series branches. Therefore, each branch of the circuit can be considered separately as a series circuit and then the effects of the branches are combined together. The alternating voltages and currents are phase quantities therefore, while making the circuit calculations their magnitudes and phase positions (phase angles) must be taken into consideration. There are two methods of parallel circuits.

(1) Admittance Method

(2) By phasor diagram or Vector Method.

Admittance Method. Admittance of a circuit is defined as the reciprocal of impedence and denoted by Y. The unit of admittance is mho and its symbol is ℧.

$$\therefore \quad Y = \frac{1}{Z} \text{ or } \frac{\text{r.m.s. value of amperes}}{\text{r.m.s. value of voltage}}$$

As the impedence Z has two rectangular components R and X. Similarly admittance Y has also two components g and b as shown in figure ACP–1.

The base of an admittance triangle is called conductance (g) as shown in fig. ACP–1. (b),

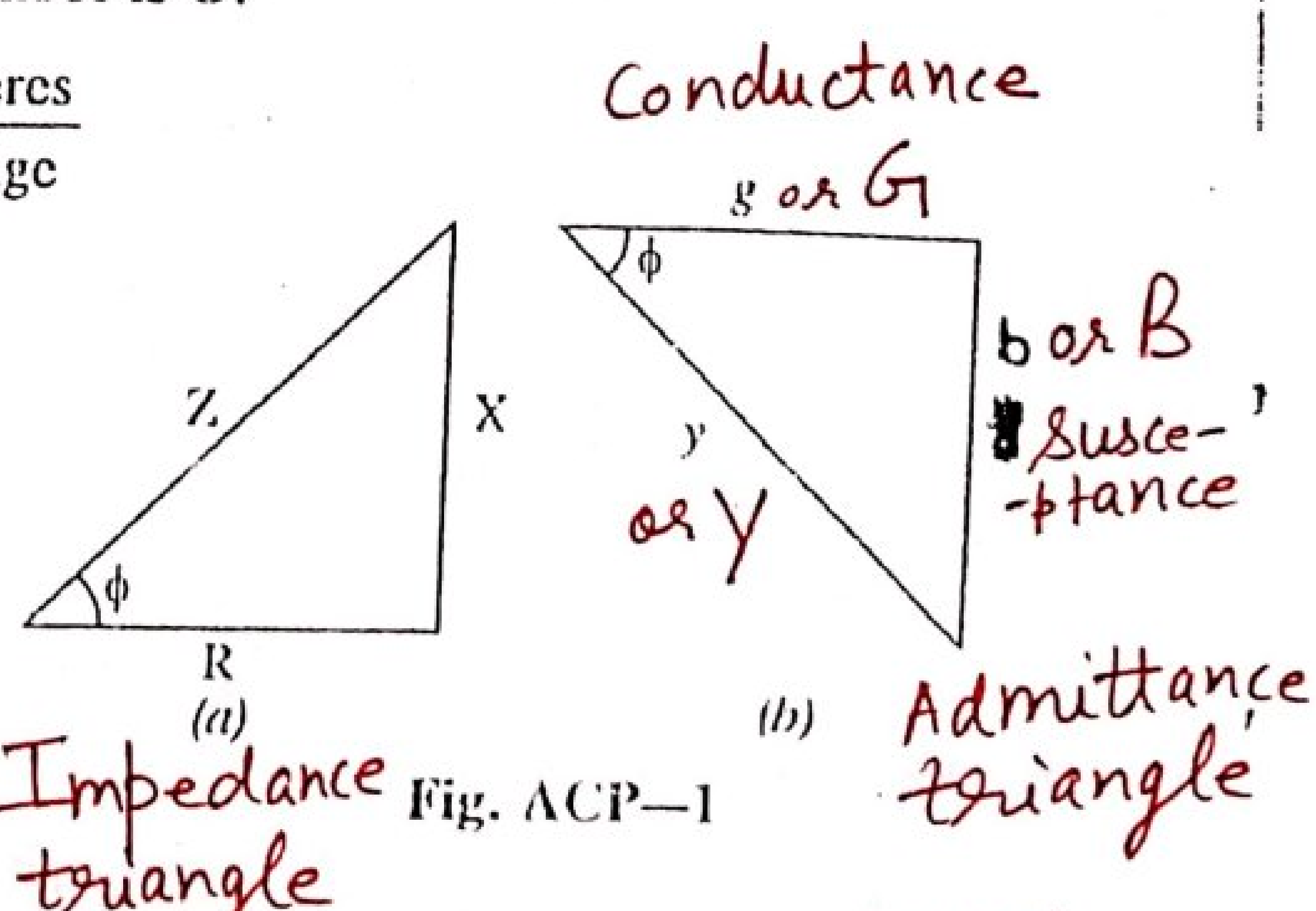

Fig. ACP–1

$$\frac{g}{Y} = \cos\phi$$

$$\text{Conductance, } g = Y\cos\phi = \frac{1}{Z}\cdot\frac{R}{Z} = \frac{R}{Z^2} = \frac{R}{R^2+X^2}$$

$$I = \frac{V}{Z} = VXY$$

Spl. case : when $X=0$, G or $g = \frac{1}{R}$, and not otherwise

Ekambir Sidhu

Capacitive Susceptance will be taken as +ve and inductive susceptance as −ve.
Though, Capacitive reactance is taken as −ve and inductive reactance is +ve.

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Conductance is always irrespective of the circuit constants. The units of conductance are mho (℧) (or Ω^{-1}).

The perpendicular of an admittance triangle is called susceptance b as shown in fig. ACP 1 (b)

Susceptance $b = Y \sin\phi = \frac{1}{Z}\cdot\frac{X}{Z} = \frac{X}{Z^2} = \frac{X}{R^2+X^2}$ (spl. case When $R=0$; B or $b = 1/X$, and not otherwise)

It will be +ve for capacitive circuit (X_C) and –ve for inductive circuit (X_L) as shown in the fig.

The units of susceptance are mho (℧).

From admittance triangle, $Y = \sqrt{g^2 + b^2}$

and Phase angle, $\phi = \tan^{-1}\frac{b}{g}$

ACP–2. Admittance Method For Parallel Circuit Solution.

Admittance method is very useful for solving parallel a.c. circuits, consisting of three or more parallel branches. Consider the parallel circuit as shown in figure ACP 2.

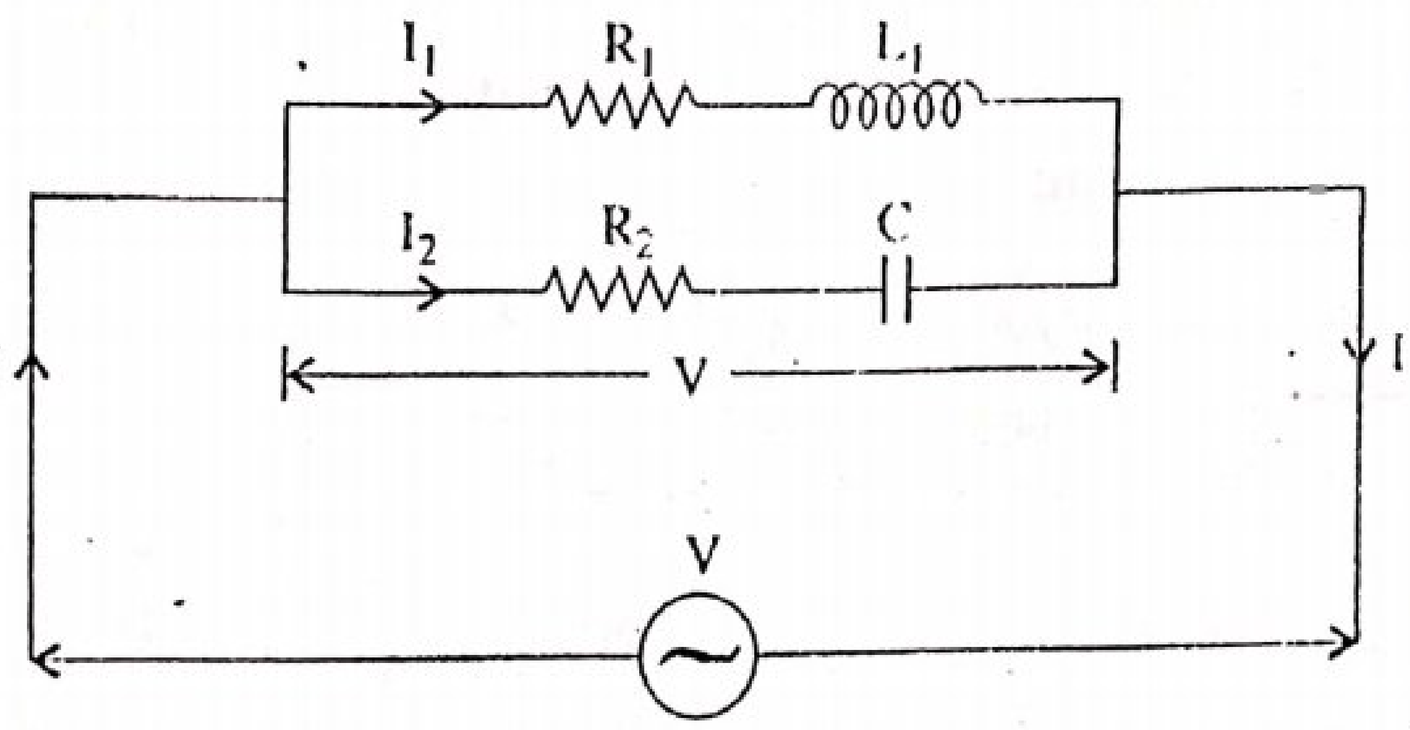


Fig. ACP—2

Equivalent conductance & susceptance of the circuit will be equal to algebraic sum of conductances and susceptances of individual branches respectively.

To solve such a parallel circuit by admittance method, proceed as under. Let the fig. ACP–2 shown above, is an a.c. parallel circuit consisting of two branches.

Step 1. Find the impedence and phase angle of each branch of the circuit separately.

$$Z_1 = \sqrt{R_1^2 + X_L^2} \quad \text{where } X_L = 2\pi f L$$

and $\tan\phi_1 = \frac{X_L}{R_1}$ $\therefore\ \phi_1 = \tan^{-1}\frac{X_L}{R_1}$

Similarly $Z_2 = \sqrt{R_2^2 + X_C^2}$

and $\tan\phi_2 = \frac{X_C}{R_2}$ $\therefore\ \phi_2 = \tan^{-1}\frac{X_C}{R_2}$

If ϕ is +ve, current will lead and if ϕ is –ve, current will lag behind applied voltage

Step 2. Find the conductance and susceptance of each branch separately

$$g_1 = \frac{R_1}{Z_1^2} \qquad b_1 = \frac{X_L}{Z_1^2} \text{ (negative)}$$

$$g_2 = \frac{R_2}{Z_2^2} \qquad b_2 = \frac{X_C}{Z_2^2} \text{ (positive)}$$

Step 3. Find the total conductance and total susceptance of the circuit by taking their algebraic sum

i.e., $G = g_1 + g_2$ and $B = + b_2 - b_1 = b_2 - b_1$

Step 4. Find the total admittance $Y = \sqrt{G^2 + B^2}$

Step 5. Find the phase angle $\phi = \tan^{-1} \frac{B}{G}$, then power factor, $\cos \phi = \frac{G}{Y}$.

Step 6. Find the total current, $I = VY$.

Problem ACP–1. An AC circuit consists of two branches. One branch consists of 8 ohms resistance and inductance of 0·015 H. The other having a resistance of 4 ohms and a capacitance of 637 μF. The two branches are connected in parallel across 230 V, 50 Hz supply. Find the current in each branch and resultant current.

Sol. The figure shows a parallel circuit.

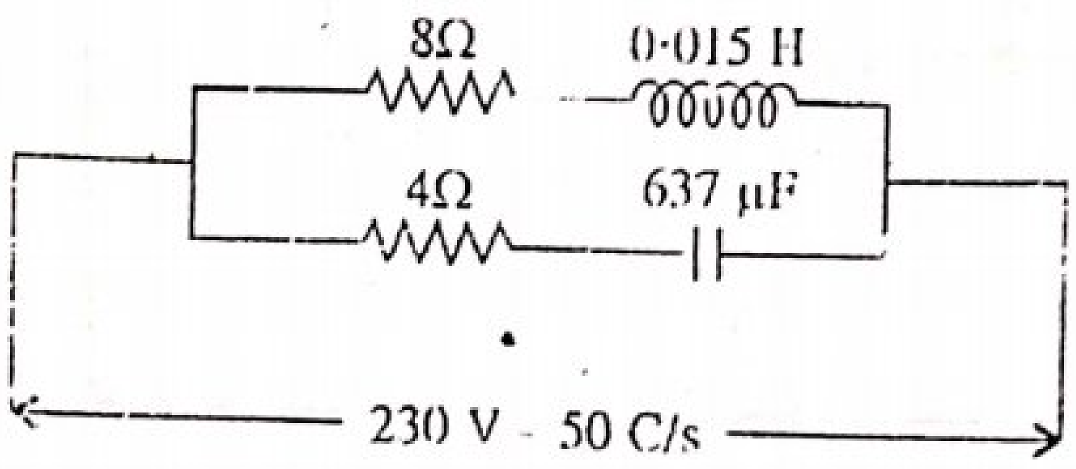


Fig. ACP—3

Supplied voltage, $V = 230$ V

Supply frequency, $f = 50$ Hz

Branch No. 1

Resistance, $R = 8\ \Omega$

Inductance, $L = 0{\cdot}015$

Inductive reactance $X_L = \omega L = 2\pi f.L = 2 \times 3{\cdot}14 \times 50 \times 0{\cdot}015 = 4{\cdot}71\ \Omega$

Impedence, $Z_1 = \sqrt{R_1^2 + X_L^2} = \sqrt{8^2 + (4{\cdot}71)^2} = 9{\cdot}28$ ohms

Conductance $g_1 = \frac{R_1}{Z_1^2} = \frac{8{\cdot}00}{(9{\cdot}28)^2} = 0{\cdot}0928\ ℧$

Susceptance, $b_1 = \frac{X_L}{Z_1^2}$ (negative)

$$b_1 = \frac{4{\cdot}71}{(9{\cdot}28)^2} = 0{\cdot}054\ ℧$$

Admittance $y_1 = \sqrt{g_1^2 + b_1^2} = \sqrt{(0 \cdot 0928)^2 + (0 \cdot 054)^2} = 0 \cdot 107$ ℧

Current in branch no. 1

$$I_1 = VY_1 = 230 \times \cdot 107 = 24 \cdot 61 \text{ A (Ans.)}$$

Branch No. 2

Resistance, $R = 4\ \Omega$

Capacitance, $C = 637\ \mu F$

$$\text{Capacitive reactance, } X_c = \frac{1}{\omega C} = \frac{10^6}{2\pi \times 50 \times 637} = 5\ \Omega$$

$$\text{Impendence, } Z_2 = \sqrt{R^2 + X_c^2} = \sqrt{4^2 + 5^2} = 6 \cdot 4\ \Omega$$

$$\text{Conductance, } g_2 = \frac{R}{Z_2^2} = \frac{4}{41} = 0 \cdot 09757 \text{ ℧}$$

$$\text{Susceptance, } b_2 = \frac{Xc}{Z_2^2} \text{ (positive)}$$

$$b_2 = \frac{5}{41} = 0 \cdot 1219 \text{ ℧}$$

$$\text{Admittance } y_2 = \sqrt{g_2^2 + b_2^2}$$

$$= \sqrt{(0 \cdot 0975)^2 + (0 \cdot 1219)^2} = 0 \cdot 156 \text{ ℧}$$

Current in branch no. 2, $I_2 = Vy_2 = 230 \times 0 \cdot 156 = 35 \cdot 88$ A

Total Conductance $G = g_1 + g_2 = \cdot 0928 + 0 \cdot 0975 = 0 \cdot 1903$ ℧

Total Susceptance $B = -b_1 + b_2 = (-\cdot 054 + 0 \cdot 1219) = \cdot 0679$ ℧,

Total Admittance $y = \sqrt{G^2 + B^2} = \sqrt{(\cdot 1903)^2 + (\cdot 0679)^2} = 0 \cdot 2020$ ℧

Total circuit current, $I = VY = 230 \times \cdot 2020 = 46 \cdot 46$ **A (Ans.)**

Problem : ACP–2. Applying Admittance method to the series parallel circuit shown in figure. Find

(i) Total impedence in ohms

(ii) Total supply current

(iii) Power factor of the entire circuit.

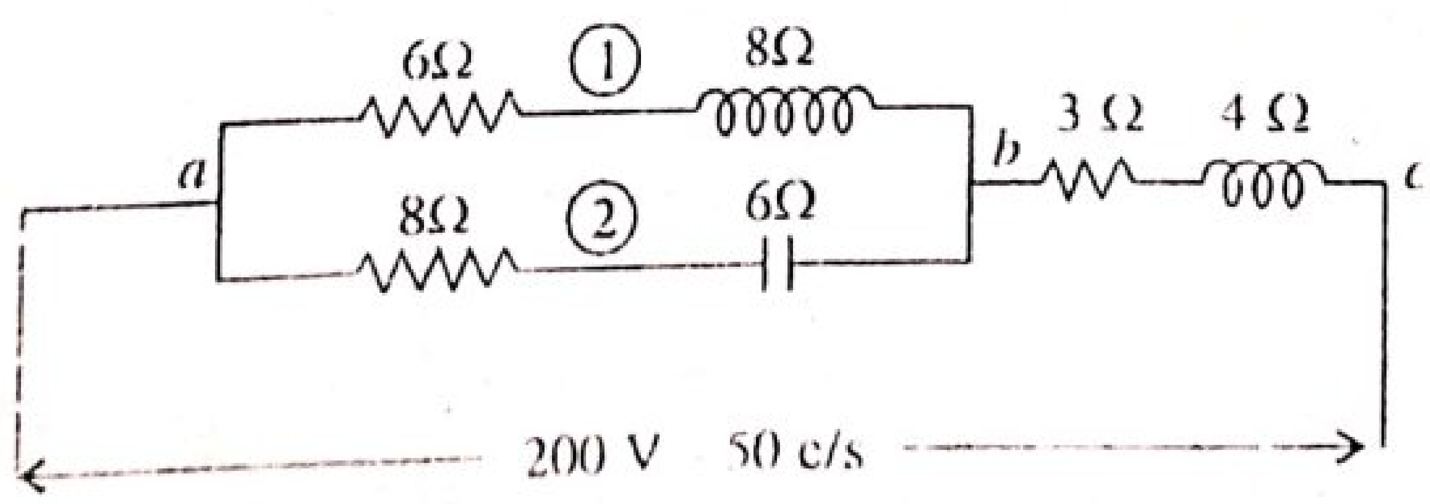


Fig. ACP—4

Sol. Voltage across the circuit $V = 200$ V

Supply frequency, $f = 50$ Hz

Branch No I.

Value of resistance, $R_1 = 6\ \Omega$

Inductive reactance, $X_L = 8\ \Omega$

$$\text{impedence } Z_1 = \sqrt{6^2 + 8^2} = 10\ \Omega$$

$$\text{Conductance, } g_1 = \frac{R_1}{Z_1^2} = \frac{6}{100} = \cdot 06\ \mho$$

$$\text{Susceptance, } b_1 = \frac{X_L}{Z_1^2} = \frac{8}{100} = \cdot 08\ \mho \text{ (negative)}$$

Branch No II.

Value of resistance, $R_2 = 8\ \Omega$

Capacitive reactance $X_c = 6\ \Omega$

$$\text{Impedence } Z_2 = \sqrt{8^2 + 6^2} = 10\ \Omega$$

$$\text{Conductance, } g_2 = \frac{R_2}{Z_2^2} = \frac{8}{100} = 0{\cdot}08\ \mho$$

$$\text{Susceptance, } b_2 = \frac{X_c}{Z_2^2} = \frac{6}{100} = 0{\cdot}06\ \mho \text{ (positive)}$$

Total conductance of circuit "ab" is

$$G_{ab} = g_1 + g_2 = \cdot 06 + \cdot 08 = 0{\cdot}14\ \mho$$

$$B_{ab} = -b_1 + b_2 = -\cdot 08 + \cdot 06 = -\cdot 02\ \mho$$

Admittance of circuit 'ab',

$$Y_{ab} = \sqrt{G_{ab}^2 + B_{ab}^2} = \sqrt{(0\cdot 14)^2 + (-\cdot 02)^2} = 0{\cdot}1414\ \mho$$

The parallel circuit may now be reduced to an equivalent series circuit shown in fig. ACP-5.

$$Z_{ab} = \frac{1}{Y_{ab}} = \frac{1}{0\cdot 1414} = 7{\cdot}07\ \Omega.$$

$$R_{ab} = G_{ab} \cdot Z_{ab}^2 = 0{\cdot}14 \times 50 = 7{\cdot}0\ \Omega$$

$$X_{ab} = B_{ab} \cdot Z_{ab}^2 = \cdot 02 \times 50 = 1{\cdot}0\ \Omega$$

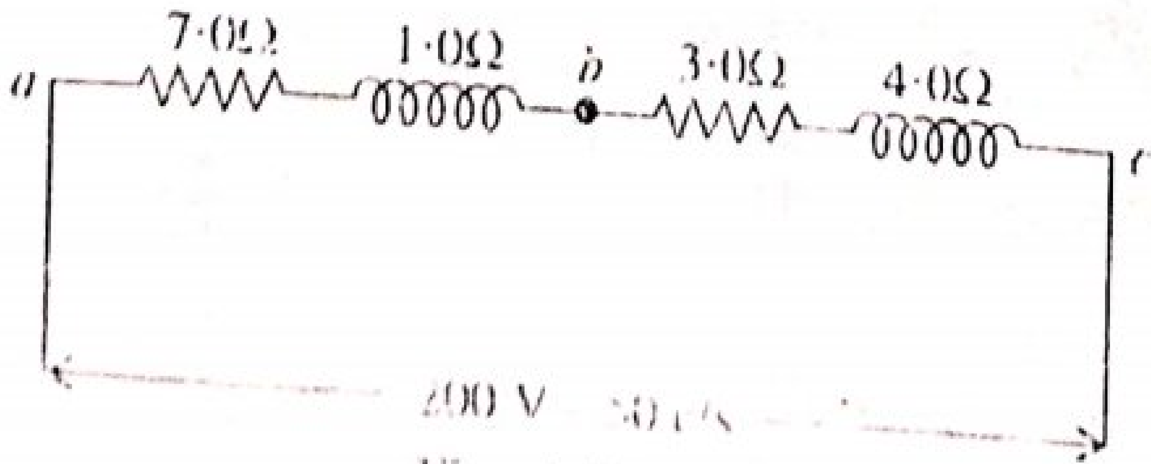


Fig. ACP-5

As seen from figure ACP-.

Total resistance of the circuit $R_{ac} = 7 \cdot 0 + 3 \cdot 0 = 10 \cdot 0\ \Omega$

Total inductive reactance $X_{L_{ac}} = 1 \cdot 0 + 4 \cdot 0 = 5 \cdot 0\ \Omega$

Total impedance, $Z_{ac} = \sqrt{10^2 + 5^2} =$ **11·18 Ω (Ans.)**

Total supply current, $I = \dfrac{V}{Z} = \dfrac{200}{11 \cdot 18} =$ **17·88 A (Ans.)**

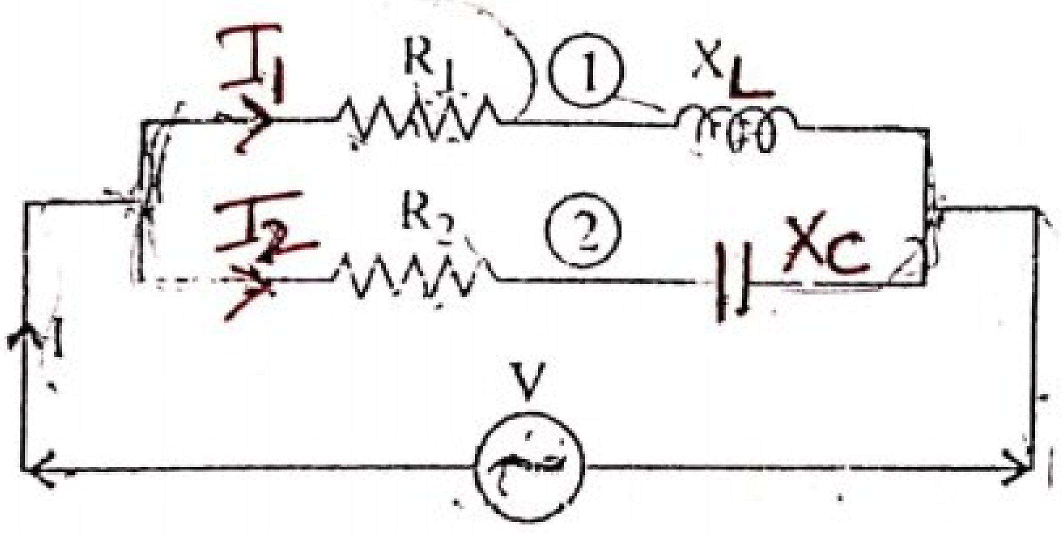
Power factor of the entire circuit,

$$\cos \phi = \frac{R_{ac}}{Z_{ac}} = \frac{10}{11 \cdot 18} = \mathbf{0 \cdot 8944\ (Ans.)}$$

ACP–3. BY PHASOR METHOD OR VECTOR METHOD

In this method, the magnitude and phase angle of each branch current are calculated. Then we draw the phasor diagram taking voltage as the reference vector. The total current or circuit current is the vector sum of all the branch currents.

Consider the circuit shown in figure ACP–6, and 7.



AC Parallel Circuit Containing Impedances

Fig. ACP–6

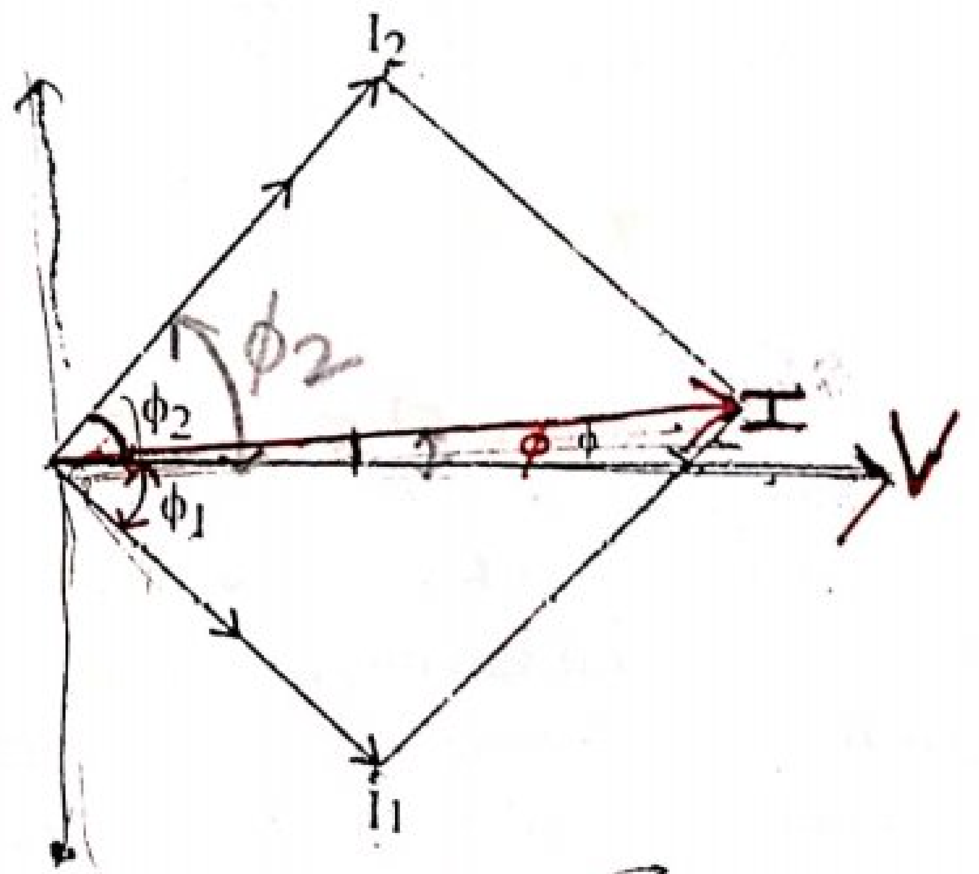


Fig. ACP–7

Here there are two branches connected in parallel. The voltage across both the branches is the same but the currents through them are different.

Branch No. I

cos 0

$$Z_1 = \sqrt{R_1^2 + X_L^2} \quad \text{where } X_L = 2\pi f L$$

$$I_1 = \frac{V}{Z_1}$$

$$\cos \phi_1 = \frac{R_1}{Z_1} \quad \text{or} \quad \phi_1 = \cos^{-1} \frac{R_1}{Z_1}$$

Current I_1 lags behind the voltage by an angle ϕ_1 (see fig. ACP-7)

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Branch No. II.

$$Z_2 = \sqrt{R_2^2 + X_C^2}$$

$$I_2 = \frac{V}{Z_2}$$

$$\cos\phi_2 = \left(\frac{R_2}{Z_2}\right) \quad \text{or} \quad \phi_2 = \cos^{-1}\left(\frac{R_2}{Z_2}\right)$$

Current I_2 leads the voltage V by an angle ϕ_2.

The line current or total current is the phasor sum of I_1 and I_2. Suppose its phase angle is $\phi°$. The values of I and ϕ can be determined by resolving the current into rectangular components.

X – Component – Algebraic sum of components of I_1 and I_2 along X-axis,

$$= I_1 \cos\phi_1 + I_2 \cos\phi_2$$

Y – Component = Algebraic sum of components of I_1 and I_2 along Y-axis,

$$= I_2 \sin\phi_2 - I_1 \sin\phi_1$$

Resultant or total current $= \sqrt{X^2 + Y^2}$

$$\tan\phi = \frac{Y\text{-component}}{X\text{-component}}$$

$$\phi = \tan^{-1}\frac{Y\text{-component}}{X\text{-component}}$$

If ϕ is positive, line current I leads the voltage and if ϕ is negative I lags the voltage.

Problem ACP-3. A coil having a resistance of 8 Ω and an inductive reactance of 6 Ω is connected in parallel with a resistance of 5 Ω and capacitive reactance of 8 Ω across 100 V, 50 Hz supply terminals. Calculate *(i)* The total current taken from the supply *(ii)* phase angle between voltage and current. Draw the vector diagram showing all the three currents.

Sol. Supply voltage, $V = 100$ volts

Supply frequency, $f = 50$ Hz

For Branch No. I.

Resistance, $R_1 = 8\ \Omega$

Inductive Reactance, $X_L = 6\ \Omega$

$$\text{Impedence, } Z_1 = \sqrt{8^2 + 6^2} = 10\ \Omega$$

$$I_1 = \frac{V}{Z_1} = \frac{100}{10} = 10\ \text{A}.$$

$$\cos\phi_1 = \frac{R_1}{Z_1} = \frac{8}{10} = 0.8 \text{ (lagging)}$$

$\therefore \quad \phi_1 = \cos^{-1} 0{\cdot}8 = 36°{\cdot}52'$

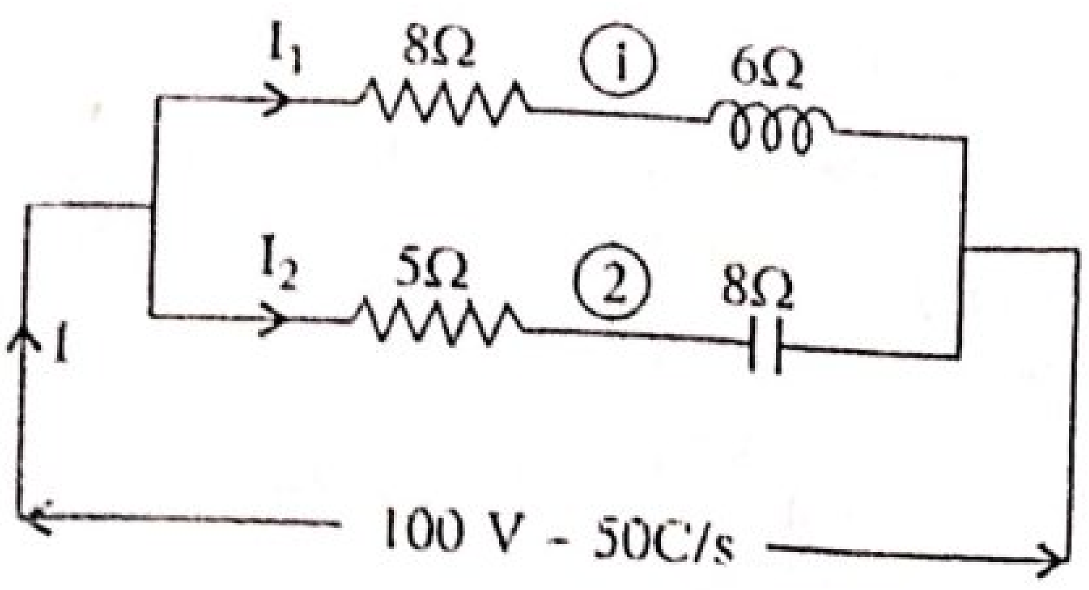

Fig. ACP—8

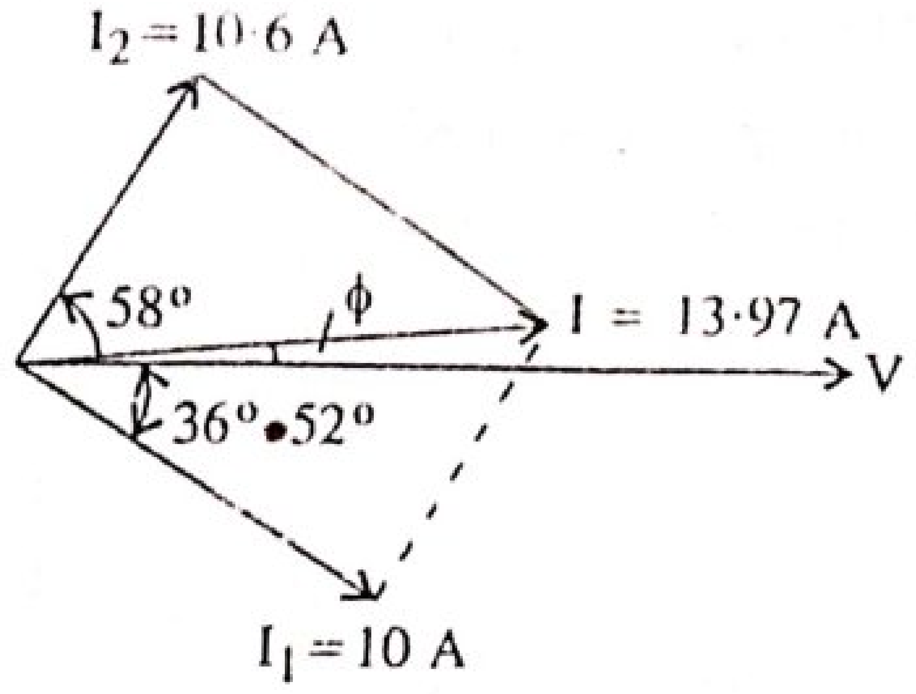

Fig. ACP—9

For Branch No. II.

Resistance, $R_2 = 5\ \Omega$

Capacitive Reactance, $X_C - 8\ \Omega$

$$\text{Impedence, } Z_2 = \sqrt{5^2 + 8^2} = 9 \cdot 44\ \Omega$$

$$\text{Current, } I_2 = \frac{V}{Z_2} = \frac{100}{9 \cdot 44} = 10{\cdot}6 \text{ A}$$

$$\cos \phi_2 = \frac{R_2}{Z_2} = \frac{5}{9 \cdot 44} = 0{\cdot}5296$$

$$\therefore \quad \phi_2 = \cos^{-1} 0{\cdot}5296 = 58° \text{ (leading)}$$

The resultant current or total current can be determined by resolving I_1 and I_2 into rectangular components.

$$\begin{aligned} X\text{– components} &= I_1 \cos \phi_1 + I_2 \cos \phi_2 \\ &= 10 \times 0{\cdot}8 + 10{\cdot}6 \times 0{\cdot}5296 \\ &= 8 + 5{\cdot}613 = 13{\cdot}613 \text{ A.} \end{aligned}$$

$$\begin{aligned} Y\text{–components} &= I_2 \sin \phi_2 - I_1 \sin \phi_1 = 10{\cdot}6 \times 0{\cdot}848 - 10 \times {\cdot}6 \\ &= 8{\cdot}98 - 6 = 2{\cdot}98 \end{aligned}$$

$$\therefore \quad \text{Resultant current, } I = \sqrt{X^2 + Y^2}$$

or

$$I = \sqrt{(13 \cdot 613)^2 + (2 \cdot 98)^2} = 13{\cdot}97 \text{ A.}$$

$$\tan \phi = \frac{Y - \text{component}}{X - \text{component}} = \frac{2 \cdot 98}{13 \cdot 613} = 0{\cdot}2189$$

$$\therefore \quad \phi = \tan^{-1} 0{\cdot}2189 = 12{\cdot}35°.$$

Problem ACP-4. A coil having a resistance of 20 ohms and an inductance of 0·1 H is connected in parallel with a capacitor having a capacitance of 50 μF across a 100 V, 50 Hz

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supply. Calculate (i) The current in each branch (ii) The resultant current and (iii) The power factor of the whole circuit.

Sol. Applied voltage $V = 100$ V

Supply frequency, $f = 50$ Hz

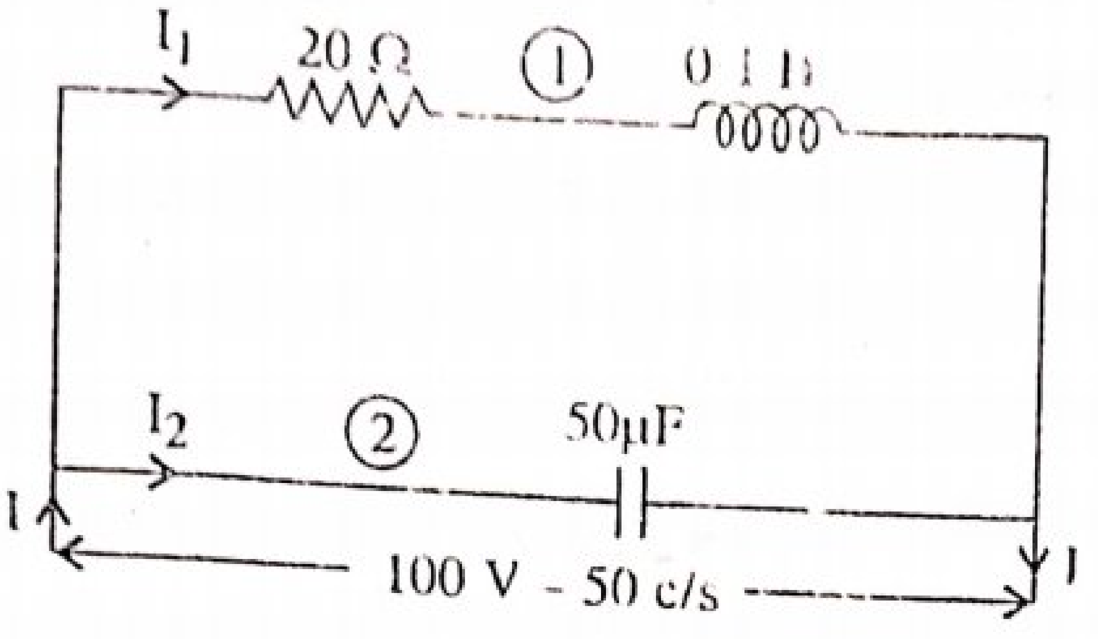

Fig. ACP—10

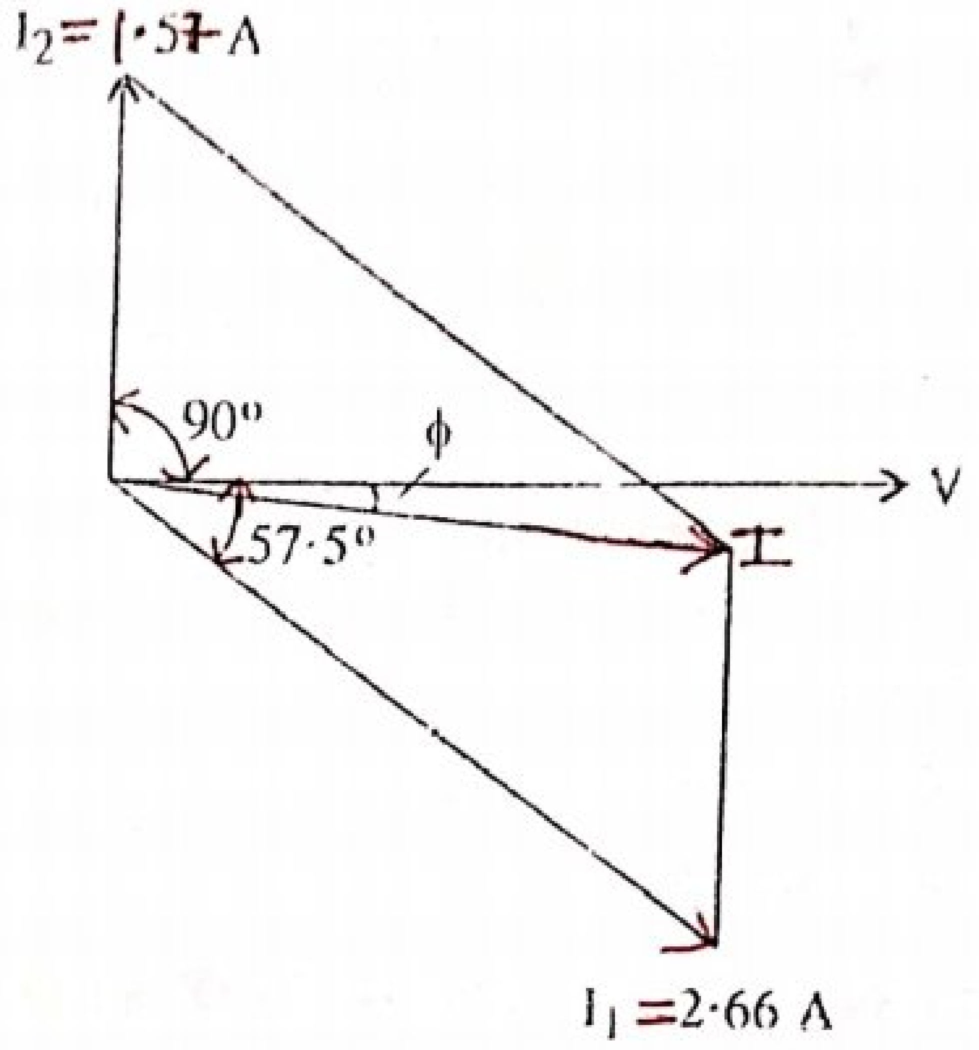

Fig. ACP—11

Branch No. 1

Value of resistance, $R = 20\ \Omega$

Value of inductance, $L = 0{\cdot}1$ H.

$\therefore$ Inductive reactance, $X_L = \omega L = 2\pi . f . L$

$$= 2\pi . 50 \times 0{\cdot}1 = 31{\cdot}4\ \Omega$$

$\therefore$ Impedence, $Z_1 = \sqrt{R^2 + X_L{}^2} = \sqrt{(20)^2 + (31\cdot4)^2} = 37{\cdot}60\ \Omega$

$$\text{Current } I_1 = \frac{V}{Z_1} = \frac{100}{37\cdot6} = 2{\cdot}66 \text{ A}$$

$$\tan\phi_1 = \frac{X_L}{R} = \frac{31\cdot4}{20} = 1{\cdot}571$$

$$\phi_1 = \tan^{-1} 1{\cdot}571 = 57\cdot5^\circ \text{ (lagging)}$$

Branch No. 2

Value of capacitance, $C = 50\ \mu\text{F} = 50 \times 10^{-6}$ F

$$\text{Capacitive Reactance } X_C = \frac{1}{\omega C} = \frac{1}{2\pi f\ C}$$

$$X_C = \frac{10^6}{2\pi \times 50 \times 50} = 63{\cdot}66 \text{ ohms.}$$

$$I_2 = \frac{V}{X_C} = \frac{100}{63\cdot66} = 1{\cdot}57 \text{ A.}$$

Since it is pure capacitive circuit, therefore, I_2 leads the voltage by 90°. Further resolving the currents I_1 and I_2 into rectangular axis.

X-component $= I_1 \cos\phi_1 + I_2 \cos\phi_2$

$= 2{\cdot}66 \times \cos(57{\cdot}5^\circ) + 1{\cdot}57 \cos 90^\circ$

$= 2{\cdot}66 \times 0{\cdot}5373 + 1{\cdot}57 \times 0 = 1{\cdot}429$ A

Y-component $= I_2 \sin 90^\circ - I_1 \sin(57{\cdot}5^\circ)$

$= 1{\cdot}57 \times 1 - 2{\cdot}66 \times 0{\cdot}8434$

$= 1{\cdot}57 - 2{\cdot}24 = -0{\cdot}6734$

Resultant current, $I = \sqrt{X^2 + Y^2}$

$$= \sqrt{(1 \cdot 429)^2 + (-0 \cdot 6734)^2} = 1{\cdot}58 \text{ A (Ans.)}$$

$$\tan\phi = \frac{Y - \text{component}}{X - \text{component}} = \frac{-0 \cdot 6734}{1 \cdot 429} = -0{\cdot}471238$$

$\therefore \quad \phi = \tan^{-1}(-0{\cdot}47) = -25{\cdot}23^\circ$ **(Ans.)**

The negative sign shows that I is lagging the voltage by an angle $25{\cdot}23^\circ$

Power factor, $\cos\phi = \cos 25{\cdot}23^\circ = 0{\cdot}9045$ **(lagging) (Ans.)**

Problem ACP–5. A coil of resistance 15 ohm and inductance 0·05 H is connected in parallel with a non–inductive resistance of 20 ohms. Find,

(i) Current in each branch circuit.

(ii) Total current supplied.

(iii) Phase angle of combination when a voltage of 200 volts at 50 Hz is applied.

Sol. The circuit diagram is shown in fig ACP–12.

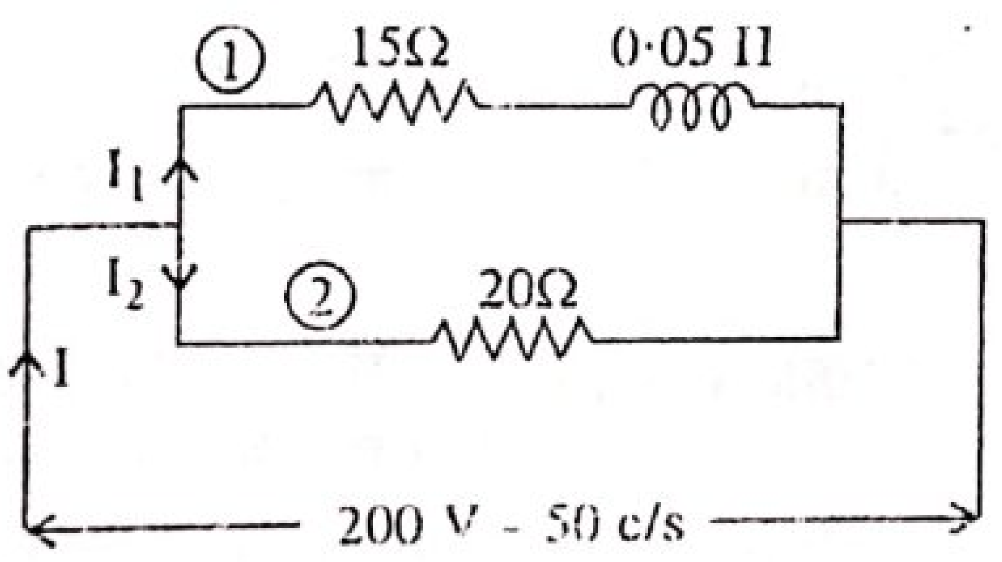

Fig. ACP—12

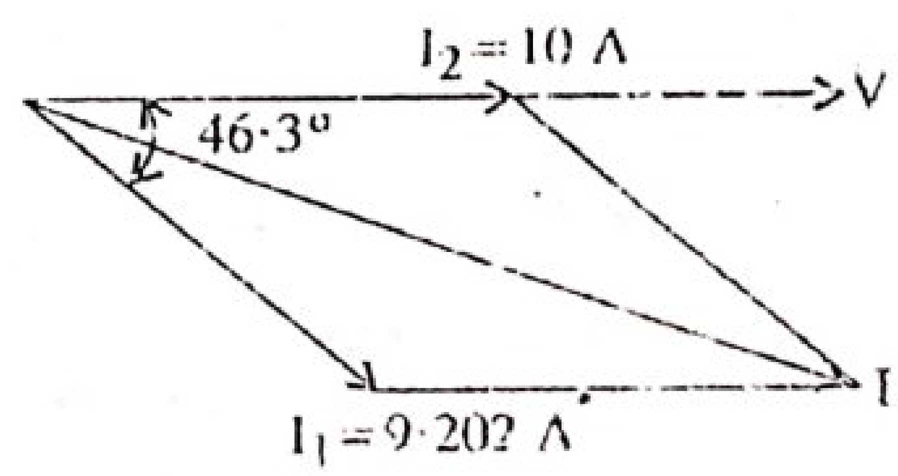

Fig. ACP—13

Applied voltage, $V = 200$ volts

Supply frequency, $f = 50$ Hz.

For Branch No. 1.

Value of resistance, $R_1 = 15\ \Omega$

Value of inductance, $L = 0{\cdot}05$ H

Inductive reactance, $X_L = 2\pi f L = 2\pi \times 50 \times {\cdot}05 = 15{\cdot}7\ \Omega$

Impedence, $Z_1 = \sqrt{R_1^2 + X_L^2} = \sqrt{(15)^2 + (15{\cdot}7)^2} = 21{\cdot}73\ \Omega$

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$$\tan \phi_1 = \frac{X_L}{R_1} = \frac{15 \cdot 7}{15} = 1 \cdot 04$$

$$\phi_1 = \tan^{-1} 1 \cdot 04 = 46 \cdot 3° \text{ (lagging)}$$

$$I_1 = \frac{V}{Z_1} = \frac{200}{21 \cdot 73} = 9 \cdot 202 \text{ A (Ans.)}$$

For Branch No. 2

Since in branch No 2, only resistance is present.

∴ Current I_2 will be in phase with the voltage.

Value of resistance, $R_2 = 20\ \Omega$

$$\text{Current, } I_2 = \frac{V}{R_2} = \frac{200}{20} = \mathbf{10\ A\ (Ans.)}$$

$$\tan \phi_2 = \frac{X}{R_2} = \frac{0}{R_2} = 0$$

∴

$$\phi_2 = \tan^{-1} 0 = 0$$

To find the total current, resolve I_1 and I_2 into rectangular components.

$$X - \text{component} = I_1 \cos \phi_1 + I_2 \cos 0$$

$$= 9 \cdot 202 \cos 46 \cdot 3° + 10 \times \cos 0$$

$$= 9 \cdot 202 \times 0 \cdot 6007 + 10 \times 1 = \mathbf{16 \cdot 354\ A}$$

$$Y - \text{component} = I_2 \sin 0 - I_1 \sin \phi_1$$

$$= 0 - 9 \cdot 202 \sin 46 \cdot 3°$$

$$= - 9 \cdot 202 \times 0 \cdot 7232 = \mathbf{- 6 \cdot 655\ A}$$

$$\text{Resultant current, } I = \sqrt{X^2 + Y^2}$$

$$= \sqrt{(16 \cdot 354)^2 + (-6 \cdot 655)^2} = \mathbf{17 \cdot 66\ A\ \ (Ans.)}$$

$$\tan\phi = \frac{-6 \cdot 655}{16 \cdot 354} = - 0 \cdot 4069$$

∴

$$\varphi = \tan^{-1} 0 \cdot 4069 = 22° - 08'$$

The negative sign shows that the resultant current lags the voltage by **22° - 08′ (Ans.)**

CP–4. By j- Notation or Complex Notation

Phasors like voltage and current of an A.C. circuit can be represented by graphical presentations. But now the engineers have developed another method to represent the phasor the algebraic form or mathematical form. Such method is known as **Phasor Algebra** or **omplex Algebra.** It simplifies the mathematical manipulations of phasors to a great extent.

In this method, a vector quantity is expressed algebraically in terms of its rectangular mponents. Hence, this form of representation is also known as Rectangular form of notation.

ACP-4.1. Notation of Phasors on Rectangular-Co-ordinate axes :

Consider a voltage vector V lying along X-axis *i.e.* reference axes as shown in the fig.ACP—14 if this vector is multiplied by (−1), then the phasor is reversed or we can say that the vector is rotated through an angle of 180° in the counter clockwise direction and attains the position OX′- axis. Let j is a factor which when multiplied by the phasor V, it rotates through 90° in C.C.W. and attains the position OY. If it is multiplied by the factor j^2 then we get $j^2.V$. and attains the position of OX′. As shown in fig. ACP—14 multiplying the phasor by j^2 rotates the phasor through 180° in C.C.W. direction. This means that multiplying the phasor by j^2 is the same as multiplying by −1, It follows, therefore, that

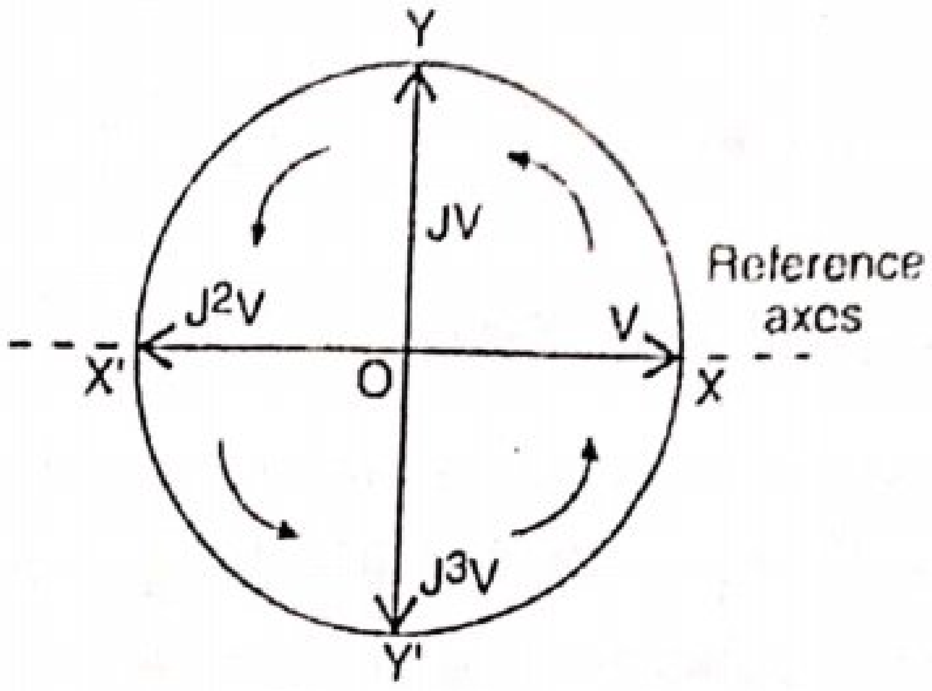

Fig. ACP—14

$$j^2 = -1$$

$$j = \sqrt{-1}$$

So it is seen that when a phasor is multiplied by $j\left(=\sqrt{-1}\right)$, the phasor is rotated through 90° in C.C.W. direction. Each successive multiplication by j rotates the phasor by 90° in C.C.W.

Then $j = \sqrt{-1}$ 90° C.C.W. rotation from OX-axis

$j^2 = -1$ 180° C.C.W. rotation from OX-axis

$j^3 = j^2 \cdot j = -j$ .. 270° C.C.W. rotation from OX-axis

$j^4 = j^2 \cdot j^2 = (-1), (-1) = 1$, 360° C.C.W. rotation from OX-axis.

Fig ACP-14 represents the effect of multiplying a phasor by j. The following points should be noted :

(i) The symbol j is an operator and is used to represent the vertical component of the quantities without change in magnitude.

(ii) Operator j will occur only when the phasor is along vertical axis *i.e.* OY.

∴ OY = $+ j.V$. and if the phasor is along OY′

Then OY′ = $- j$V.

(iii) The value of $j = \sqrt{-1}$ as already stated and its value cannot be determined and therefore, it is called imaginary number and any phasor associated with j is also called the imaginary part. A phasor along X-axis is called real part.

Consider a phasor V rotated through an angle θ in anti-clock wise direction as shown in fig.ACP—15. The phasor has two components.

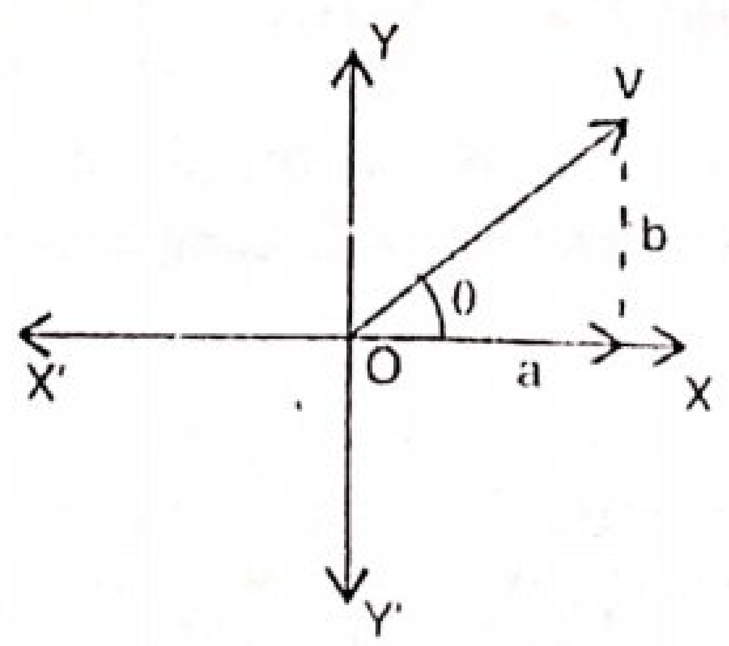

Fig. ACP—15

(i) Horizontal Component "a" along OX.

(ii) Vertical Component "b" rotated through 90° in C.C.W. direction from OX-axis and is represented by $j.b$.

$\therefore$

$$V = a + jb$$

$$\text{Magnitude of } V = \sqrt{a^2 + b^2}$$

$$\text{Angle } \theta \text{ with X-axis, } \theta = \tan^{-1}\left(\frac{b}{a}\right)$$

The quantity (a + jb) is called complex number or complex quantity. This quantity consists of two parts real part (a) and imaginary part. (jb)

If the phasor V is rotated clock wise as shown in fig. ACP—16. The vertical component is expressed as ($-jb$)

$\therefore$ The complex number will be

$$V = a - jb$$

$$\text{Magnitude of } V = \sqrt{a^2 + b^2}$$

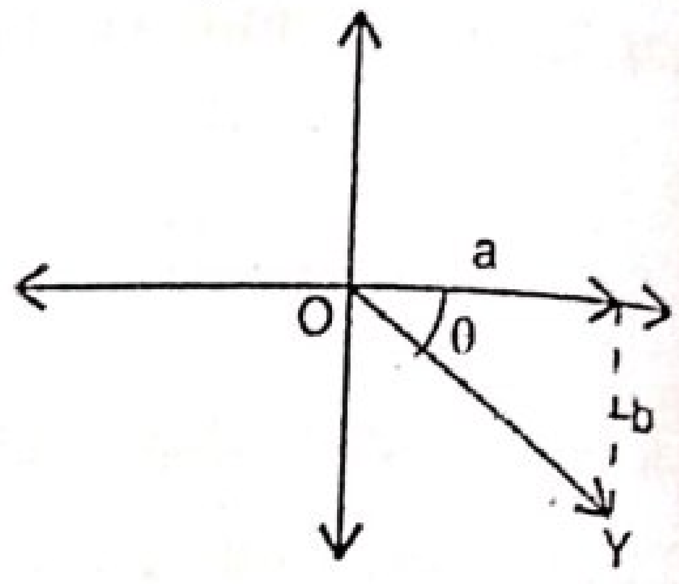

Fig. ACP—16

$$\text{angle } \theta \text{ with X-axis, } \theta = \tan^{-1}\left(\frac{-b}{a}\right) \text{ (-ve)}$$

Mathematical representation :— A phasor can be represented in three ways in the mathematical form

(i) Rectangular form.

(ii) Trigonometrical form.

(iii) Polar form.

***(i)* Rectangular form :—** This method is also known as Symbolic Notation. In this method, the phasor is resolved into horizontal and vertical components and is expressed in the complex form *i.e.* $V = a + jb$

$$\text{Magnitude of phasor, } V = \sqrt{a^2 + b^2}$$

$$\text{Its angle } \theta \text{ with X-axis, } \theta = \tan^{-1}\left(\frac{b}{a}\right)$$

Fig. ACP—17 shows the phasors in the various quadrants. The phasors have been resolved into horizontal and vertical components. The various phasors are expressed in the rectangular form as,

$$V_1 = a_1 + jb_1$$
$$V_2 = -a_2 + jb_2$$
$$V_3 = -a_3 - jb_3$$
$$V_4 = a_4 - jb_4$$

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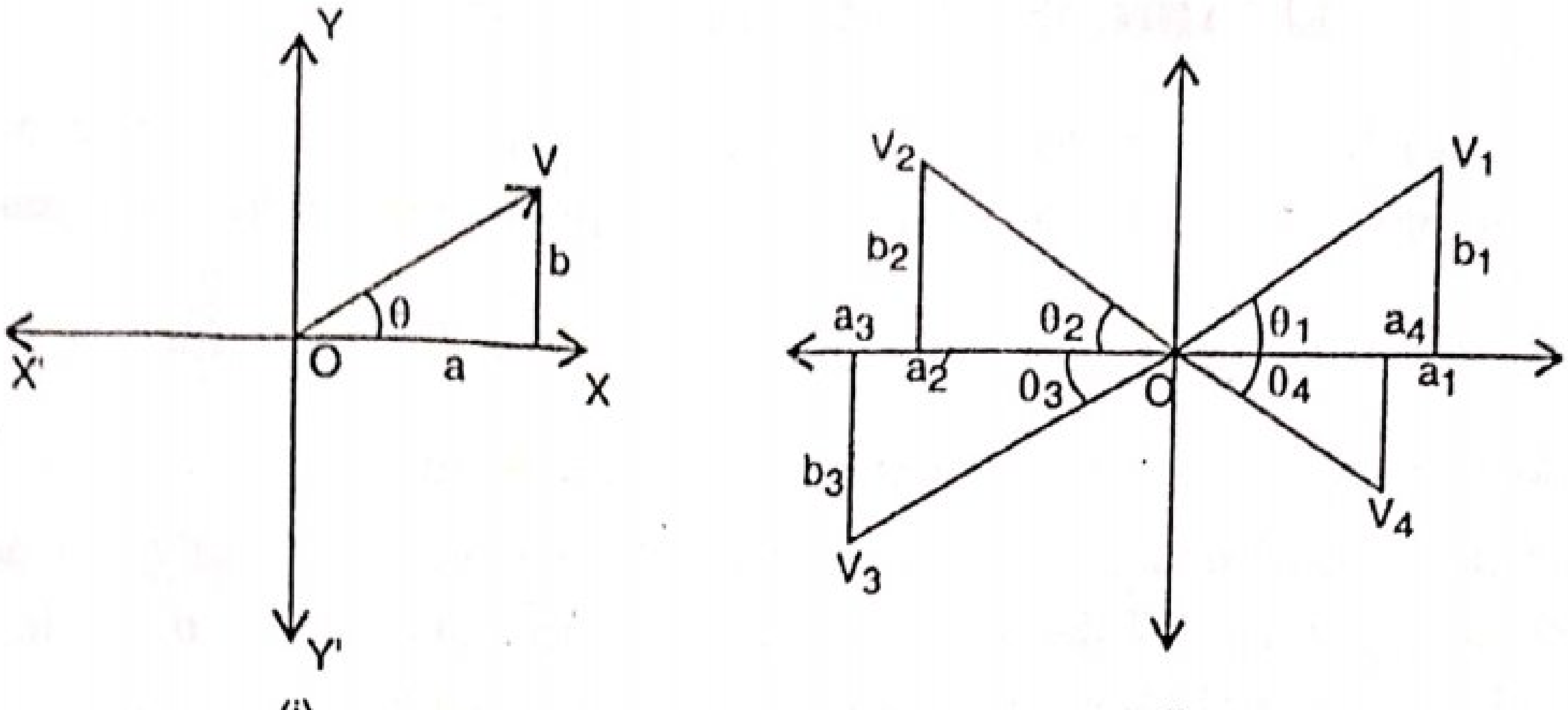


(i) (ii)

Fig. ACP—17

The phase angles and magnitude of these phasors are found out as explained above. If the angle of rotation is in C.C.W. direction, from the reference axis, then it is assigned +ve sign and if it is measured in close wise direction, it is assigned –ve sign.

***(ii)* Trigonometrical form :—** In this method, the horizontal and vertical components of the phasor are expressed in the trigonometrical form. Such as,

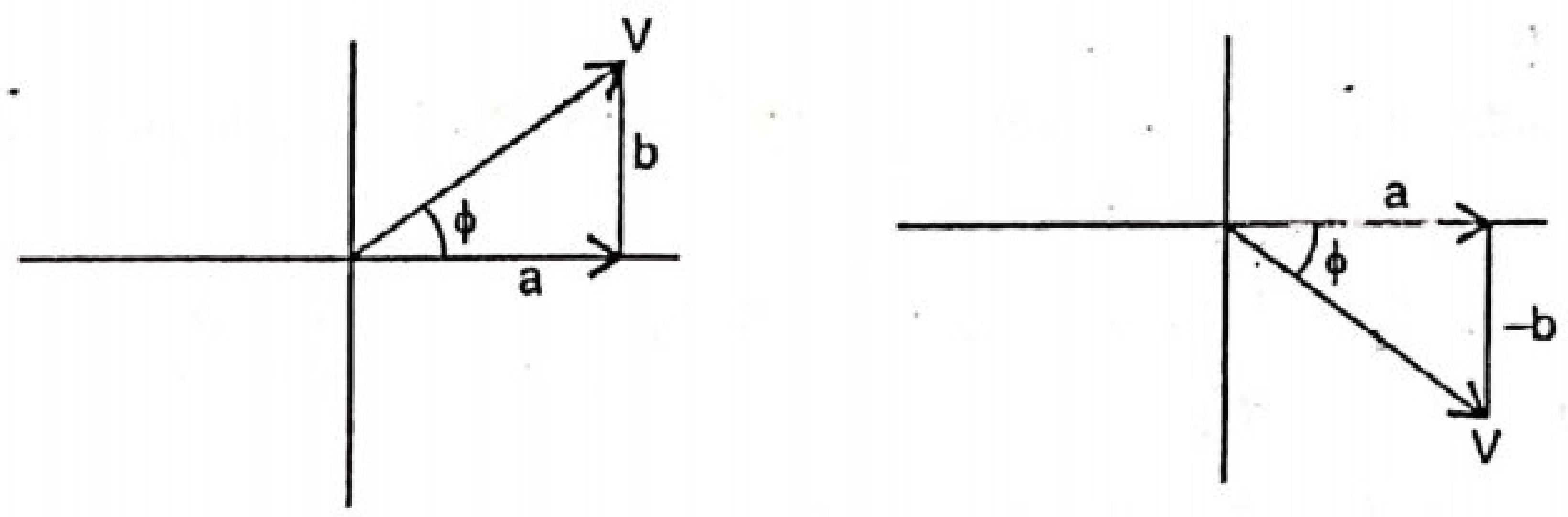
$$a = V \cos \phi$$

$$b = V \sin \phi$$

$$\therefore \quad V = (V \cos \phi + j \, V \sin \phi)$$

and on the other hand, if ϕ is negative, as shown in Fig. ACP—18(ii). Then

$$V = (V \cos \phi - j \, V \sin \phi)$$



(i) (ii)

Fig. ACP—18

***(iii)* Polar Form :—** In this method, the expression V cos ϕ +jV sin ϕ is written in the simplified form of V. $\angle\theta$. Here V is the magnitude of the phasor and θ is the angle measured along C.C.W. from the reference axis. *i.e*, OX-axis. A negative angle in the polar form indicates measurement of angle along clockwise direction. Hence polar form can be written in general as,

$$V = V \angle \pm \theta$$

It is seen that all the three mathematical forms of representing phasors give the same information regarding the magnitude of the phasor and its direction of measurement from the reference axis. Therefore, conversion is made from one form to the other rapidly as per the requirement of the calculations to get quick result

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For example, consider a phasor V having in phase and quadrature component as 3 and respectively. Write the three equations in three different forms. Magnitude of phasor V = $\sqrt{3^2+4^2}$ = 5. Its angle w.r.t. OX-axis $\theta = \tan^{-1}\frac{4}{3} = 53.1°$. In rectangular form V = (3 + 4) Trigonometrical form V = 5 (cos 53.1° +j sin 53.1). Polar form, V = 5 ∠ 53.1°.

Addition and subtraction of phasors :— Rectangular form is the best suited for addition and subtraction of vector quantities. If the phasors are given in polar from, they should be converted into rectangular forms and then addition or subtraction should be carried out.

Suppose we are given two vector quantities $E_1 = a_1 + jb_1$ and $E_2 = a_2 + jb_2$ and it is required to find their sum and difference

$$\therefore \quad E_1 = a_1 + jb_1$$

$$E_2 = a_2 + jb_2$$

Then their sum or resultant is,

$$E_1 + E_2 = (a_1 + jb_1) + (a_2 + jb_2)$$

$$= (a_1 + a_2) + j(b_1 + b_2)$$

Magnitude of resultant, $E = \sqrt{(a_1+a_2)^2 + (b_1+b_2)^2}$,

Its angle from OX-axis, $\theta = \tan^{-1}\frac{b_1+b_2}{a_1+a_2}$

Subtraction :— The subtraction of phasors is done by using ordinary rules of phasor algebra

$$E_1 = a_1 + jb_1$$

$$E_2 = a_2 + jb_2$$

Their difference, $E_1 - E_2 = (a_1 + jb_1) - (a_2 + jb_2)$

$$= (a_1 - a_2) + j(b_1 - b_2)$$

$$\text{Magnitude } V = \sqrt{(a_1-a_2)^2 + (b_1-b_2)^2}$$

$$\text{Phase angle, } \theta = \tan^{-1}\left(\frac{b_1-b_2}{a_1-a_2}\right)$$

Multiplication and Division of Phasors

Polar form is the best suited for multiplication and division of phasor quantities. Again, suppose we are given two phasors $E_1 = a_1 + jb_1$, $E_2 = a_2 + jb_2$,

Multiplication

(i) Rectangular Form :—

$$E_1 \times E_2 = (a_1 + jb_1) \times (a_2 + jb_2)$$
$$= (a_1a_2 + j\,a_1b_2 + j\,a_2b_1 + j^2\,b_1b_2)$$
$$= (a_1a_2 - b_1b_2) + j\,(a_1b_2 + a_2b_1)$$

$a_1a_2 + ja_1b_2 + jb_1a_2 + j^2b_1b_2$

$a_1a_2 + j(a_1b_2 + a_2b_1) - b_1b_2$

Magnitude of Phasor, $= \sqrt{(a_1a_2 - b_1b_2)^2 + (a_1b_2 + a_2b_1)^2}$

Its angle from OX-axis, $\theta = \tan^{-1}\dfrac{a_1b_2 + a_2b_1}{a_1a_2 - b_1b_2}$

(ii) **Polar form :—** $E_1 = E_1\angle\theta_1,\ E_2 = E_2\angle\theta_2$

$$E_1 \times E_2 = E_1E_2\angle\theta_1 + \theta_2$$

Division of Phasors :— Division of phasors again be

(i) Rectangular form

(ii) Polar form.

Rectangular Form :— If phasor $E_1 = a_1 + jb_1$ ...(i)

and $E_2 = a_2 + j\,b_2$ ...(ii)

Dividing (i) by (ii)

$$\frac{E_1}{E_2} = \frac{a_1 + jb_1}{a_2 + jb_2} \times \frac{a_2 - jb_2}{a_2 - jb_2}$$
$$= \frac{(a_1a_2 + b_1b_2) + j(b_1a_2 - a_1b_2)}{a_2^2 + b_2^2}$$

Polar Form :— While dividing the phasors that are in polar form, divide their magnitudes and subtract their angles (algebraically)

$$E_1 = a_1 + jb_1$$
$$E_2 = a_2 + jb_2$$

$$\therefore \quad \frac{E_1}{E_2} = \frac{E_1\angle\theta_1}{E_2\angle\theta_2} = \frac{E_1}{E_2}\angle\theta_1 - \theta_2$$

Example ACP—6. Two current-phasors are given in the rectangular form and illustrate the process graphically.

Solution :

$$I_1 = 16 + j\,12$$
$$I_2 = -6 + j\,10.4$$

Let
$$I_3 = I_1 + I_2$$
$$= (16 + j\,12) + (-6 + j10.4)$$
$$= 10 + j\,22.4$$

Magnitude of $I_3 = \sqrt{(10)^2 + (22.4)^2} = 25.5$

The phasor addition by graphical methods is shown in fig. ACP—19.

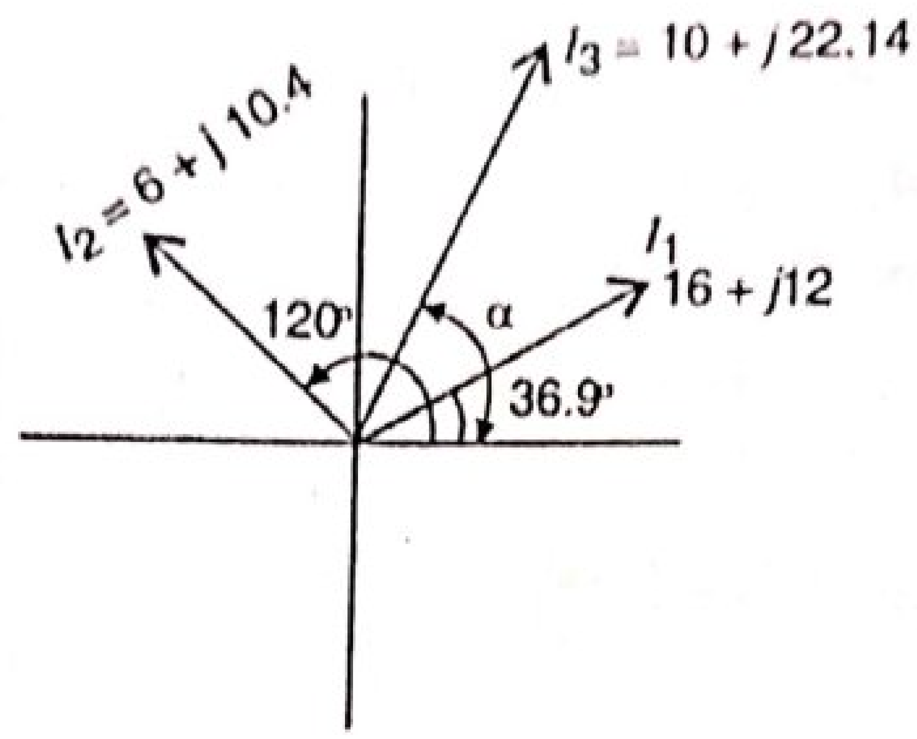

Fig. ACP—19

$$\tan\alpha = \frac{22.4}{10} = 2.24$$

$$\therefore \quad \alpha = \tan^{-1} 2.24 = 65.95^\circ$$

Let θ_1 be the slope of first current $\theta_1 = \tan^{-1}\dfrac{12}{16} = 36.9^\circ$

θ_2 be the slope of second current $\theta_2 = -\dfrac{10.4}{6} = \tan^{-1} -240^\circ$

or 120°

The resultant phasor is found by using parallelogram law of vectors.

Example ACP—7. Two voltage vectors (phasors) are ...ven in the rectangular form as $V_1 = 15 + j\,10$ and ... $= 12 + j6$. Perform the operation.

(i) $V_1 + V_2$ and (ii) $V_1 - V_2$.

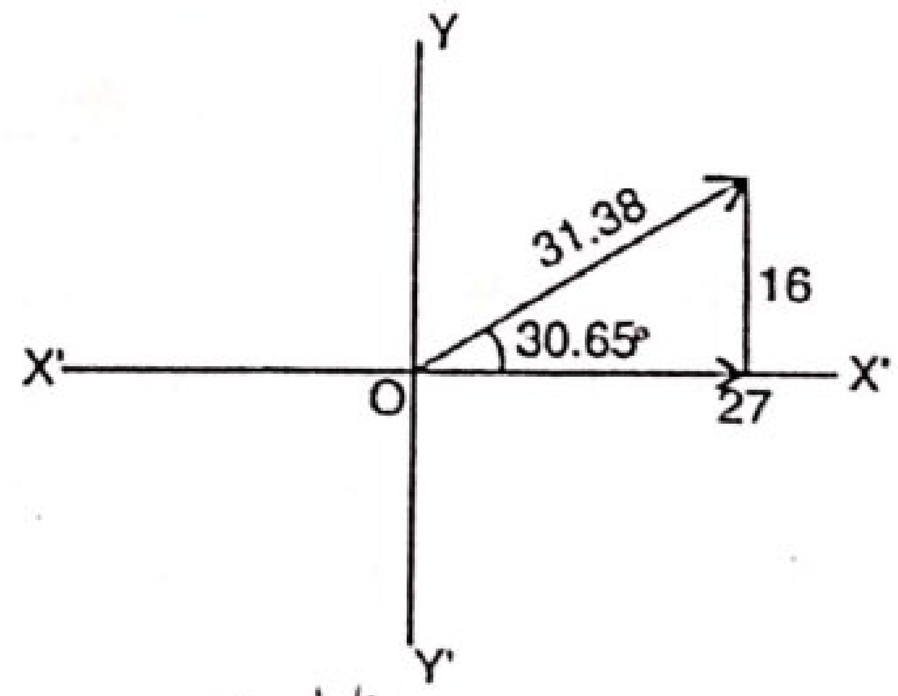

Fig. ACP—20

Solution : (i) Addition

$$V_1 = 15 + j\,10$$

$$V_2 = 12 + j\,6$$

Resultant voltage, $V = V_1 + V_2$

$$= (15 + j\,10) + (12 + j\,6)$$

$$= (27 + j\,16)$$

Magnitude of resultant voltage $V = \sqrt{(27)^2 + (16)^2} = 31.38$

Its angle with OX-axis, $\theta = \tan^{-1}\dfrac{16}{27}$

$$\therefore \quad \theta = 30.65^\circ$$

(ii) **Subtraction**

Let $V_1 = 15 + j10$

$V_2 = 12 + j6$

Resultant. voltage, $V = V_1 - V_2$

$= (15 + j\,10) - (12 + j\,6)$

$= (3 + j\,4)$

Magnitude of resultant voltage is $= \sqrt{3^2 + 4^2} = 5$

It makes an angle $\theta = \tan^{-1}\frac{4}{3} = 53.1°$ with OX-axis as shown in the fig. ACP—21.

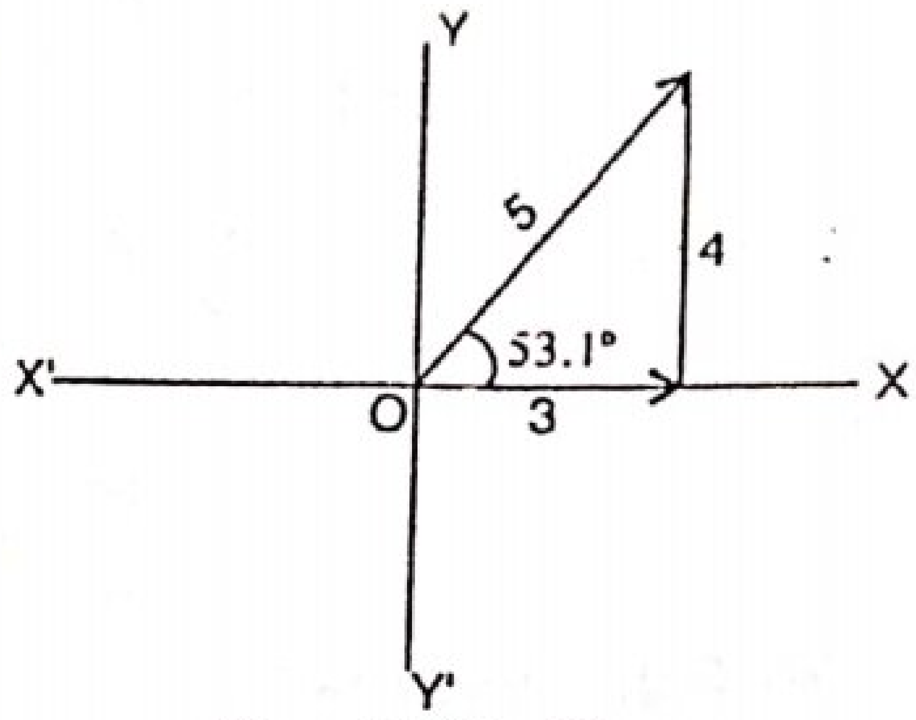

Fig. ACP—21

Example ACP—8. Two alternators having their voltages $V_1 = 25\angle 15°$ and $V_2 = 15\angle 60$ are in series, find the resultant voltage of two alternating quantities.

Solution : Since these values are given in polar form therefore first convert polar form into rectangular form.

$\therefore \quad V_1 = 25(\cos 15° + j \sin 15°) = 24.15 + j\,6.47$

$V_2 = 15(\cos 60° + j \sin 60°) = 7.5 + j\,12.99$

Resultant voltage, $V = 31.65 + j\,19{\cdot}46$

Magnitude of resultant voltage $= \sqrt{(31.65)^2 + (19.46)^2} = 37.15$

Its angle from OX-axis, $\theta = \tan^{-1}\frac{19.46}{31.65} = 31.6°$

$\therefore \quad$ Voltage $V = 31.15\angle 31.6°$ volts

Example ACP—9. Given following two phasors. $A = 4 + j\,3$, $B = 5 + j6$

Perform the following indicated operations

(i) $A \times B$ and (ii) $\frac{A}{B}$

Solution : *(a)* Rectangular Form :—

(i) Given , $A = 4 + j\,3$

$B = 5 + j\,6$

$A \times B = (4 + j\,3) \times (5 + j6)$

$= (20 + j\,24 + j\,15 + j^2\,18)$ (Since $j^2 = -1$)

$= (2 + j\,39)$ **(Ans)**

(ii) $A = 4 + j3$

$B = 5 + j\,6$

$$\frac{A}{B} = \frac{4+j3}{5+j6} \times \frac{5-j6}{5-J6}$$

$$= \frac{20-j24+j15-j^2 18}{5^2-j^2(6)^2} = \frac{38 \quad j9}{61}$$

$$= \frac{38}{61} - j\frac{9}{61} = \textbf{(0.6 23 - } \boldsymbol{j} \textbf{ 0.147) Ans.}$$

(b) Polar form :—

(i) $A = 4 + j3 = 5 \angle 36.87°$

and $B = 5 + j6 = 7.81 \angle 50.19°$

$$A \times B = 5 \angle 36° \times 7.81 \angle 5.19°$$
$$= 5 \times 7.81 \angle 36.87° + 50.19$$
$$= 39.05 \angle 87.06°$$

(ii) $A - 4 + j3 = 5 \angle 36.87°$

$B = 5 + j6 = 7.81 \angle 50.19°$

$$\therefore \quad \frac{A}{B} = \frac{5\angle 36.87°}{7.81\angle 50.19}$$

$$\frac{A}{B} = \frac{5}{7.81} \angle 36.87° - \angle 50.19$$
$$= 0.64 \angle -13.32°$$

The negative sign indicates that the phasor is below X-X′ axis.

Example ACP—10. A voltage expressed by the equation $e = 141.4 \sin (314t + \frac{\pi}{4})$ is applied across a circuit having resistance of 8 ohms in series with an inductive reactance of 5 ohms. Hence, find the following.

(i) Expression for current (ii) r.m.s. value for current

(iii) Power factor (iv) Power consumed by the circuit

Solution : Given Resistance, $R = 8\Omega$

Inductive Reactance, $X = 6\ \Omega$

Impedence, $Z = (8 + j6)$

R.M.S. value of voltage. $V = \frac{141.4}{\sqrt{2}} = 100V.$

Converting into polarform,

Voltage, $V = 100 \angle 45°$

Impedence, $Z = (8 + j6) = 10 \angle 36.9°$ $\quad (\because \tan\phi = \frac{6}{8})$

Value of current, $I = \frac{V}{Z} = \frac{100\angle 45^\circ}{10\angle 36.9^\circ} = 10\ \angle 45^\circ - \angle 36.9^\circ = 10\ \angle 8.1^\circ$

From the value of current, it is seen that current leads the reference axis OX-by 8.1° and lags the voltage by (45° − 8.1°) = 36.9°

Max. value of current, $I_m = \sqrt{2} \times 10 =$ **14.14A**

(i) $i = 14.14 \sin(314\,t + 8.1^\circ)$ (Ans.)

(ii) $I_{r.m.s.} =$ **10** A. (Ans.)

(iii) Power factor $\cos\phi = \cos 36.9^\circ =$ **0.8 (Ans.)**

(iv) Power $= V.I.\cos\phi$.

$= 100 \times 10 \times 0.8 = 800$ watts **(Ans.)**

Example ACP—11. The current in circuit is given by $(4.5 + j12)$ A, when the applied voltage is $(100 + j150)$ volts. Determine (i) the magnitude of impedence and (ii), phase angle.

Solution : Given value of current, $I = (4.5 + j\,12)$

Voltage, $V = (100 + j150)$

$$\text{Impedence. } Z = \frac{V}{I} = \frac{(100+j150)}{(4.5+j12)} = \frac{180.28\angle 56.31^\circ}{12.82\angle 69.44^\circ}$$

$= 14.06\ \angle -13.13^\circ$

$\therefore$ Impedence, Z = **14.06 (Ans.)**

(ii) Phase angle, ϕ = 13.13° (leading)

It is clear that the current is leading the voltage by 13.13, and hence the circuit is capacitive.

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Example ACP—12. For the circuit shown in fig. ACP—22, find out the current drawn from the source and branch currents. Also draw the phasor diagram.

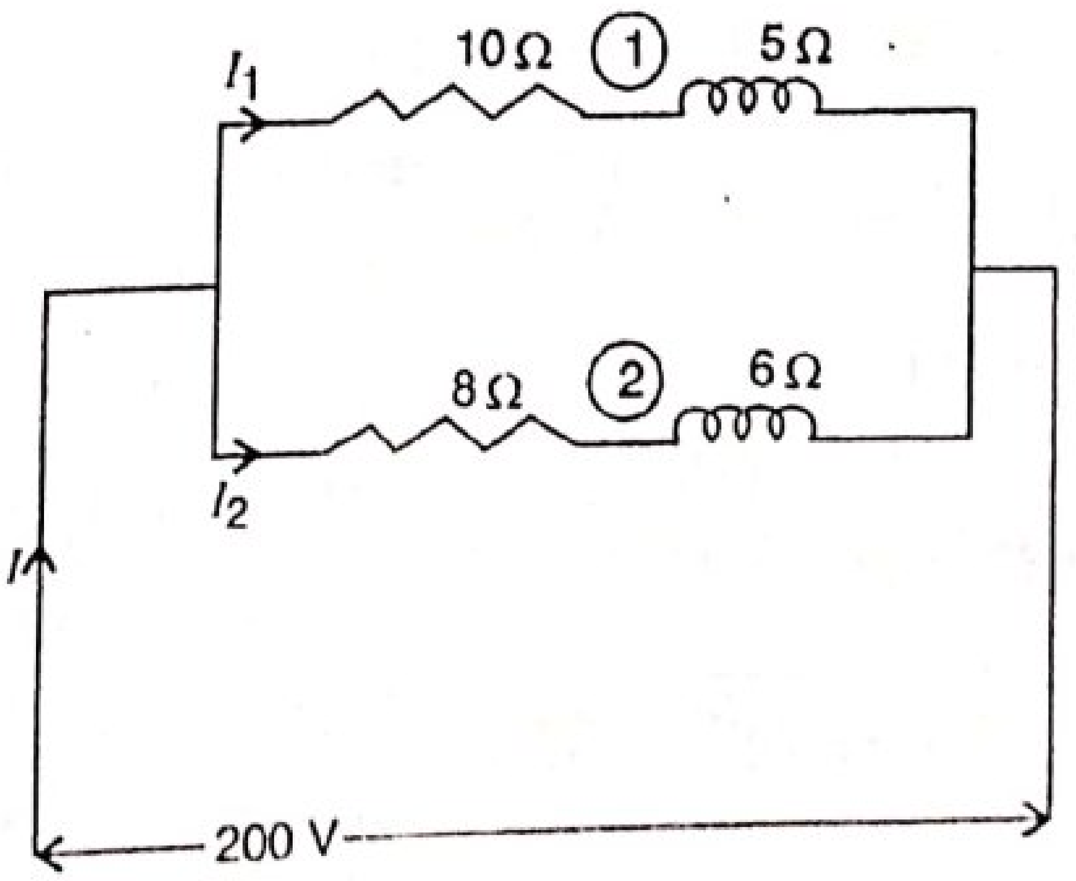

Fig. ACP—22

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Solution : **For Branch no. 1**

Given Resistance, $R_1 = 10\ \Omega$

Reactance, $X_1 = 5\ \Omega$

Impedence, $Z_1 = (10 + j\,5)$

$$\text{Admittance, } Y_1 = \frac{1}{(10+j5)} \times \frac{(10-j5)}{(10-j5)} = \frac{10-j5}{100+25}$$

$$= (0.08 - j\,0.04) \text{ mho.}$$

For Branch no. 2

Resistance, $R_2 = 8\ \Omega$

Reactance, $X_2 = 6\ \Omega$

Impedence, $Z_2 = (8 + j6)$

$$\text{Admittance } Y_2 = \frac{1}{(8+j6)} \times \frac{(8-j6)}{(8-j6)} = \frac{8-J6}{64+36} = (.08 - j\,.06) \text{ mho.}$$

Branch current. $I_1 = V.Y_1 = (200 + j0) \times (.08 - j\,0.04)$

$= (16 - j\,8)$

Magnitude of current, $I_1 = \sqrt{(16)^2 + (8)^2} = 17.88$ A

Phase. angle, $\phi_1 = \tan^{-1}\left(\frac{-8}{16}\right) = -26.56°$

Branch current I_1 has the magnitude of 17.88 A and it lags the voltage by 26.56° as shown in the phasor diagram.

Branch current, $I_2 = V.Y_2$

$= (200 + j\,0) \,.\, (.08 - j\,.06) = (16 - j12)$

Magnitude of current, $I_2 = \sqrt{(16)^2 + (12)^2} = 20\text{A}$

Phase-angle $\phi_2 = \tan^{-1} \frac{-12}{16} = -36.86°$

Total current I drawn by the circuit

$I = I_1 + I_2 = (16 - j\,8) + (16 - j12)$

$= (32 - j20)$

Magnitude of current, $I = \sqrt{(32)^2 + (20)^2} = 37\ 73$

phase angle, $\phi = \tan^{-1} \frac{-20}{32} = -32°$ (Ans.)

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The phasor diagram shows the position of three currents, I_1, I_2 and I.

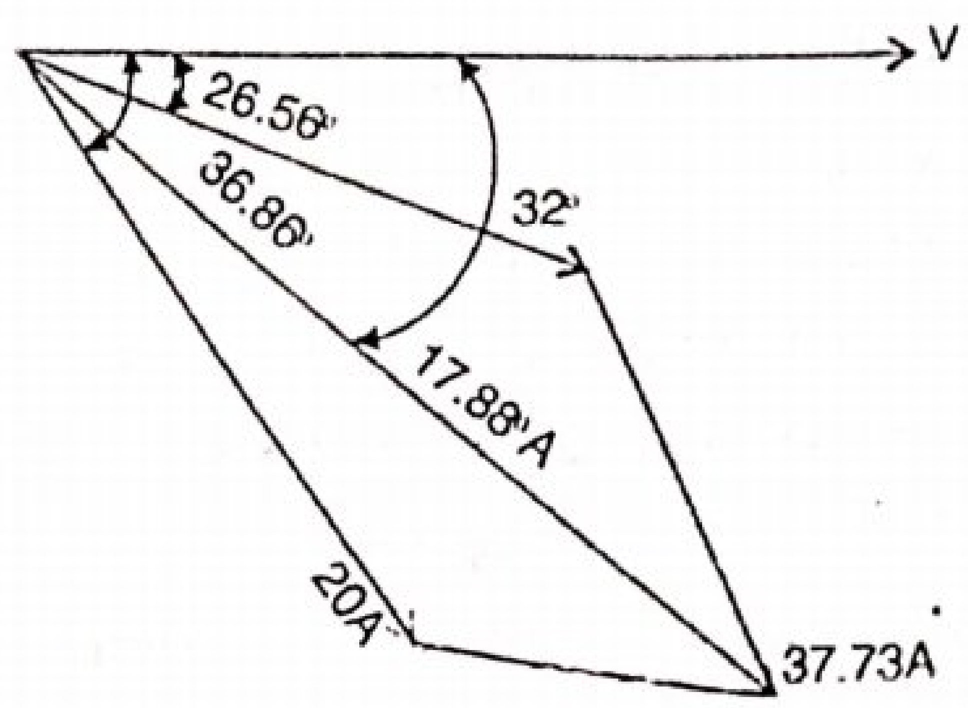

Fig. ACP—23

The total current I can also the determined by the alternate method

i.e. Total admittance

$$Y = Y_1 + Y_2$$

$$\therefore Y = (0.08 - j\,0.04) + (0.08 - j\,.06)$$

$$= (0.16 - j0.1)$$

$\therefore$ Total current, $I = V.Y$

$$= (200 + j\,0)\,(0.16 - j\,0.1)$$

$$= (32 - j20)$$

$$\text{Magnitude, } I = \sqrt{(32)^2 + (20)^2} = 37.73$$

$$\tan\phi = \frac{-20}{32} \quad \therefore \phi = \tan^{-1}\frac{-20}{32} = \mathbf{-32°.\ (Ans.)}$$

Example ACP—13. A coil of resistance 12 Ω and inductive reactance of 25 Ω is connected in series with a capacitive reactance of 41 Ω. The entire combination is connected to a supply of 230 V, 50 Hz. Find (i) circuit impedence (ii) Current and (iii) Power consumed.

Solution : Given :

Resistance, $R = 12\ \Omega$

Inductive reactance, $X_1 = j\,25$

Capacitive reactance, $X_C = -j\,41$

Net. reactance, $X = (j\,25 - j\,41) = -j16$

(i) $\therefore$ Circuit impedence, $= (R - j\,16)$

$$Z = R + (-j\,16)$$

$$= (12 - j\,16) = 20\,\angle{-53.13°}$$

$\therefore$ Magnitude of impedence = **20 ohms. (Ans.)**

(ii) Taking voltage as reference phasor.

$$V = (230 + j\,0) = 230\,\angle 0$$

$$\therefore \quad \text{Current, } I = \frac{V}{Z} = \frac{230}{20\angle -53.13} = \mathbf{11.5\,\angle 53.13°}\ \text{(Ans.)}$$

Power factor, $\cos\phi = \cos 53.13° = 0.6$ leading

Power consumed, $P = V.I.\cos\phi$

$= 230 \times 11.5 \times 0.6 =$ **15.87 watts (Ans.)**

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ACP–5. RESONANCE IN A.C. PARALLEL CIRCUITS

A parallel circuit is said to be in electrical resonance when the reactive (or wattless) component of line current is zero. The frequency at which this happens is known as resonant frequency (f_r).

Figure ACP–24 *(a)* shows a parallel circuit having a coil in parallel with a capacitor.

The vector diagram is shown in fig. ACP–14 *(b)*.

Net reactive component $= I_C - I_L \sin\phi_L$

At resonance this reactive component should be zero.

$\therefore$ $$I_C - I_L \sin\phi_L = 0$$

or $$I_C = I_L \sin\phi_L \qquad \ldots(i)$$

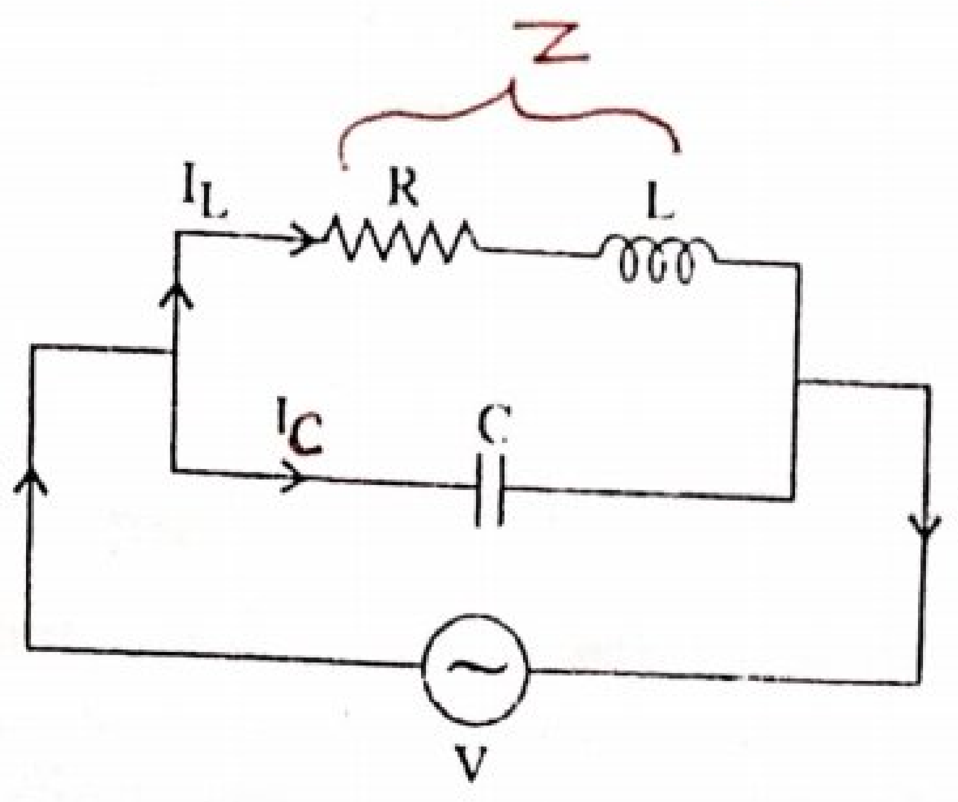

Fig. ACP—24*(a)*

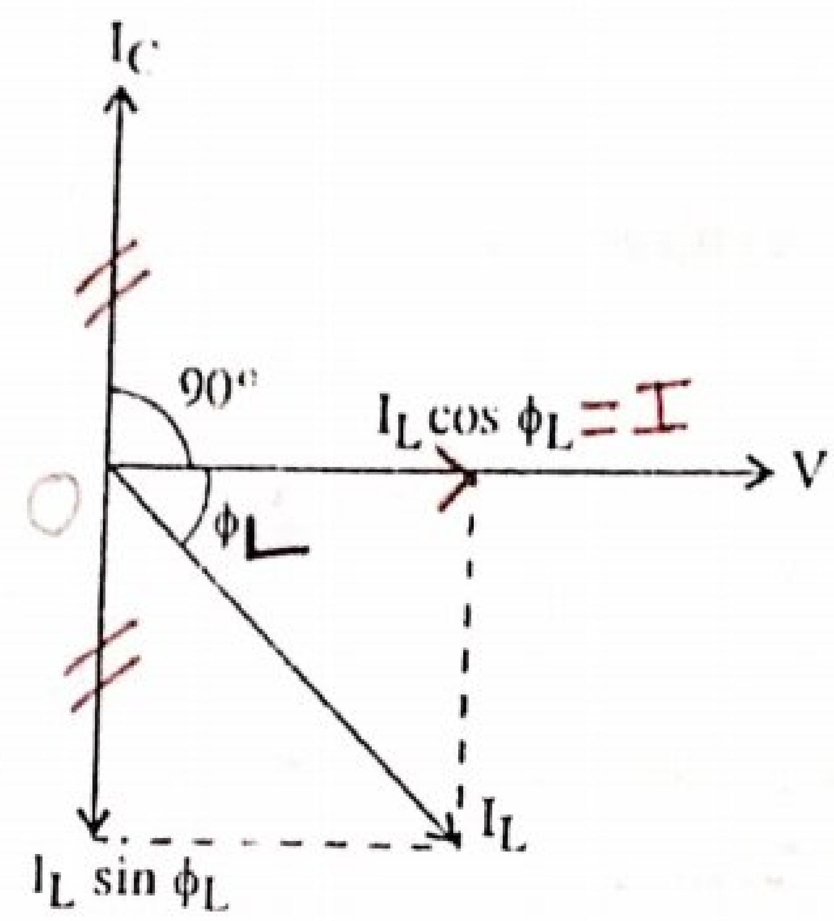

Fig. ACP—24*(b)*

Now $\quad I_L = \dfrac{V}{Z}$ where Z is the impedence of the coil $\quad$ or $I_L = \dfrac{V}{\sqrt{R+X_L^2}}$

$$\sin\phi_L = \frac{X_L}{Z} \quad \text{and} \quad I_C = \frac{V}{X_C} = \frac{V}{X_C}$$

By putting these values in equation *(i)*,

$$\frac{V}{X_C} = \frac{V}{Z}\cdot\frac{X_L}{Z}$$

Also $\quad X_L = \omega L \quad$ and $\quad X_C = \dfrac{1}{\omega C}$

$\therefore$ $$\omega C = \frac{\omega L}{Z^2} \quad \text{or} \quad \frac{L}{C} = Z^2$$

or $$\frac{L}{C} = R^2 + X_L^2 \quad \text{or} \quad \frac{L}{C} = R^2 + (2\pi\, fr\,.\,L)^2 \qquad \ldots(ii)$$

$\therefore$ $$\frac{L}{C} - R^2 = (2\pi\, fr\, L)^2$$

$$(2\pi f_r)^2 = \frac{1}{LC} - \frac{R^2}{L^2}$$

$$\therefore \quad f_r = \frac{1}{2\pi}\sqrt{\frac{1}{LC} - \frac{R^2}{L^2}}$$

This is the resonant frequency and its units are Hz, if R is in ohms, L in Henry and C in farads. Also if R is negligible, then,

$$\boxed{f_r = \frac{1}{2\pi\sqrt{LC}}}$$

ACP–6. VALUE OF CURRENT AT RESONANCE

As shown in the vector diagram and discussed in the above article that is, at resonance, the reactive component or wattless component is zero therefore, the circuit current is minimum *i.e.* $I_{min} = I_L \cos\phi_L$:

$$\therefore \quad I_{min} = \frac{V}{Z} \cdot \frac{R}{Z} = \frac{VR}{Z^2}$$

From the equation *(ii)*, the value of $Z^2 = \dfrac{L}{C}$

$\therefore$ Putting the value, we get,

$$I_{min} = \frac{VR}{L/C} = \frac{VRC}{L} = \frac{V}{L/CR}$$

$I_{min} = \dfrac{V}{\frac{L}{CR}}$

equivalent impendence] – $\dfrac{L}{CR}$.

The quantity $\dfrac{L}{CR}$ is known as the equivalent impedence of the parallel circuit at resonance. since current is minimum at resonance, hence such a circuit is sometimes called rejector circuit because it rejects that frequency to which it resonates. This resonance is also sometimes referred as current resonance.

This phenomenon of parallel circuit resonance is of much importance because this forms the basis of tuned circuits in electronics.

ACP–7. GRAPHICAL REPRESENTATION OF PARALLEL RESONANCE

We will discuss the effect of variation of frequency on the susceptances of the two parallel branches. The variations are shown in figure ACP–25.

(i) **Inductive Susceptance.** It is reciprocal of inductive reactance *i.e.*,

$$b = \frac{1}{X_L} = \frac{1}{\omega L} = \frac{1}{2\pi f \cdot L}.$$

It is obvious that inductive susceptance is inversly proportional to the frequency of the supply. It is represented by a rectangular hyperbola in the fourth quadrant because it is regarded negative.

(ii) **Capacitive Susceptance.** It is reciprocal of capacitive reactance *i.e.*

$$b = \frac{1}{X_C} = \omega C = 2\pi f \cdot C.$$

It increases with the increase in frequency of the supply voltage. Hence it is represented by a straight line drawn in the first quadrant because it is regarded positive.

(iii) **Net Susceptance B.** It is the difference of the two susceptances and has been shown by dotted hyperbola, then at point A, net susceptance is zero.

(iv) **Admittance.** Admittance is reciprocal of impedence *i.e.*

$Y = \frac{1}{Z} = \frac{1}{\sqrt{R^2 + X^2}}$ variations of admittance (Y) are shown in fig. ACP–25. It has large values for frequencies, lower as well as higher than the resonant frequency f_r. If $f < f_r$, Y is inductive and for $f > f_r$, it is capacitive in nature. It has minimum value of $Y = G$ at $f = f_r$. Its value would have been zero if R had been zero (because then $G = 0$).

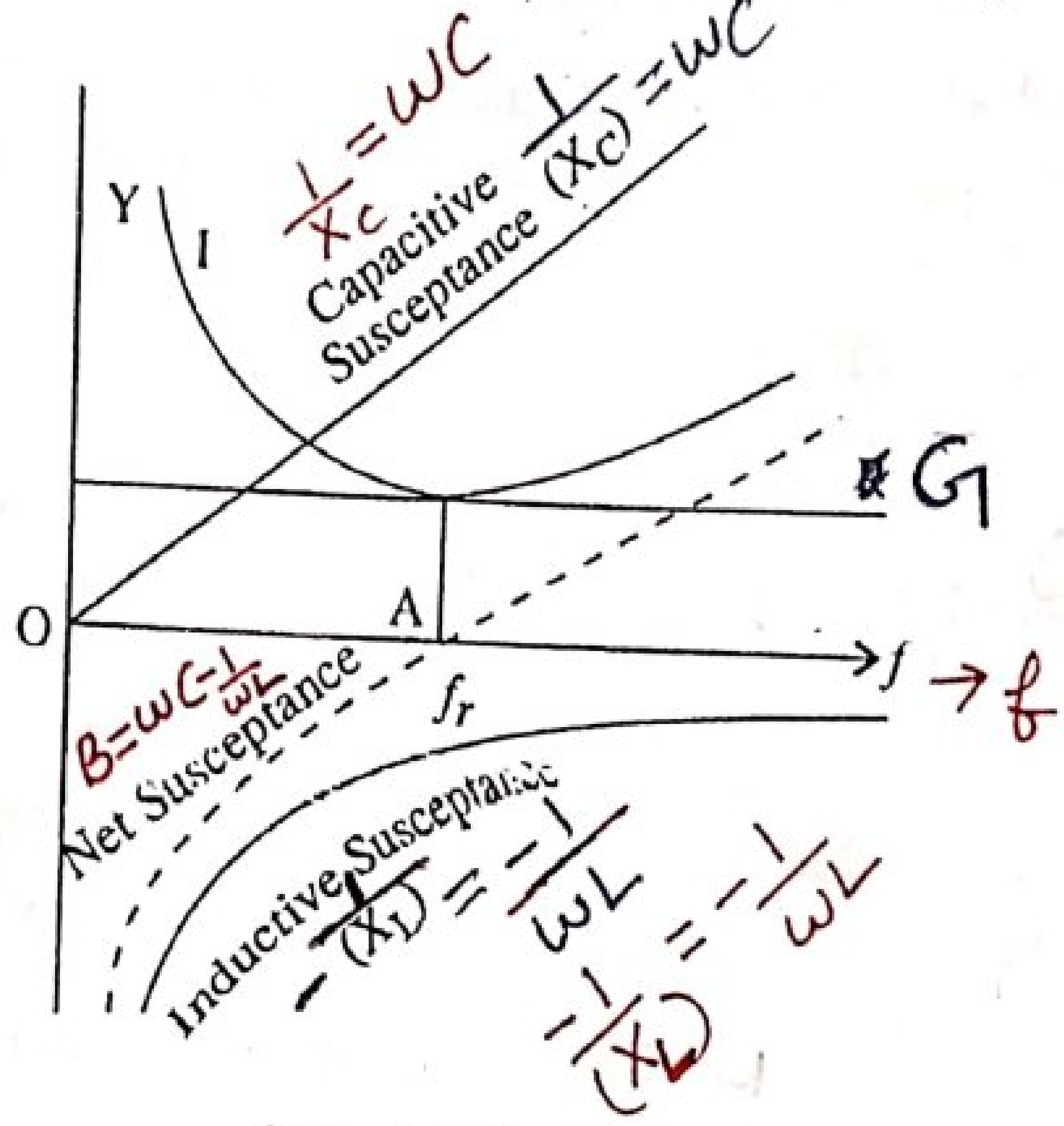

Fig. ACP—25

(v) **Current.** Current varies exactly in the same manner as Y. It has minimum value at A where $f = f_r$) and is equal to $VY = VG$.

From the figure ACP–25, it is clear that below resonant frequency (corresponding to point), inductive susceptance predominates, and so the line current lags behind the applied voltage ut for frequencies more than resonant frequency, capacitive susceptance predominates and so the urrent leads. The following points are worth noting about parallel resonance and be compared ith those about series resonance.

(i) Net susceptance is zero *i.e.* $\frac{1}{X_C} = \frac{X_L}{Z^2}$

or $X_L \times X_C = Z^2$ or $Z^2 = \frac{L}{C}$

(ii) Reactive component of current is zero.

(iii) Dynamic impedence $= \frac{L}{CR}$ ohms.

(iv) The value of line current is minimum and is equal to $\frac{V}{L\,/\,CR}$ and in phase with the voltage.

(v) Power factor is unity.

Resonance Curve. It is a graph which is obtained by plotting a relation between current and frequency. A typical resonance curve is shown in figure ACP–26.

The value of line current is minimum $\left(\frac{V}{L\,/\,CR}\right)$ at resonance.

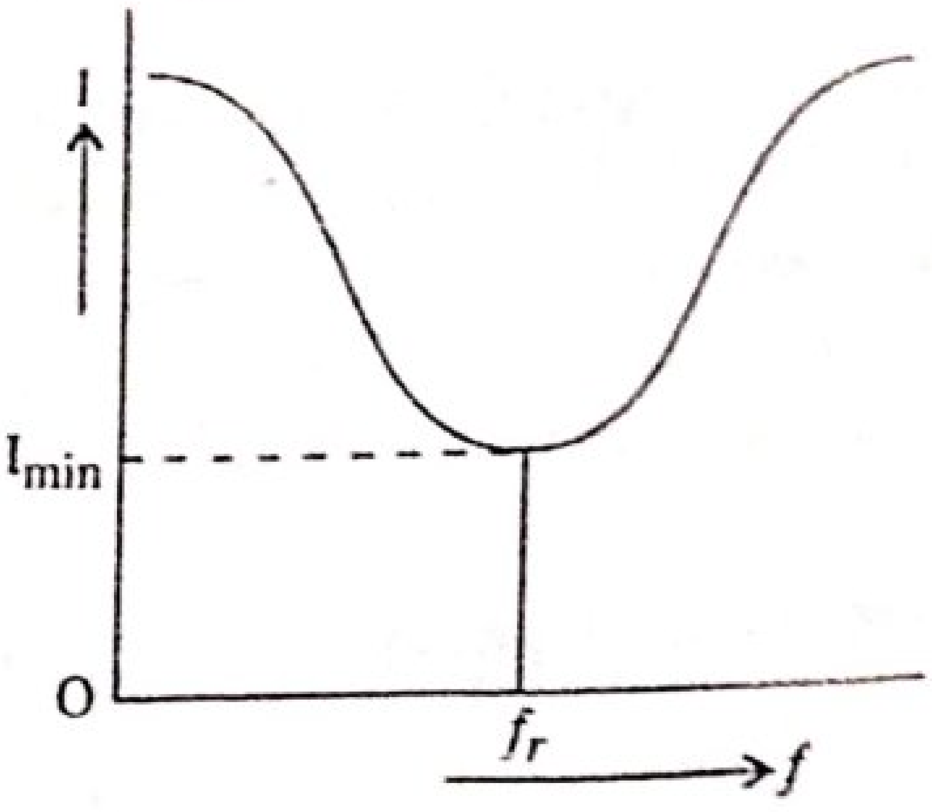

Fig. ACP—26

ACP-8. Q-FACTOR OF A PARALLEL CIRCUIT

In parallel resonance, the current circulating between the two branches is many times more than the line current drawn from the mains. This current amplification produced by the resonance is called Q – factor of the parallel resonant circuit.

Let the circulating current between capacitor and coil is I_C.

I_{min} is the line current during resonance

Then $$Q\text{–factor} = \frac{I_C}{I_{min}}$$

Now $$I_C = \frac{V}{X_C} = \frac{V}{\frac{1}{\omega C}} \cancel{V\omega C} = (V \cdot \omega C)$$

and $$I_{min} = \frac{V}{\frac{L}{CR}} = \frac{VCR}{L}$$

$$\therefore \quad Q\text{–factor} = \frac{V\omega C}{\frac{VCR}{L}} = \frac{\omega L}{R} = \frac{2\pi fr\,L}{R} \quad \text{(Same as for series circuit)}$$

If R is negligible, then $$f_r = \frac{1}{2\pi\sqrt{LC}}$$

Putting this value, we get, $$Q\text{–factor} = \frac{2\pi . L}{R} \cdot \frac{1}{2\pi\sqrt{LC}}$$

or $$Q\text{–factor} = \frac{1}{R}\sqrt{\frac{L}{C}}$$

It should be noted that, in series circuit, Q – factor gives the voltage magnification, whereas in parallel circuits, it gives the current magnification.

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Problem ACP-14. A resonant circuit consists of an 8μF capacitor in parallel with an inductor of 0·15 H having a resistance of 25 Ω. Calculate the frequency of resonance.

Sol. Value of capacitor, $C = 8\mu F = 8 \times 10^{-6}$ F

Value of inductor, $L = 0\cdot15$ H.

Value of resistance, $R = 25\ \Omega$

The resonant frequency, $f_r = \dfrac{1}{2\pi}\cdot\sqrt{\dfrac{1}{LC} - \dfrac{R^2}{L^2}}$

or $$f_r = \frac{1}{2\pi}\cdot\sqrt{\frac{10^6}{0\cdot15 \times 8} - \frac{(25)^2}{(\cdot15)^2}}$$

$$f_r = \frac{1}{2\pi}\cdot\sqrt{83333\cdot3 - 27777\cdot7}$$

$\therefore$ $$f_r = \frac{1}{2\pi} \times 235\cdot7 = \mathbf{37\cdot5\ Hz\ Ans.}$$

Problem ACP-15. A coil of resistance 50 Ω and inductance 30 mH is connected in parallel with a variable capacitor across a supply of 25 V and frequency 1000/π Hz. The capacitance of the capacitor is then varied untill the current taken from supply is minimum (at unity p.f.). For this condition, find,

(i) The capacitance of the capacitor.

(ii) The value of the current.

Sol. Value of resistance, $R = 50\Omega$

Value of inductance, $L = 30$ mH $= 30 \times 10^{-3}$ H

Applied Voltage, $V = 25$ Volts

Supply frequency, $f = \dfrac{1000}{\pi}$ Hz.

Value of inductive reactance

$$X_L = \omega L = 2\pi.\frac{1000}{\pi} \times 30 \times 10^{-3} = \mathbf{60\ \Omega.}$$

Impedence, $Z = \sqrt{R^2 + X_L{}^2} = \sqrt{(50)^2 + (60)^2} = \mathbf{78\cdot10\ ohms.}$

(i) At resonance, $Z^2 = \dfrac{L}{C}$

$\therefore$ $$C = \frac{L}{Z^2} = \frac{30 \times 10^{-3} \times 10^6}{6100} = \mathbf{4\cdot91\ \mu F\ (Ans.)}$$

(ii) $$I_{min} = \frac{V}{L / CR}$$

Dynamic impedence $= L/CR = \dfrac{30 \times 10^{-3} \times 10^6}{4\cdot91 \times 50} = \dfrac{30 \times 10^3}{4\cdot91 \times 50} = 122\ \Omega$

$$I_{min} = \frac{25}{122} = 0\cdot2049 \text{ A (Ans.)}$$

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SUMMARY

1. (a) Admittance $= \dfrac{1}{\text{Impedence}} = \dfrac{\text{R.M.S. value of amperes}}{\text{R.M.S. value of voltage}}$

(b) Conductance $= \dfrac{1}{\text{Resistance}}$ *i.e.*, $g = \dfrac{1}{R}$

(c) Susceptance $= \dfrac{1}{\text{Reactance}}$ *i.e.*, $b - \dfrac{1}{X}$

b is + ve for capacitive circuit and – ve for inductive circuit.

2. The method to express the phasor in the algebric or mathematical form is known as phasor algebra or complex algebra. In this method a vector quantity is expressed algebraically in terms of its rectangular components. This form is also known as rectangular form of notation.

3. **(a)** j is a factor which when multiplied by a phasor V, rotates through 90° in C.C.W. (*Counter clockwise*)

$$j = \sqrt{-1}, \; j^2 = -1$$

(b) In rectangular form, voltage is represented by ;

$$V = a \pm jb$$

where a and b are the horizontal and vertical components of voltage.

(c) In polar form, $V = V(\cos \phi \pm j \sin \phi)$

(d) In polar form, $V = \underline{|+\theta}$ and $\theta = \tan^{-1}\left(\dfrac{b}{a}\right)$

4. For addition and subtraction of vector quantities, rectangular form is best and for multiplication and division, polar form is best.

5. (a) The condition for parallel resonance is that reactive (*wattless*) component of current is zero.

(b) The value of current at resonance is minimum.

(c) The equivalent impedance of a parallel circuit at resonance $= \dfrac{L}{CR}$ and such circuit is also known as rejector circuit.

6. Q-factor of a parallel circuit $= \dfrac{1}{R}\sqrt{\dfrac{L}{C}}$

Q factor gives the voltage magnification in series circuit but in parallel circuit, it gives the current magnification.

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REASONING QUESTIONS

(i) Domestic appliances are always connected in parallel with the supply lines, why ?

(ii) For three phase balanced load, the power factor for the three phases are the same, why ?

(iii) By adding more resistance to an RC circuit, the true power increases, why ?

(iv) The power dissipated in a pure capacitor is zero, why ?

(v) For three phase balanced load, the current in the neutral is zero, why ?

(vi) Parallel resonant circuit is sometimes called the rejecter circuit, why ?

(vii) For R-L-C parallel circuit, the current at parallel resonance is minimum at unity power factor, why ?

(viii) Two watt meters are used for the measurement of power factor of a 3-phase balanced load. If one wattmeter reads positive, the other reads negative, why ?

EXAMINATION QUESTIONS

(Appeared & Expected)

Q. 1. In an alternating circuit, the impressed voltage, V = (100 – j50) volts and the current in the circuit is I = (3 – j4). Determine the real and reactive power in the circuit.

Ans. Given,

$$\overline{V} = (100 - j50)\text{V},$$

$$\overline{I} = (3 - j4)\text{A},$$

$$\text{Power, } S = P + jP_r = \overline{V} \times \overline{I} = (100 - j50)\,(3 + j4)$$

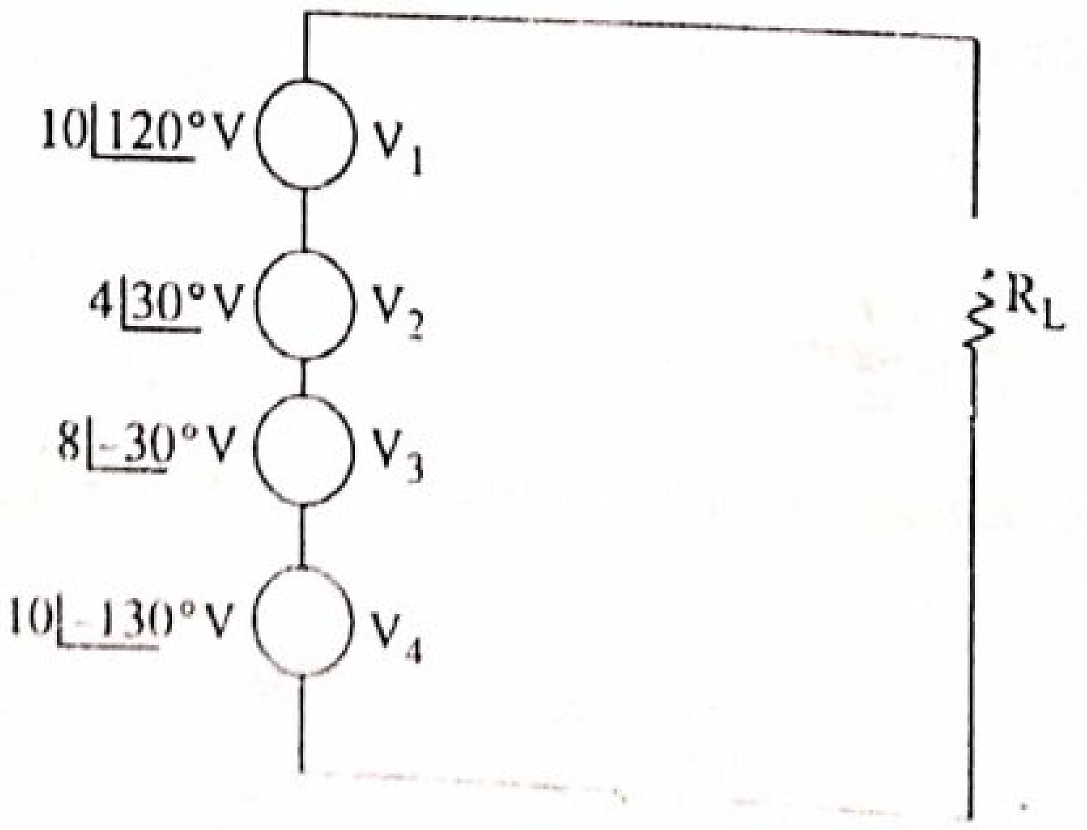
$$= 300 + j400 - j150 + 200$$

$$= 500 + j150$$

Hence active power, P_c = 500 watts or 0·5 kW **(Ans.)**

Reactive power, P_r = 150 VAR **(Ans.)**

Q. 2. Find the total voltage across R_2 as shown in Fig.



Ans. $\vec{V} = \vec{V_1} + \vec{V_2} + \vec{V_3} + \vec{V_4}$ = Voltage across R_L

$$= 10\angle 120^\circ + 4\angle 30^\circ + 8\angle -30^\circ + 6\angle -130^\circ$$

Changing into rectangular form, we get,

$$\vec{V} = 10(\cos 120^\circ + j\sin 120^\circ) + 4(\cos 30^\circ + j\sin 30^\circ) + 8[\cos(-30^\circ) + j\sin(-30^\circ)] + 6[\cos(-130^\circ) + j\sin(-130^\circ)]$$

$$= (-5 + j\,8{\cdot}66) + (3{\cdot}464 + j2) + (6{\cdot}928 - j4) + (-3{\cdot}8568 - j\,4{\cdot}596)$$

$$= 1{\cdot}5352 + j\,0{\cdot}2064$$

$$|V| = \sqrt{(1{\cdot}5352)^2 + (2{\cdot}064)^2}\ \tan^{-1}\frac{2{\cdot}064}{1{\cdot}5352}$$

$$= 2.572\angle 53{\cdot}36^\circ \text{ volts (Ans.)}$$

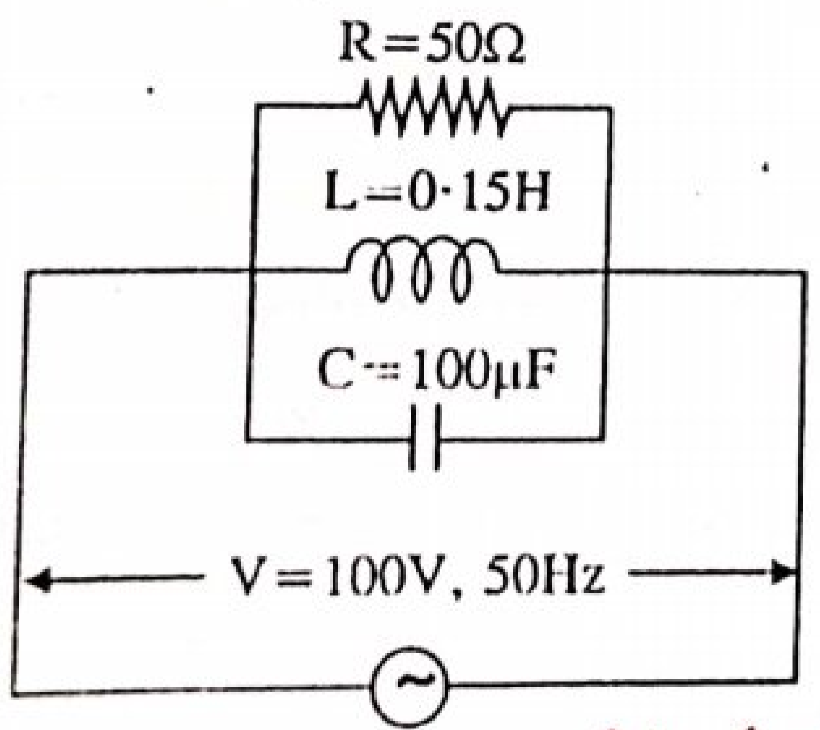
Q. 3. Three branches possessing a resistance of 50Ω and inductance of 0.15H and a capacitance of 100μF respectively are connected in parallel across 100V, 50 Hz supply. Calculate :

(a) current in each branch

(b) supply current

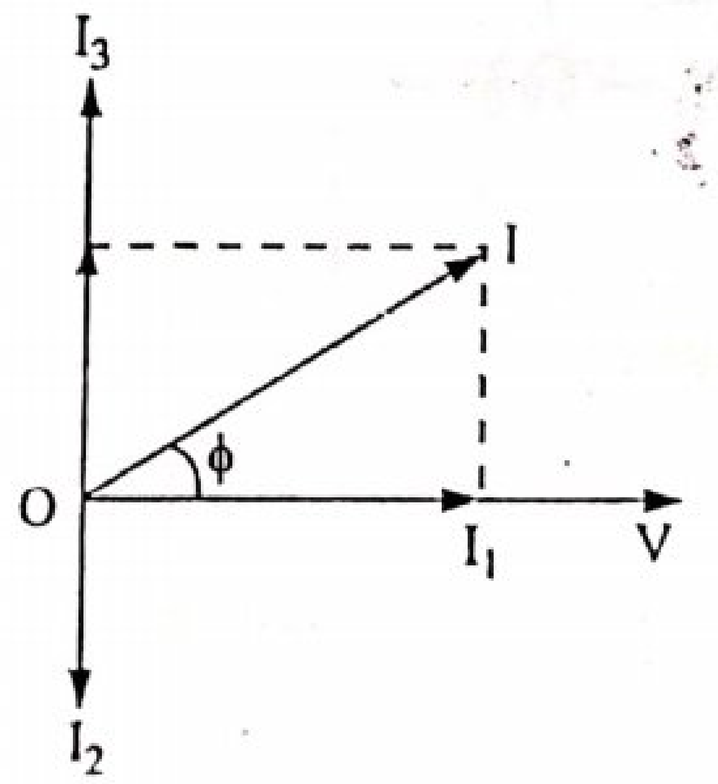
(c) phase angle between supply voltage and current.

Ans. The circuit is shown in fig.



An RLC parallel Circuit

(a)



(b)

Opposition of branch I, $R = 50\ \Omega$

Current, $I_1 = \dfrac{V}{R} = \dfrac{100}{50} = 2\text{A}$ (*in phase with V*) (Ans.)

Opposition of branch II, $X_L = 2\pi fL = 2\pi \times 50 \times 0.15 = 47.12\Omega$

$$\text{Curremt, } I_2 = \frac{V}{X_L} = \frac{100}{43.12} = 2.122 \text{ A } (\textit{lags V by } 90^\circ) \text{ (Ans.)}$$

$$\text{Opposition of branch III, } X_C = \frac{1}{2\pi fC} = \frac{1}{2\pi \times 50 \times 100 \times 10^{-6}} = 31.83\Omega$$

$$\text{Current, } I_3 = \frac{V}{X_C} = \frac{100}{31.83} = 3.14 \text{ A } (\textit{leads V by } 90^\circ) \text{ (Ans.)}$$

The phasor diagram is shown in fig. *(b)*

$$\text{Total current, } I = \sqrt{(I_1)^2 + (I_3 - I_2)^2} = \sqrt{(2)^2 + (3.14 - 2.122)^2}$$

$$= \mathbf{2.245 \text{ A}} \text{ (Ans.)}$$

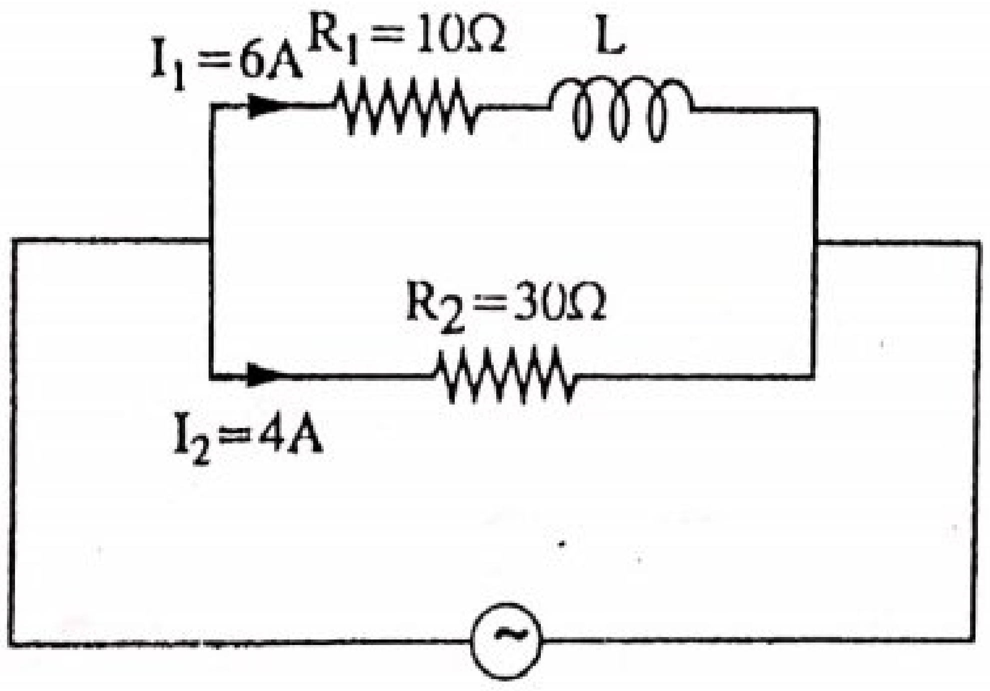
$$\text{Phase angle, } \phi = \tan^{-1}\frac{I_3 - I_2}{I_1} = \tan^{-1} 0.509 = \mathbf{26.97^\circ} \text{ (Ans.)}$$

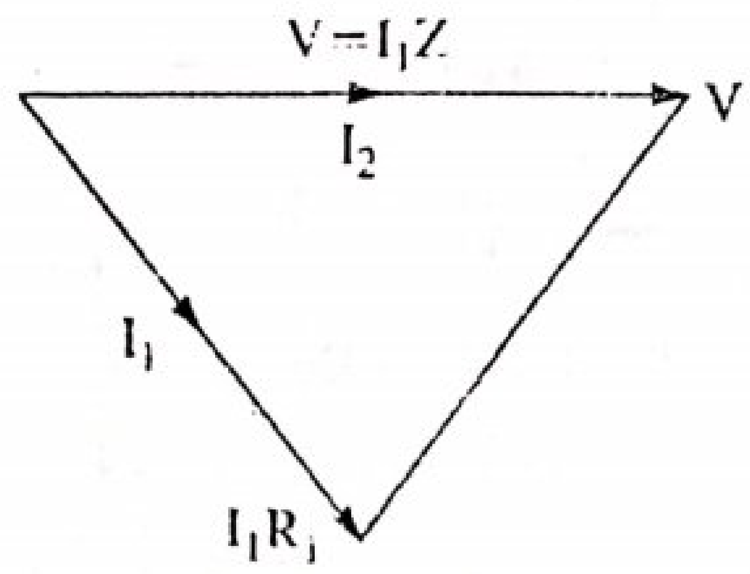
Q. 4. An inductive coil having resistance 10 Ω is connected in parallel across a non-inductive resistance of 30 Ω and this parallel circuit is connected across ac 50 Hz mains. Total current in non-inductive resistance is 4A and that in the inductive coil is 6A. Calculate :

***(i)* inductance of the coil *(ii)* P.F. of the circuit**

***(iii)* Power absorbed by the parallel circuit. Draw the vector diagram also.**

Ans.





The circuit is shown in fig.

$$\text{Applied voltage, V} = I_2R_3 = 4 \times 30 =$$

$$\text{Impedance of the coil, } Z = \frac{V}{I_1} = \frac{120}{6} = 20\Omega$$

$$\text{Inductive reactance of the coil, } X_L = \sqrt{Z^2 - R_1^2} = \sqrt{(20)^2 - (10)^2} = 17.32\Omega$$

$$\text{Inductance, } L = \frac{X_L}{2\pi f} = \frac{17.32}{2\pi \times 50} = 55.13 \text{ mH (Ans.)}$$

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Let, current I_1 lags behind the voltage vector by an angle ϕ_1, then

$$\phi_1 = \tan^{-1}\frac{X_2}{R_1} = \tan^{-1}\frac{17.32}{10} = 60°$$

$$\text{Resultant current, } I = \sqrt{(I_2 + I_1\cos\phi_1)^2 + (I_1\sin\phi_1)^2}$$

$$= \sqrt{(4 + 6\cos 60)^2 + (6\sin 60)^2} = 8.72 \text{ A}$$

$$\text{Power factor, } \cos\phi = \frac{I_2 + I_1\cos\phi_1}{I} = \frac{4 + 6\cos 60°}{8.72} = \mathbf{0.803 \text{ lag (Ans.)}}$$

Power absorbed by the circuit, $P = VI\cos\phi = 120 \times 8.72 \times 0.803 = 837$ W **(Ans.)**

Q. 5. What do you mean by *Q*-factor of a parallel circuit ? What is its importance ?

Ans. In parallel resonance, the current circulating between the two branches is many times more than the line current drawn from the mains. This current amplification produced by the resonance is called Q – factor of the parallel resonant circuit.

It should be noted that, in series circuit, Q–factor gives the voltage magnification, whereas in parallel circuits, it gives the current magnification.

Q. 6. Answer the following :

Fill in the blanks

(i) A circuit has impedence of $(3 + j4)$. If a voltage $(100 + j50)$ is applied, the real power in the circuit will be ....

(ii) Admittance is reciprocal of ....

(iii) The unit of inductive susceptance is ....

(iv) A parallel resonance circuit is known as a .... circuit.

(v) The condition for parallel resonance is that .... component of current should be zero.

(vi) The current is .... under parallel resonance condition.

(vii) If two waves are expressed as ;

$$e_1 = 10\sin\omega t + 25\sin[3\omega t + 30°]$$

and

$$e_2 = 10\sin\omega t + 25\sin[3\omega t + 30°]$$

The *r.m.s.* values of the tw[illegible] induc[illegible] are ....

(viii) The third wire of a 3-pin plug is us[illegible] as an ....

(ix) An alternating voltage of 220 V is given, it gives the .... of voltage.

(x) The formula for calculating the inductive reactance; X_L ....

(xi) The formula for calculating capacitive reactance, $X_C = \dfrac{1}{\text{.........}}$.

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ELECTRICAL ENGINEERING (E)

Type B. Multiple-choice Questions

Select the correct answers

. What is meant by "Current and voltage are in a phase" ?
(a) Voltage and current are of the same magnitude
(b) Voltage and current have the same frequency
(c) The voltage leads the current in time
(d) Voltage and current reach their maximum and zero value at the same time.

. In electric circuit, the lagging power factor is found to be 0·5. This might be a circuit with
(a) parallel connection of ohmic and inductive resistance
(b) resistance connection of ohmic and inductive resistance
(c) pure ohmic resistance
(d) as much ohmic as inductive resistance.

Kirchoff's laws are applicable to
(a) d.c. circuit only
(b) a.c. circuit only
(c) both d.c. and d.c. circuits
(d) None of these.

. The direction of current in an a.c. circuit
(a) is always in one direction
(b) cannot be determined
(c) varies from instant to instant
(a) is from positive to negative.

The maximum and minimum values of power factor can be
(a) 1 and 0
(b) + 1 and – 1
(c) + 1 and – 5
(d) + 5 and – 5.

Two impedences (3 + j 4) and (6 + j 9) are connected in parallel. The combined impedence is
(a) (2 + j 2·67)
(b) (0·2 + j 0·5)
(c) (10 + j 15)
(d) (9 + j 12).

Conductance is the reciprocal of
(a) inductive reactance
(b) capacitive reactance
(c) resistance
(d) impedence.

Admittance is the reciprocal of
(a) inductive reactance
(b) capacitive reactance
(c) resistance
(d) impedence.

A resistance and an inductance are connected in parall' ' across a.c. source. The currents through the two parallel branches will be out of phase '
(a) 90°
(b) 0°
(c) 180°
(d) None of these.

When a parallel circuit is at resonance ;
(a) the circuit is maximum
(b) the circuit impedence is maximum
(c) the circuit phase angle is maximum
(d) the circuit admittance is maximum

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C. State true or false :

(i) The power factor of the load $(8 + j6)$ is zero.

(ii) Admittance $= \dfrac{I}{V} = \dfrac{\text{R.M.S. value of amperes}}{\text{R.M.S value of voltage}}$

(iii) Q-factor of a parallel circuit $= \dfrac{1}{R}\sqrt{\dfrac{L}{C}}$

(iv) Dynamic impedence of parallel resonance $= \dfrac{L}{CR}$ ohm.

ANSWERS

A. (i) 300 watts (ii) impedence (iii) mho (iv) rejector (v) reactive

(vii) equal (viii) earth wire (ix) effective value (x) ωL (xi) $2\pi fc$.

B. 1. (d) 2. (d) 3. (c) 4. (c) 5. (a)

6. (a) 7. (c) 8. (d) 9. (a) 10. (b).

C. (i) False (ii) True (iii) True (iv) True

TEXT QUESTIONS

1. Why is it more economical to use a reactor instead of a resistor for dimming a bank of lamp ?
2. What are active and reactive components of the current and hence define kV A, kW and kVAR ?
3. Derive the relationship between the voltage and current for a purely inductive circuit. Also show that the average power consumed by a circuit is zero.
4. Develop the expression for the mean power consumed over a cycle of a single phase sinusoidal supply delivering power to a load comprising of resistance "R" in series with an inductance L.
5. Explain the following terms as applied to a circuit.
   (i) Impedence (ii) Power (iii) Power factor.
6. Explain, why the phase of voltage across the inductor leads the current phasor by 90° and the phase of voltage across capacitor lags its current by 90°.
7. Explain the terms (a) apparent power (b) active power (c) reactive power and (d) power factor.
8. Define inductive reactance [illegible] acitive reactance.
9. Draw the phasor diagram of an R.L.C. series circuit.
10. Define conductance, susceptance and admittances.
11. What are the methods of solving a parallel circuit, explain with the help of an example ?
12. Define admittance, and prove that in a parallel circuit, I = V Y What are their units ?
13. Explain, the use of capacitor for power factor improvement.

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